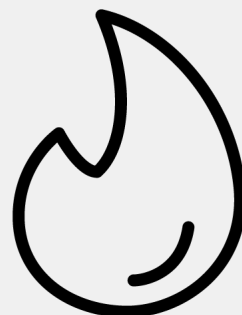
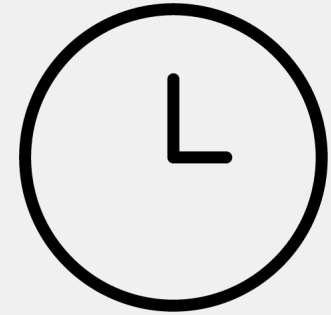
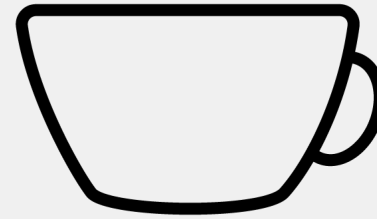
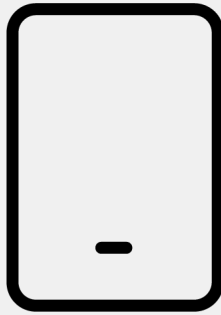


Network Infrastructure

Day 1



Housekeeping



How to complete this module

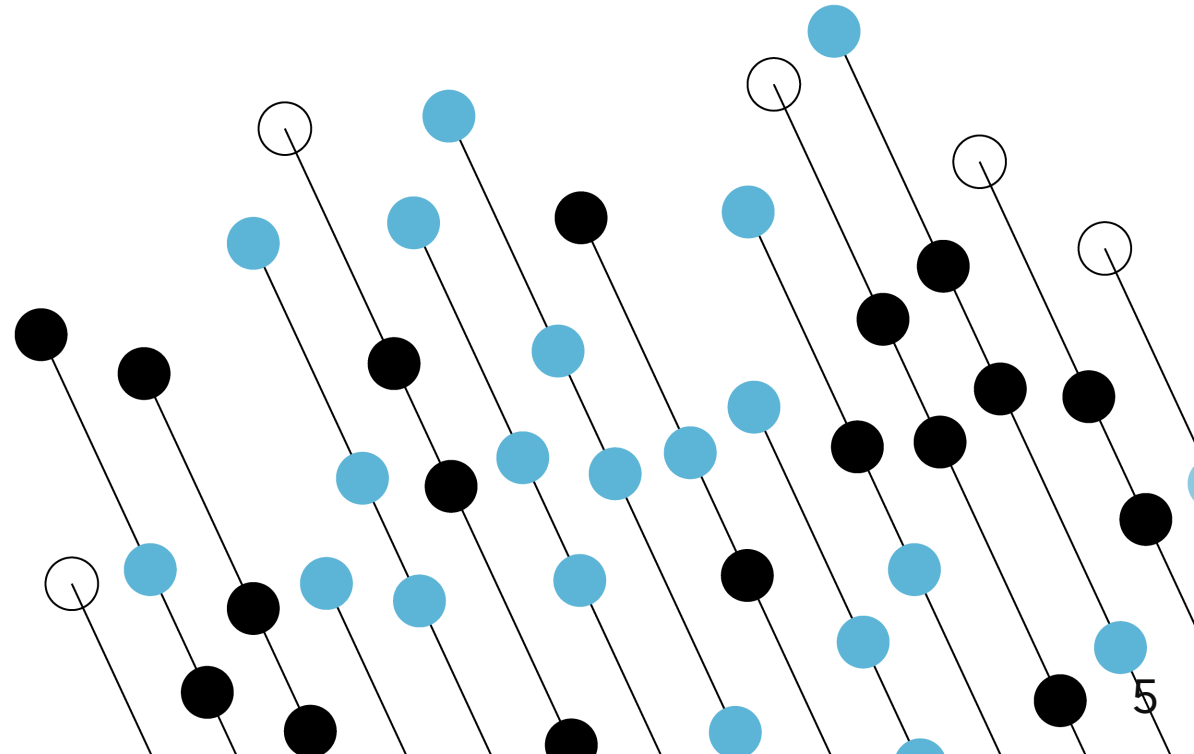
As part of your live event you will have the activity book and labs to complete:

- Open your '**Learner Activity Document**' for this course and use it to document your answers.
- Add your **full name and date** in the '**header**' to your **Learner Activity Document**.
- You may need to ask questions / do some research to complete some of the activities.
- Once your **Learner Activity Document** is completed, upload it to **Bud** along with your **Certificate of completion**.
- And most importantly, if you need **help, message or email your tutor**.

Learning outcomes

- Compare and contrast various devices, their features, and their appropriate placement on the network.
- Explain the characteristics of network topologies and network types.
- Summarise the types of cables and connectors and explain which is the appropriate type for a solution.
- Compare and contrast the Open Systems Interconnection (OSI) model layers and encapsulation concepts.
- Configure a subnet and use appropriate IP addressing schemes.
- Explain common ports and protocols and their application.
- Explain the use and purpose of network services.
- Summarise types of virtualisation and usage.
- Install and configure basic wireless standards and technologies.

Types of network



Types of network: Session objectives

- Describe network types.
- Explain the types of topologies.
- Differentiate between physical and logical topologies.
- Setup a client / server network.
- Analyse a case-study scenario and provide appropriate recommendations.

Network types

Peer-to-peer vs. client / server networks

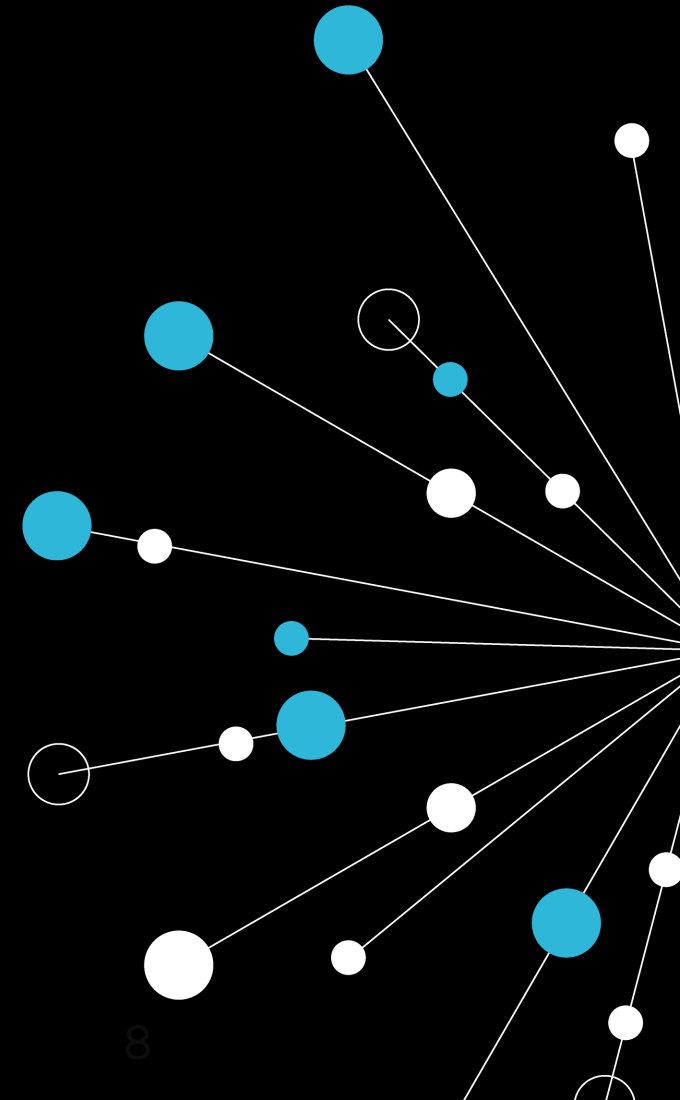
Comparing network models

- How do peer-to-peer networks differ from client / server networks in terms of architecture and functionality?
- What are the advantages and disadvantages of each model?
- Why is a client / server model preferred in enterprise environments?
- What are some examples of peer-to-peer networks, and why are they still relevant today?
- How does the centralised nature of client / server networks enhance security?
- What are the challenges of maintaining performance in large peer-to-peer networks?

Activity 1: Network types

Find the activity in your **Learner Activity Document**, and read the instructions:

- Evaluate a network design scenario and recommend a network type.



Network zones

Network zones: (Perimeter Network, extranet, intranet)

- What is the purpose of a Perimeter Network in network security, and how does it protect internal systems?
- What services are typically placed in a Perimeter Network, and why?
- How do extranets and intranets differ in terms of access and functionality?
- What security measures should be implemented for extranets to prevent unauthorised access?
- In what scenarios would a business deploy an extranet to collaborate with partners or customers?
- How can intranet access improve productivity within an organisation?

Network topologies

Physical topologies: (bus, star, mesh, ring)

- What are the key characteristics of each physical topology: bus, star, mesh, and ring?
- How does the topology choice affect network performance and reliability?
- In what scenarios would you choose a bus topology over a star topology?
- How is a mesh topology used in modern wireless networks and IoT applications?

Network topologies (continued)

- How does a single point of failure impact a bus or ring topology?
- What makes a star or mesh topology more fault-tolerant compared to other designs?
- How does the cost of implementation differ across various topologies?
- What are the challenges of scaling a ring or mesh topology as the network grows?

Network implementation types

Network implementation types: (LAN, WAN, WLAN, MAN)

- How do LAN, WAN, WLAN, and MAN differ in terms of scale, speed, and purpose?
- What factors determine the choice of network type for a specific organisation or use case?
- Can you provide examples of how LAN, WAN, WLAN, and MAN are used in different industries?

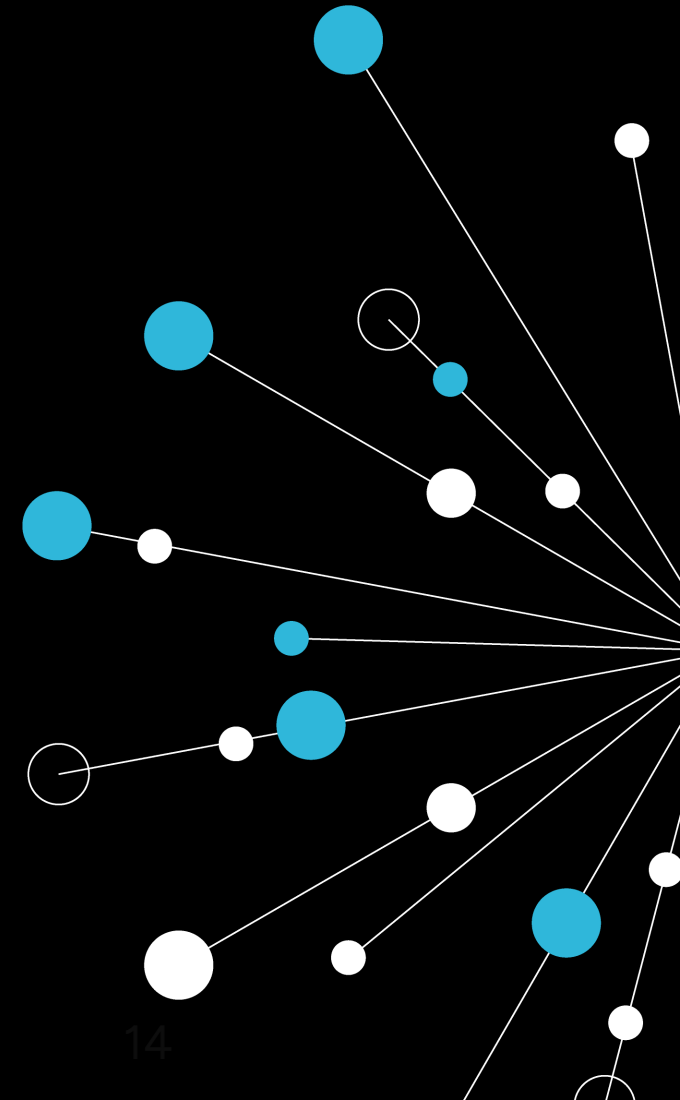
Network implementation types (continued)

- How does the role of MAN differ in metropolitan areas compared to rural settings?
- What are the challenges of integrating WLAN into an existing LAN or WAN infrastructure?
- How do organisations manage security and bandwidth in WAN environments with global users?

Activity 2: Network zones

Find the activity in your **Learner Activity Document**, and read the instructions:

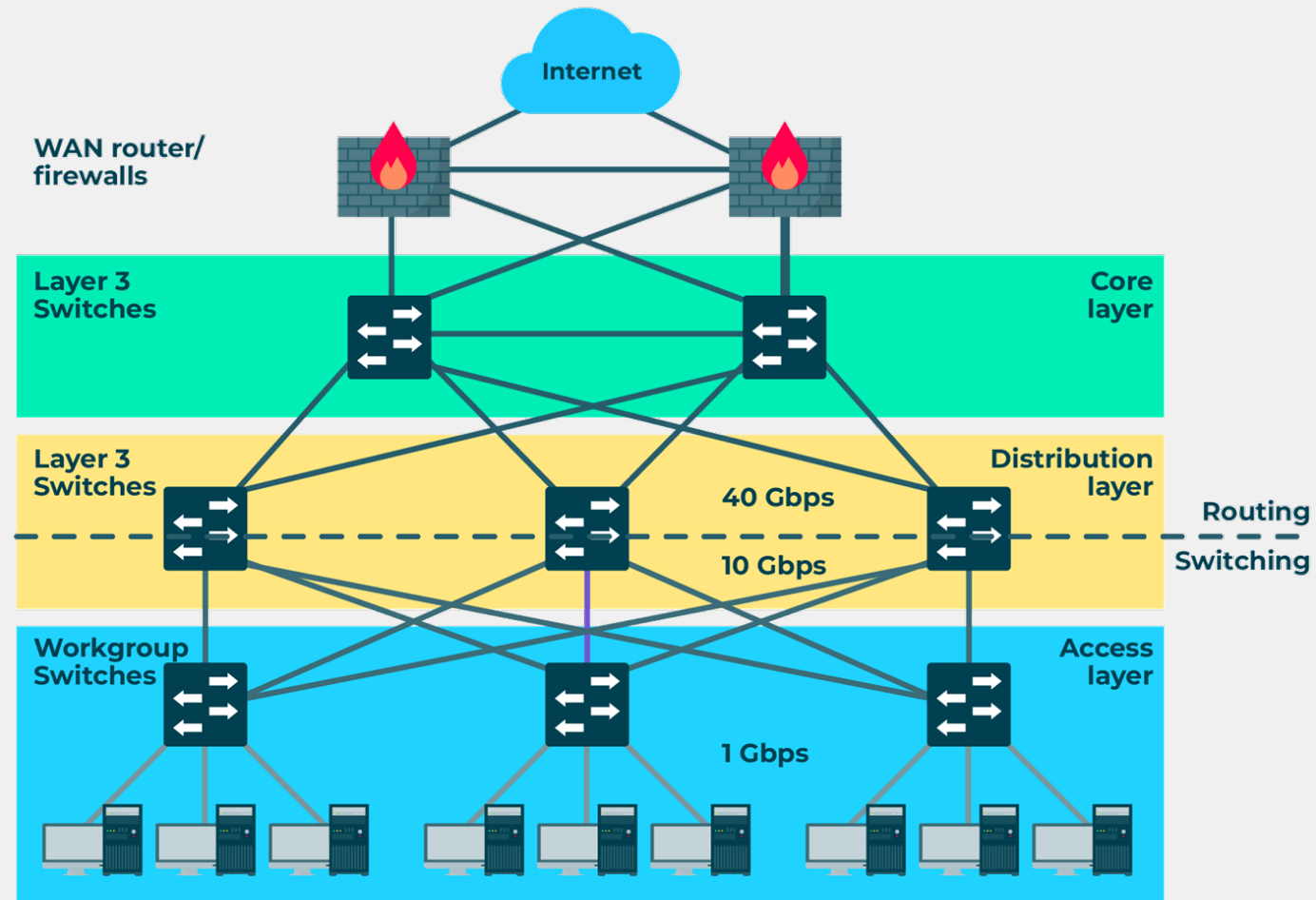
- Evaluate a network design scenario and recommend a network zone type.



Three-tiered network hierarchy

- What are the three layers of Cisco's hierarchical network model?
- Why is the three-tiered architecture considered to be an industry best practice?
- How does this model improve network scalability and performance?

Three-tiered network hierarchy (continued)



Three-tiered network hierarchy: The core layer

The core layer:

- What is the primary function of the core layer in a three-tier network?
- How does redundancy and high-speed connectivity impact core layer design?
- Why is it recommended to minimise packet filtering and complex processing at the core layer?

Three-tiered network hierarchy: The distribution layer

The distribution layer:

- What role does the distribution layer play in the hierarchical network model?
- How does the distribution layer enhance network security and policy enforcement?
- What is the significance of Layer 3 switching in the distribution layer?
- How does load balancing between core and access layers improve network efficiency?

Three-tiered network hierarchy: The access layer

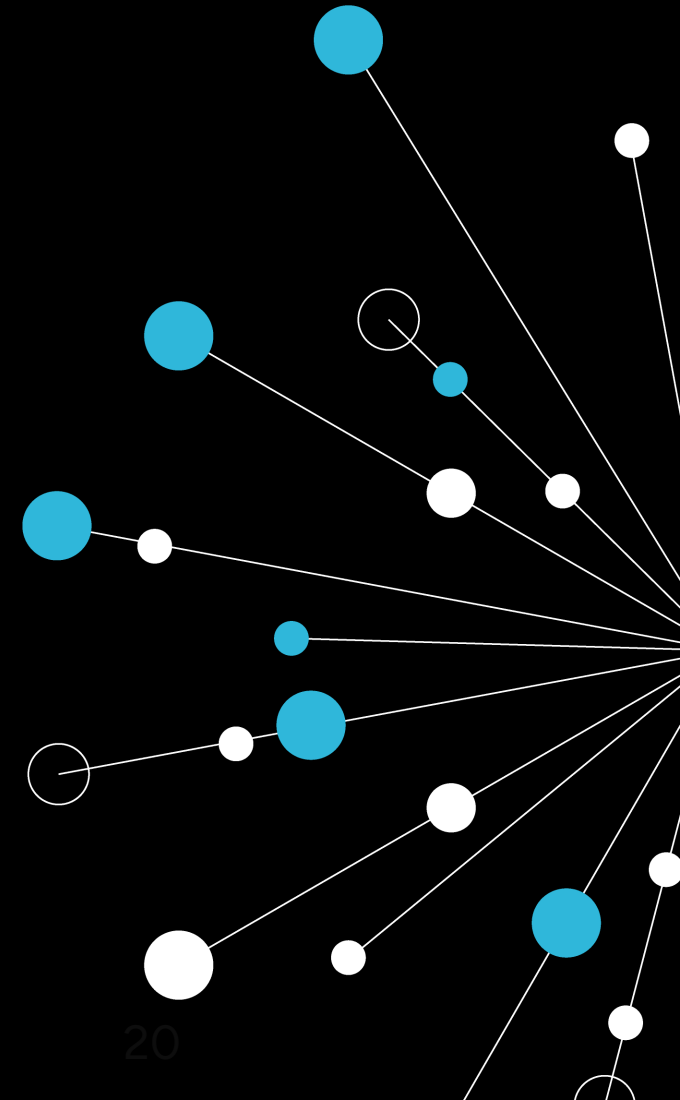
The access layer

- How does the access layer facilitate end-user connectivity?
- What are the best practices for implementing VLANs at the access layer?
- How does port security at the access layer contribute to network security?
- What are the differences between Layer 2 and Layer 3 access layer designs?

Activity 3: Network design activity

Using Packet Tracer, design your proposed network solution from activities 1 and 2.

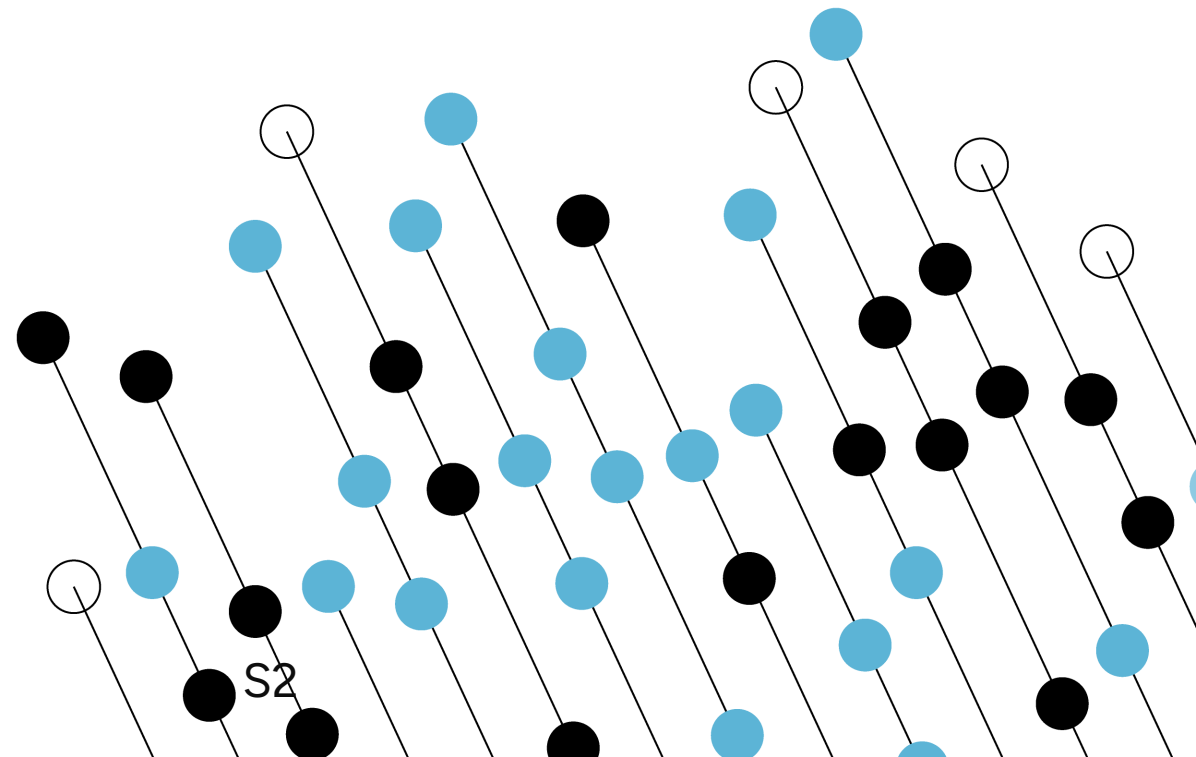
Be ready to justify your solution to the rest of the group.



Types of network summary

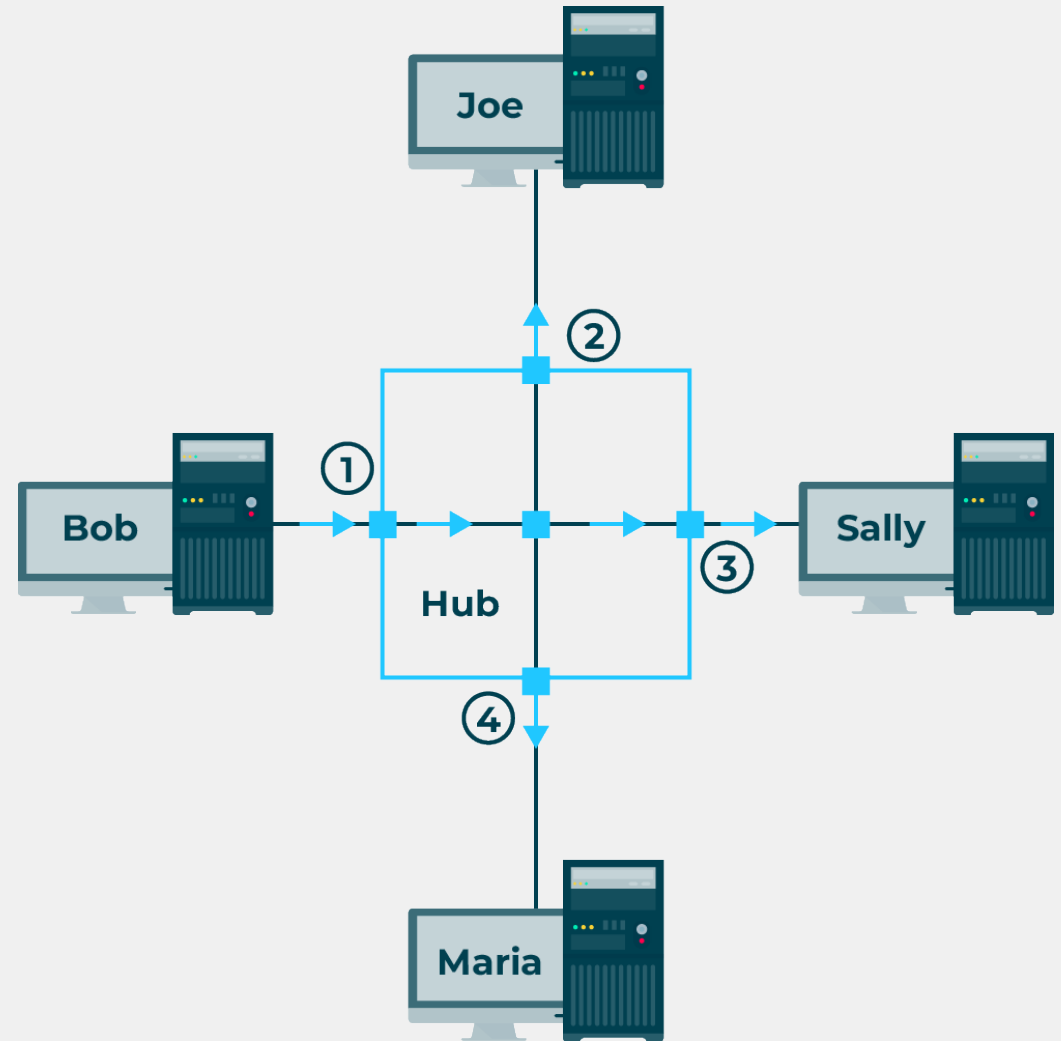
- Describe network types:
 - Peer-to-peer
 - Client / server
- Explain the types of topologies:
 - Star
 - Mesh
 - Bus
- Differentiate between physical and logical topologies.
- Setup a client / server network.
- Analyse a network design scenario and provide appropriate recommendations.

Network components



Hubs

- How do hubs, switches, and routers differ in their core functionality?
- Why is a switch considered more efficient than a hub in managing network traffic?
- In what scenarios would a hub still be a practical choice, despite its limitations?
- How does a hub handle data traffic compared to a switch?



Hubs versus switches

- How does the scalability of a network differ when using hubs versus switches?
- What security risks are associated with using hubs in a network?
- Why are hubs becoming obsolete in modern networking?
- How do cost considerations influence the choice between hubs and switches in a network setup?

Layer 2 switches

- Why is a switch considered more efficient than a hub in managing network traffic?
- Why are switches preferred in modern LAN setups over hubs?
- What are the advantages of using a switch's MAC address table for forwarding data?

Layer 2 switches (continued)

- How can managed switches enhance network security?
- What are some security features commonly found in switches, and how do they protect networks?
- How have switches evolved with the advent of technologies like VLANs and PoE (power over Ethernet)?
- Is it worth investing in managed switches for small businesses, or are unmanaged switches sufficient?

Virtual local area networks (VLANs)

- What is the purpose of VLANs in modern networks?
- How do VLANs improve network segmentation and reduce broadcast traffic?
- How do VLANs enhance network security and efficiency?
- What role do VLANs play in isolating traffic for different departments or groups?

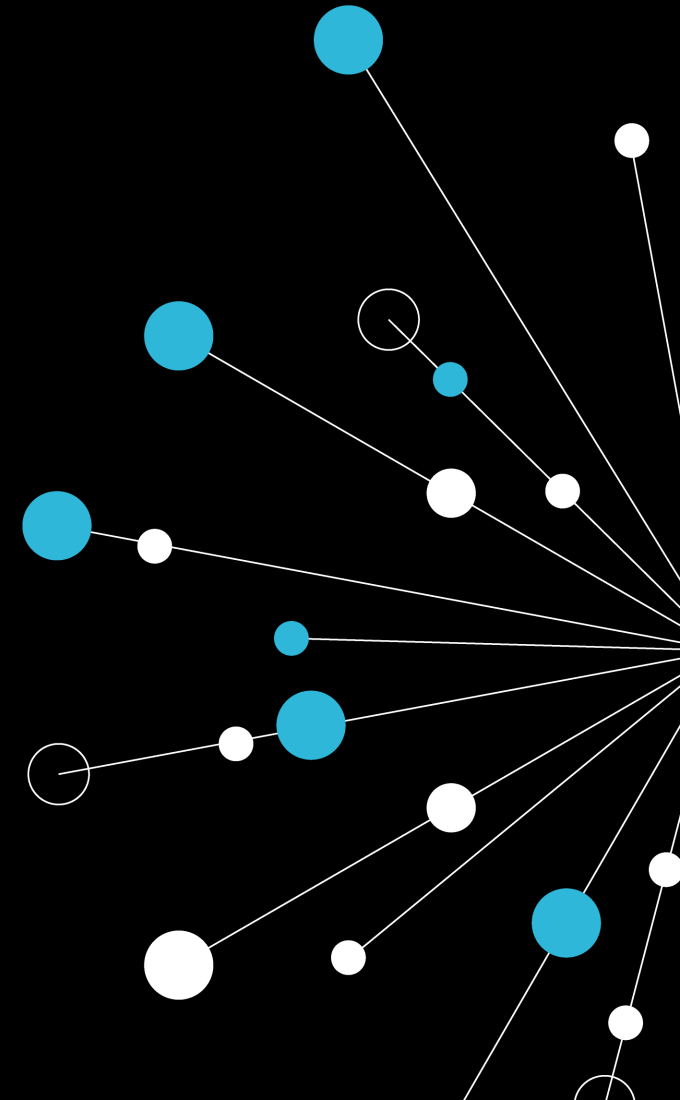
VLANs (continued)

- What factors should be considered when designing a VLAN structure for an organisation?
- How can VLANs support quality of service (QoS) in VoIP and video conferencing?
- What are the challenges in managing VLANs in large-scale networks?
- How does the use of VLANs impact network troubleshooting and maintenance?

Cisco commands demo

Follow your tutor as they demonstrate Cisco commands for:

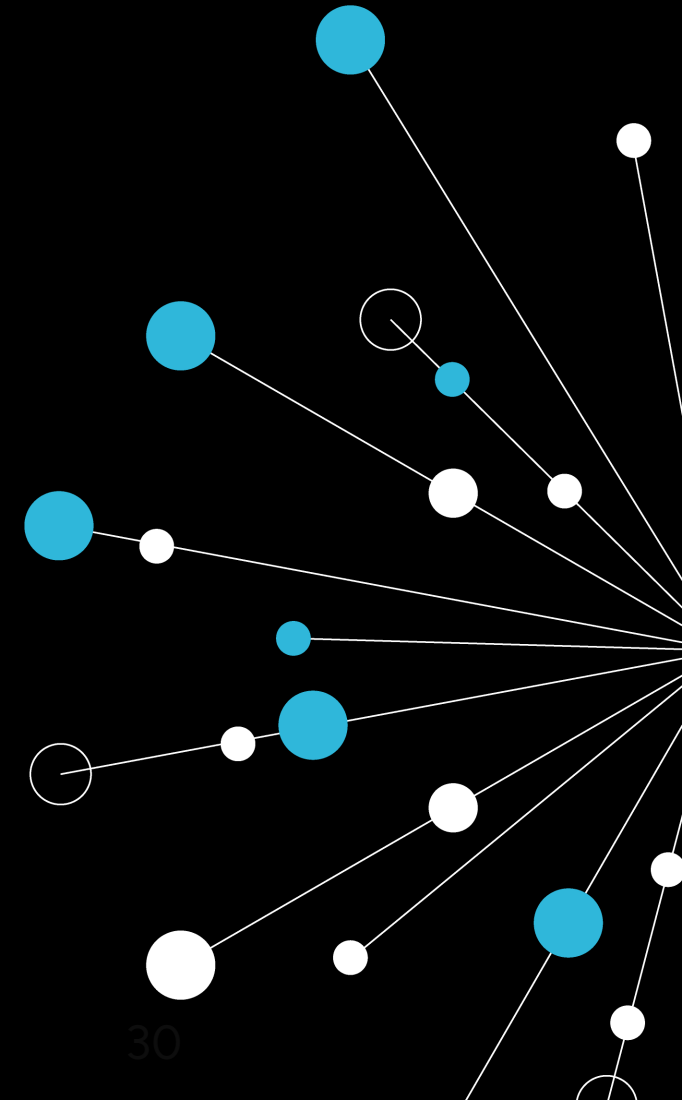
1. Entering the Global Configuration mode.
2. Setting a hostname.
3. Configure VLANs.
4. Assigning a VLAN to a port.
5. Configuring VLAN trunk ports.
6. Setting security passwords.
7. Saving configurations.
8. Viewing configurations.



Activity 4: VLAN configuration

Open the '**Activity 4 VLAN configuration.pka**' file in Packet Tracer:

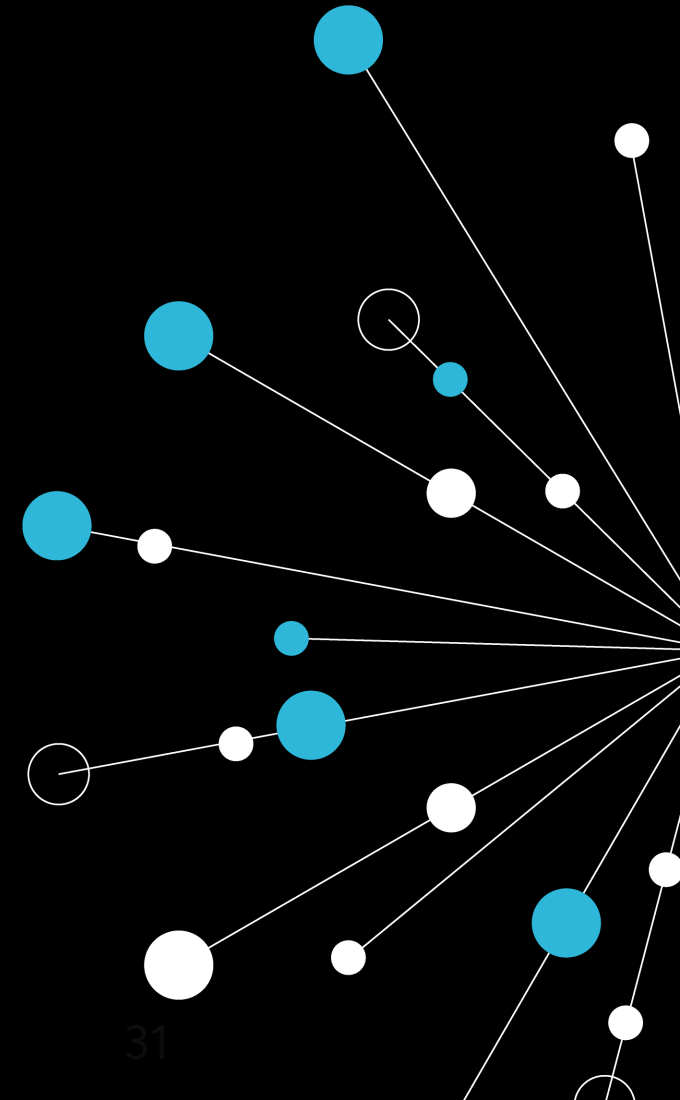
- Use the **Check Results** button at the bottom of the Packet Tracer **Activity Instructions** window to check your progress.
- Take a screenshot of the final assessment items to show you have completed the activity successfully.
- The instructions are within the PKA file and repeated in the **Learner Activity Document**.



Activity 5: Identify collision domains and broadcast domains

Find the activity in your **Learner Activity Document**, and read the instructions:

- Examine a network diagram and identify the amount of collision domains and broadcast domains that exist in the topology.



VLAN trunking

- What is the role of trunking in VLAN communication across multiple switches?
- Why is IEEE 802.1Q more commonly used than ISL in modern networks?
- What are the key considerations when configuring a trunk link between switches?
- How can VLAN mismatches on trunk links affect network connectivity?

VTP (VLAN trunking protocol)

- What is the purpose of VTP in managing VLAN configurations across multiple switches?
- How does VTP simplify VLAN management in large networks?
- What are the differences between VTP server, client, and transparent modes?

VTP (VLAN trunking protocol) (continued)

- When would you use VTP transparent mode instead of server or client mode?
- How can VTP misconfigurations lead to unintended VLAN changes in the network?
- What measures can be taken to secure VTP against unauthorised updates?

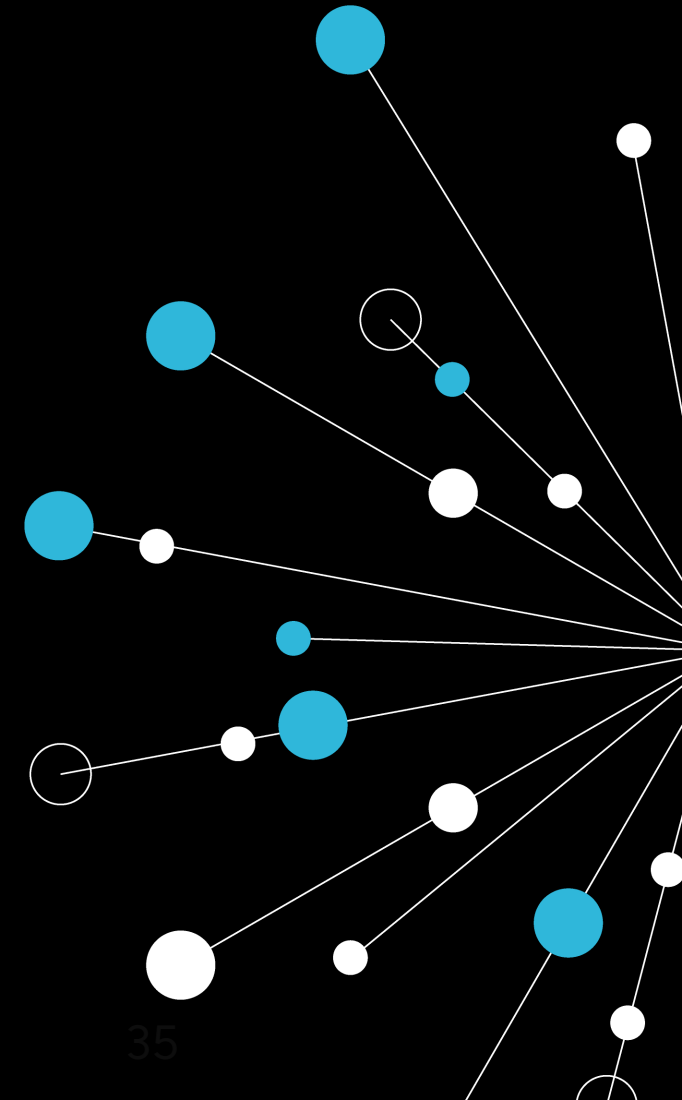
Activity 6: VLAN trunking protocol configuration

Open the 'Activity 6 VLAN Trunking Protocol (VTP).pka' file in Packet Tracer:

Use the Check Results button at the bottom of the Packet Tracer Activity Instructions window to check your progress.

Take a screenshot of the final assessment items to show you have completed the activity successfully.

The instructions are in the PKA file and repeated in the Learner Activity Document.



Layer 3 switches

- Operate at both the Data Link Layer (Layer 2) and the Network Layer (Layer 3) of the OSI model.
- Combine the switching capabilities of a Layer 2 switch with the routing capabilities of a router.
- Uses both MAC address tables and IP routing tables to forward data packets.

Layer 3 switch (continued)

- Can segment the network into multiple broadcast domains using VLANs and route traffic between them.
- Can perform routing functions, allowing it to forward packets between different IP subnets.
- Suitable for larger networks where routing between different subnets is necessary, providing better performance and scalability.

Routers

- How does a router enable communication between different networks?
- What role does a router play in home networks compared to corporate networks?
- How does a router use IP addresses to route data between networks?
- Why might a router become a bottleneck in a large-scale network?

Router security and troubleshooting

- What are some security features commonly found in routers, and how do they protect networks?
- What are the implications of layer-specific operations for troubleshooting network issues?
- What advancements in router technology have improved connectivity and security in recent years?

Types of router

- Core router: Used by service providers or large enterprises to connect multiple routers and manage high-bandwidth traffic.
- Edge router: Also known as a gateway router, it connects internal networks to external networks, such as the internet.
- Distribution router: Receives data from edge routers and distributes it to end users, often via Wi-Fi or Ethernet connections.

Key features of routers

- Routing: Determines the best path for data packets to travel across networks.
- Security: Many routers include built-in firewalls and content filtering to protect against malicious traffic.
- Network management: Routers can manage network traffic, prioritise certain types of data, and provide network diagnostics.

Inter-VLAN routing

- Why is inter-VLAN routing necessary in VLAN-based networks?
- How does inter-VLAN routing allow devices in different VLANs to communicate?
- What are the differences between router-on-a-stick and Layer 3 switch-based inter-VLAN routing?
- When would you choose a Layer 3 switch over a traditional router for inter-VLAN routing?

Inter-VLAN routing (continued)

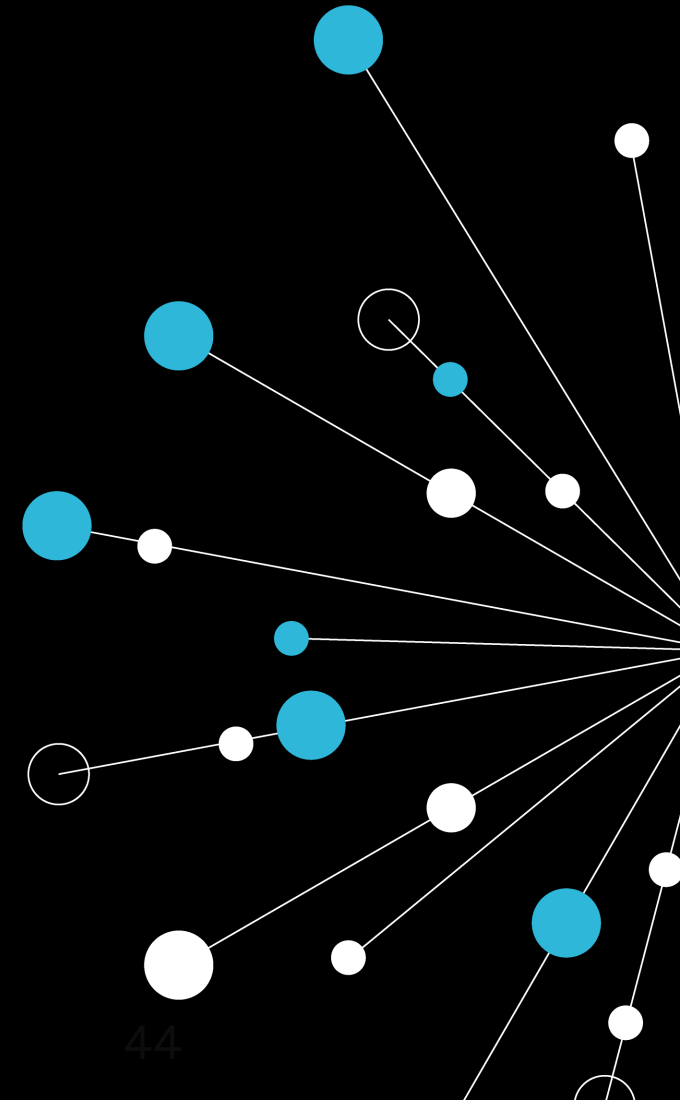
- How does inter-VLAN routing impact network performance in high-traffic environments?
- What are the best practices for optimising inter-VLAN routing in enterprise networks?
- What are common issues encountered during inter-VLAN routing configuration?
- How do routing misconfigurations affect communication between VLANs?

Activity 7: Inter-VLAN configuration

Open '**Activity 7: Inter-VLAN configuration.pkt**' file in Packet Tracer.

Follow your trainer as they demonstrate the configuration steps.

You can also follow the instructions in the **Learner Activity Document**.



Servers

A device designed to manage, store, process, and deliver data, resources, or services to other computers (clients) within a network:

- Provides centralised network access, operations management, and shared services.
- Equipped with specialised hardware to provide high performance and respond to heavy workloads.
- Uses specialised server operating systems, such as Windows Server and Ubuntu Sever.

Types of server

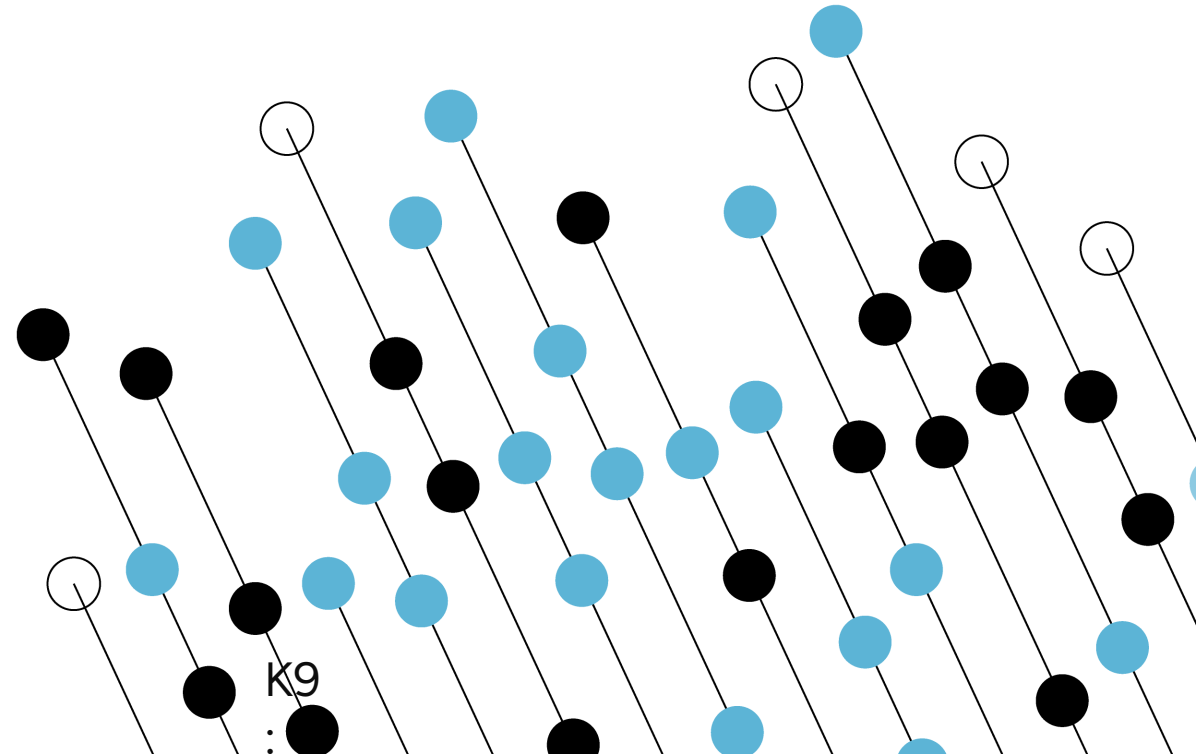
Types of servers include:

- Database
- Authentication
- File
- Mail
- Print

Network components summary

- Basic functionality of:
 - Hub
 - Switch
 - Router
 - Server
- Deployment scenarios
- Collision and broadcast domains

Open Systems Interconnection (OSI) model

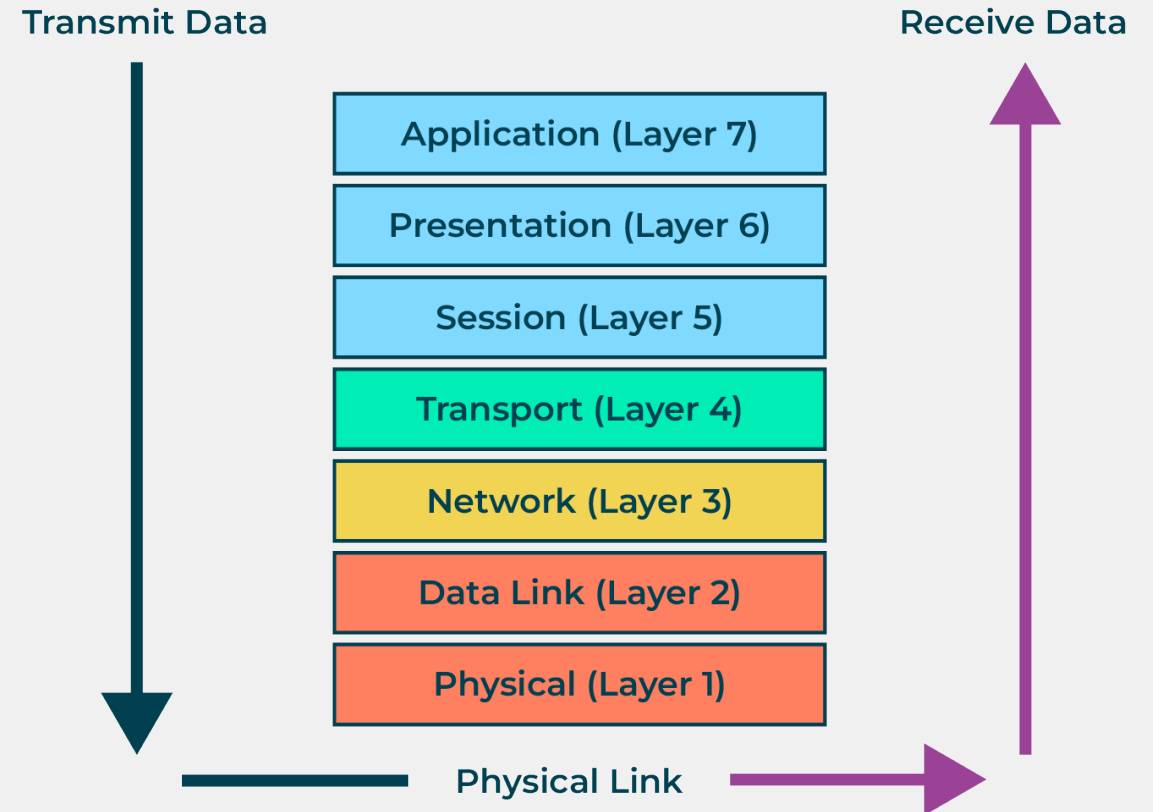


OSI model: Session objectives

- OSI model recap
- Function, protocol, and devices at each layer
- Encapsulation
- Transmission medium
- Cable types and connectors
- Layer 1 recap
- Layer 2 Recap

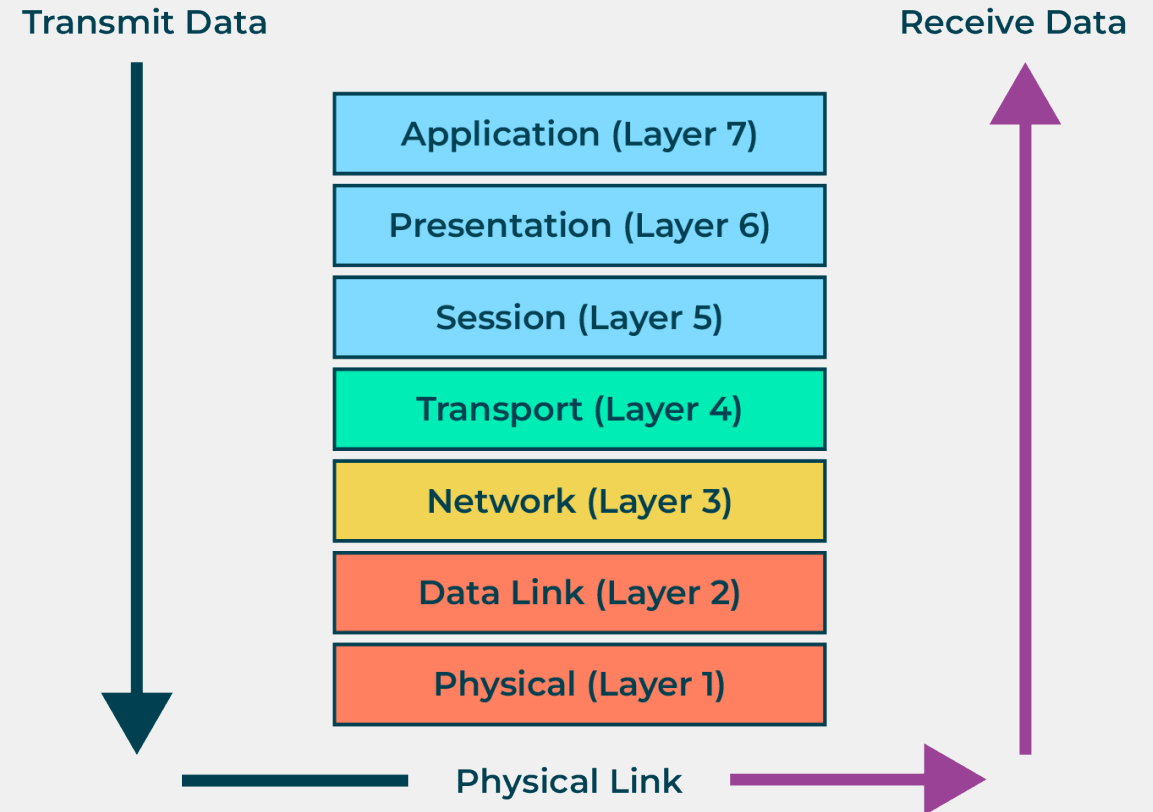
OSI model

- What are the key differences in the number of layers between the OSI and TCP / IP models?
- How does this impact their functionality?
- Why does the OSI model have separate layers for presentation and session, while the TCP / IP model combines these into its application layer?
- The OSI model was developed as a theoretical framework, while TCP / IP was designed for practical implementation. How does this influence their adoption and use?



OSI model (continued)

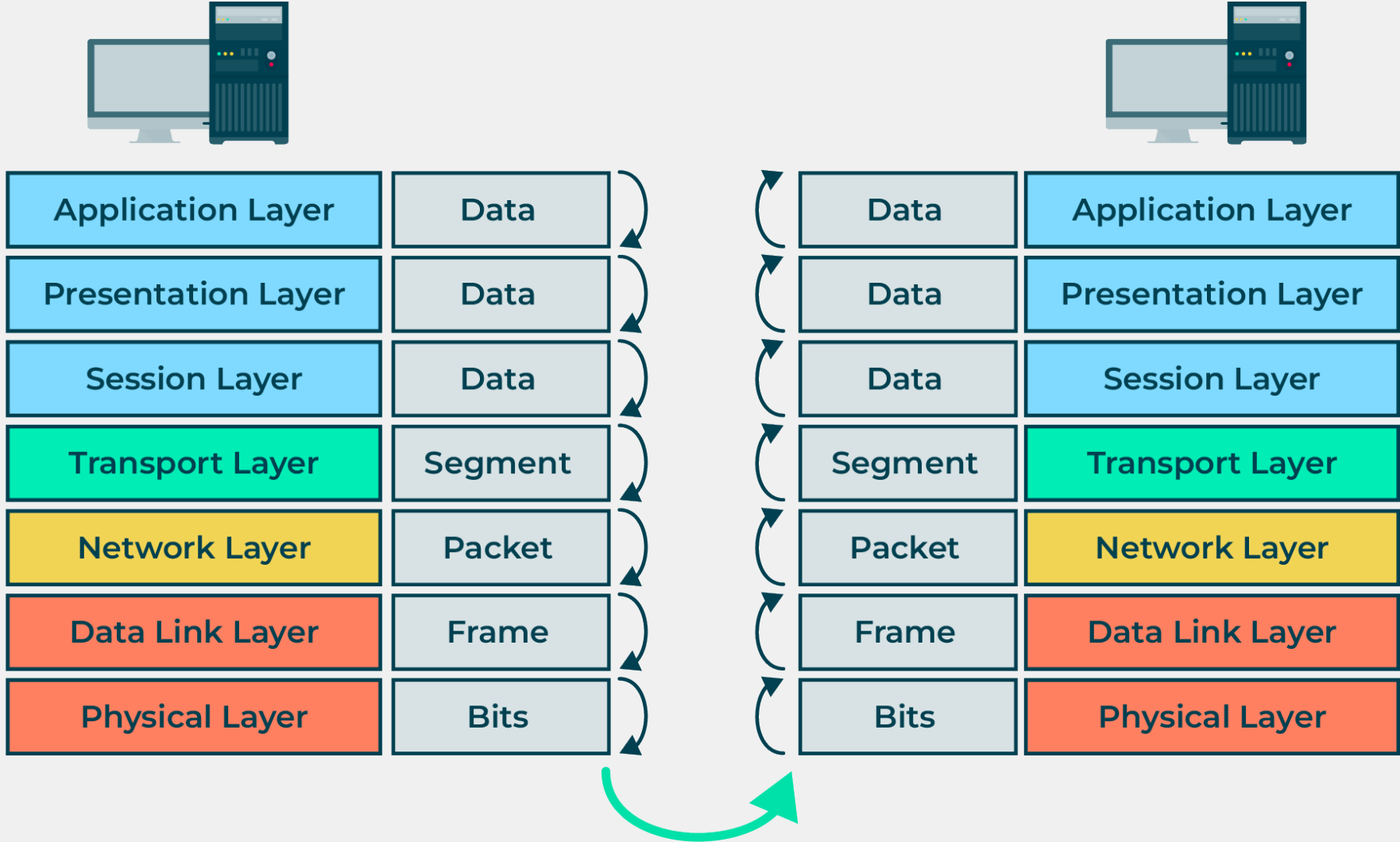
- Why is TCP / IP the dominant model in modern networking despite the OSI model's comprehensive structure?
- How do the layers of the OSI model map to those of the TCP / IP model? Are there any direct overlaps or gaps?
- Which OSI layers are consolidated in the TCP / IP model, and what are the advantages or disadvantages of this consolidation?



Function, protocol, and devices of each layer

Layer	Function	Example
Application (7)	Services that are used with end-user applications.	SMTP, POP3, FTP, HTTP, TELNET
Presentation (6)	Formats the data so that it can be viewed by the user. Encrypts and decrypts data.	JPG, GIF, HTTPS, SSL, TLS
Session (5)	Establishes and ends connections between two hosts.	NetBIOS, PPTP
Transport (4)	Responsible for the transport protocols and error handling.	TCP, UDP
Network (3)	Reads the IP address from the packet.	Routers, Layer 3 switches,
Data Link (2)	Reads the MAC address from the data packet.	Switches, NICs
Physical (1)	Sends data on to the physical wire.	Hubs, cables

Encapsulation



Layer 1: Physical

- What is the primary function of the Physical Layer in the OSI model?
- How does data transmission occur at this layer (electrical signals, radio waves, or light pulses)?
- What are the differences between wired and wireless physical layer technologies?
- What is the role of bandwidth, latency, and noise in physical layer performance?
- How do signal degradation and attenuation impact data transmission?

Topologies

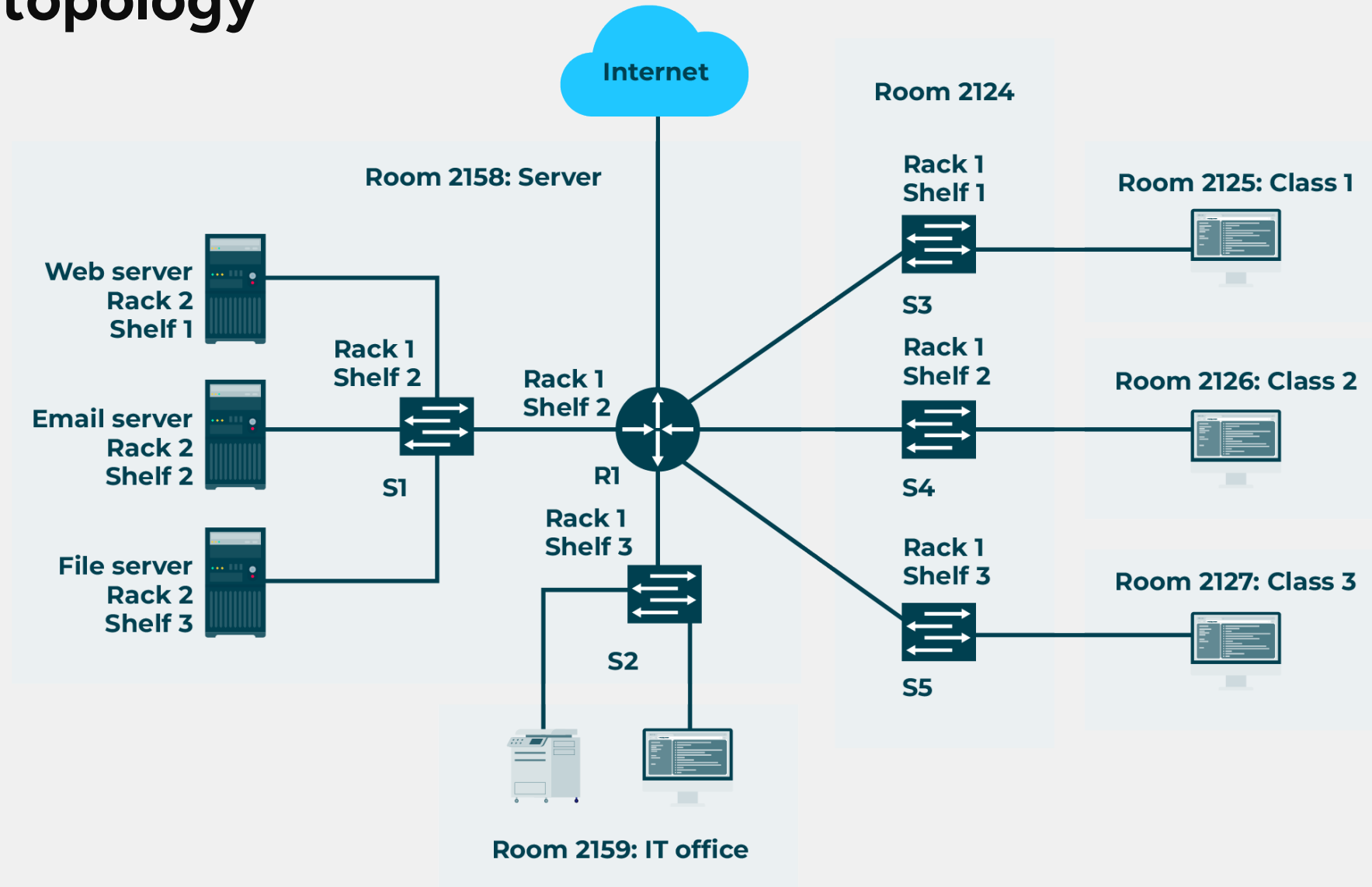
Physical

- Star
- Bus
- Ring
- Mesh

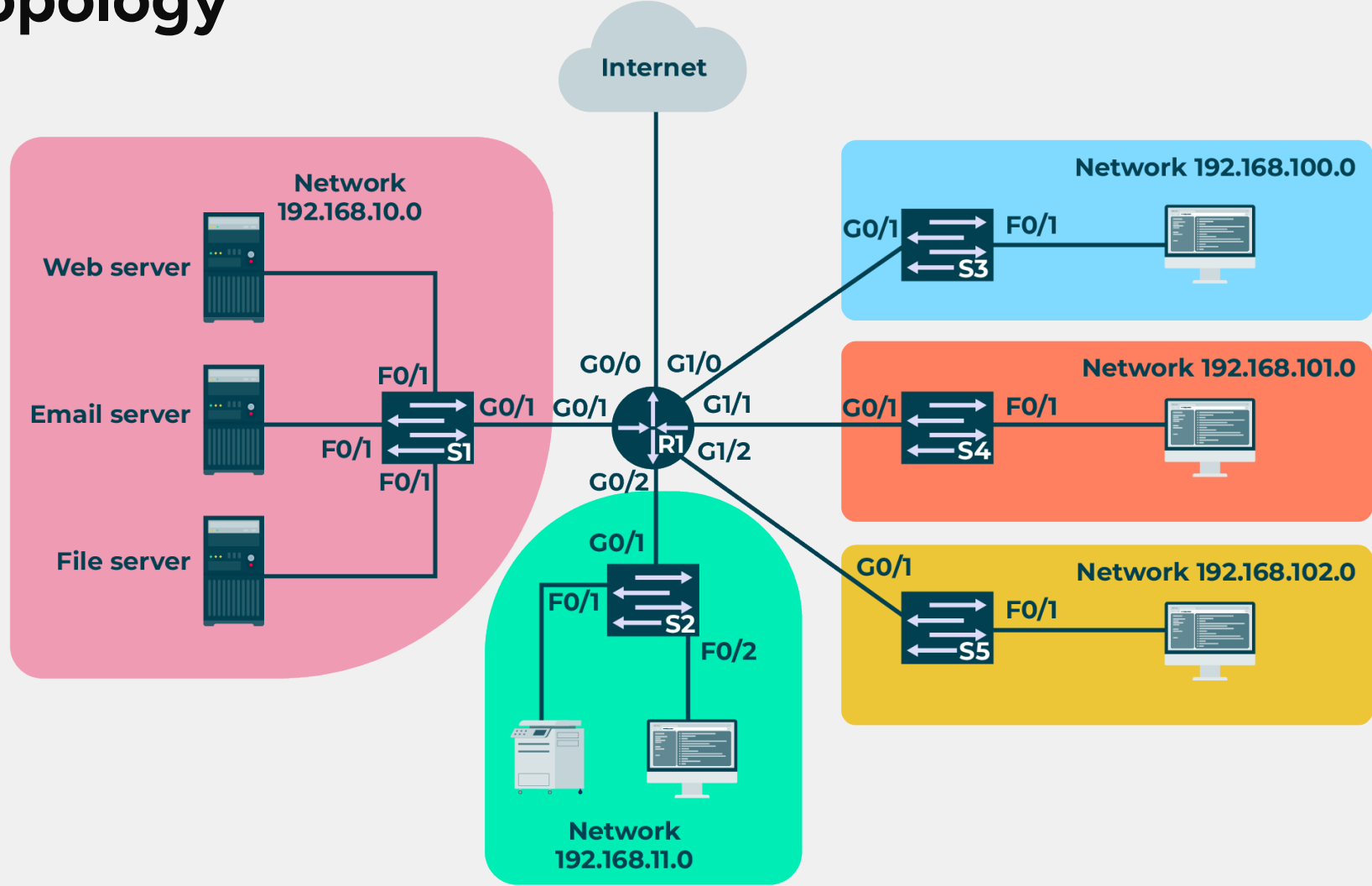
Logical

- Ethernet 802.3
- Token passing 802.5

Physical topology



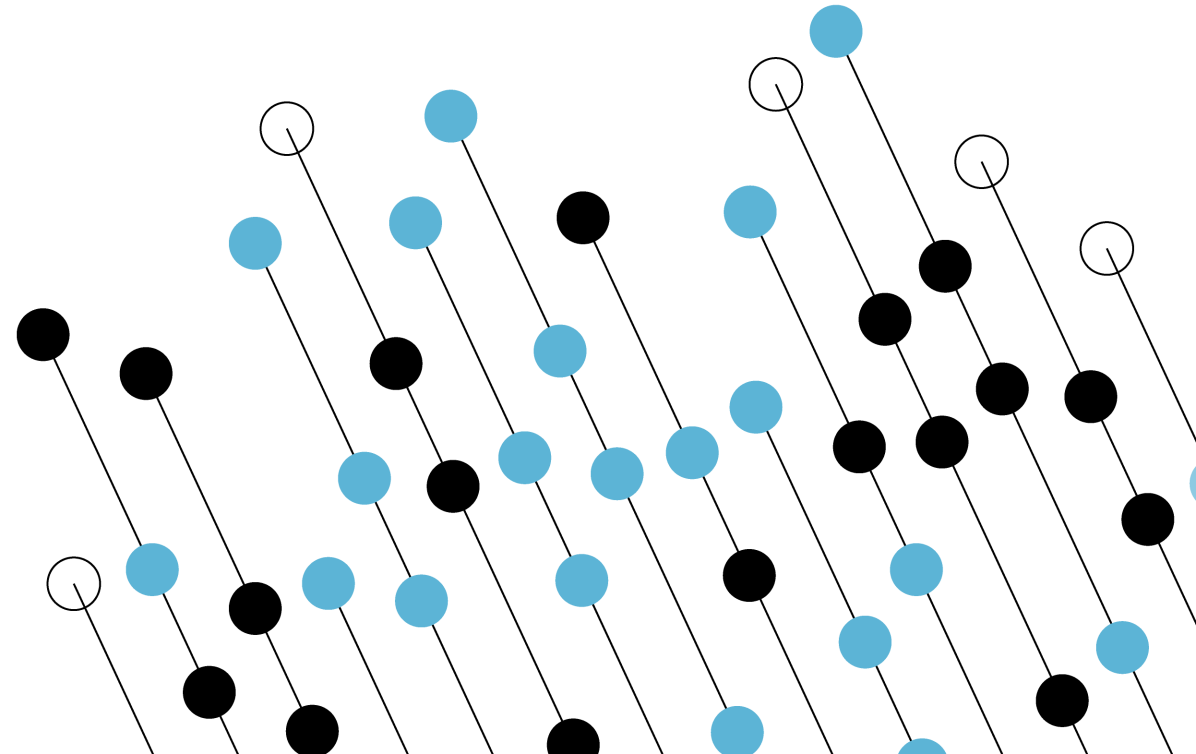
Logical topology



Legacy fast Ethernet standards

- Hubs and bridges increasingly replaced by switches.
- Auto-negotiation of link properties (half vs full duplex).
- MDIX.
- Auto-MDI / MDI-X (medium dependent interface).

Transmission medium



Network transmission medium

- Transmission medium is used to transfer data between devices.
- It can be guided (wired) or unguided (wireless).
- It transmits data as 1s and 0s.

Network transmission medium (continued)

Specific factors that are used to select the right medium include:

- Cable speed
- Distances coverage
- Environment (interference)

Specific transmission medium can include:

- Copper
- Fibre optic
- Wireless

802.3 Ethernet cabling standards

Standard	Cable	Speed	Use case
10Base-T	Unshielded twisted pair (UTP)	10 Mbps	Legacy networks, basic telephone wiring.
100Base-T	Cat5e or higher	100 Mbps	Enhanced Cat 5; supports Gigabit Ethernet and reduces crosstalk (STP).
1000Base-T Gigabit Ethernet	Cat6 or higher	1000 Mbps	Gigabit and 10-Gigabit Ethernet; suitable for small to medium networks.
10GBase-T 10 Gigabit Ethernet	Cat6 or higher	10 Gbps	High-speed networks; improved shielding for reduced interference.

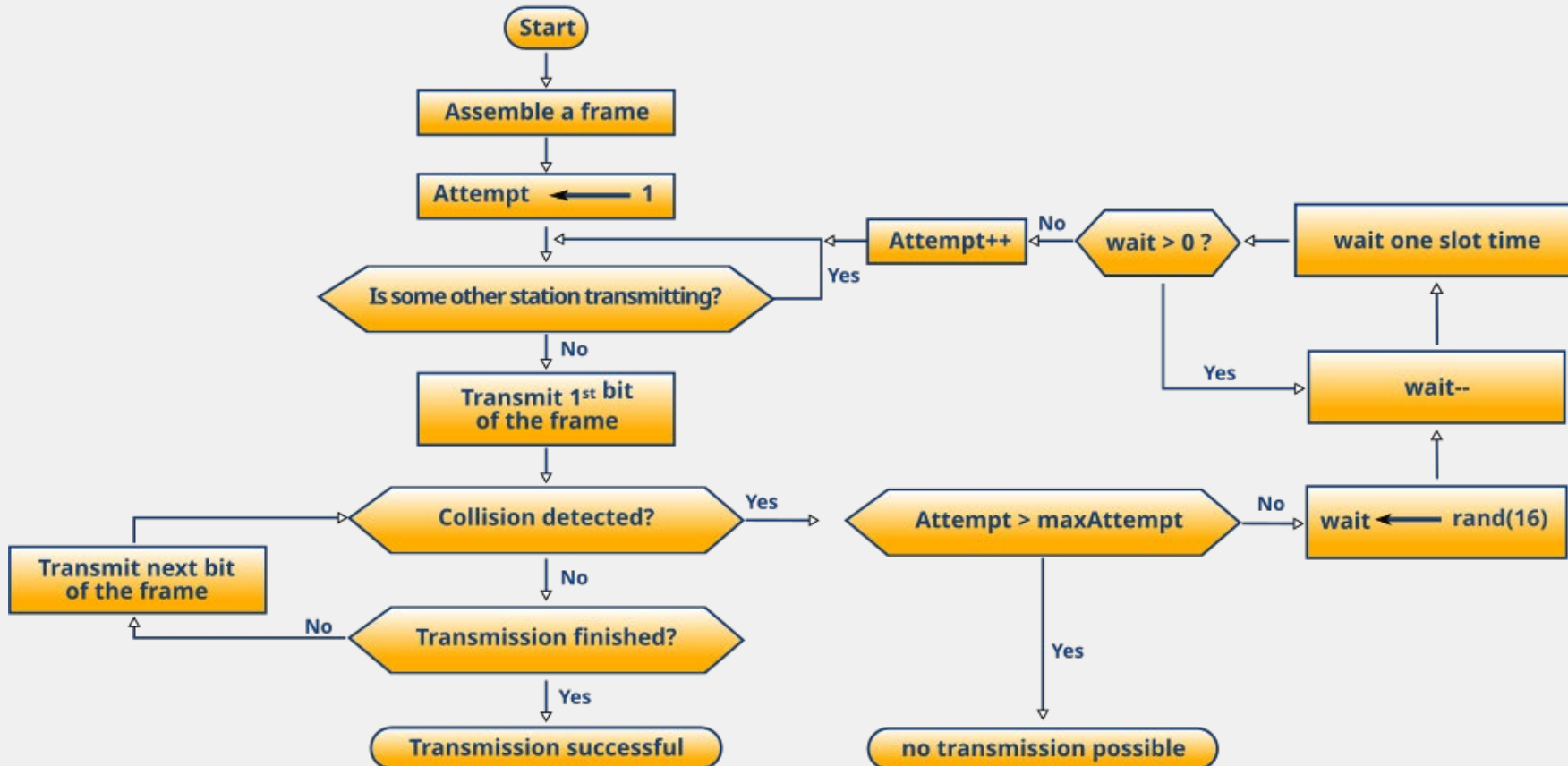
Fibre cable standards

Standard	Cable	Speed	Use case
100BASE-FX	Multi-mode fibre	100 Mbps	Medium-distance fast Ethernet using fibre.
1000BASE-SX	Multi-mode fibre	1 Gbps	Short-range Gigabit Ethernet using fibre.
1000BASE-LX	Single-mode fibre (multi-mode fibre compatible)	1 Gbps	Long-distance Gigabit Ethernet over fibre.
10GBASE-SR	Multi-mode fibre	10 Gbps	Short-range 10 Gigabit Ethernet using fibre.
10GBASE-LR	Single-mode fibre	10 Gbps	Long-range 10 Gigabit Ethernet.
10GBASE-ER	Single-mode fibre	10 Gbps	Extended-range 10 Gigabit Ethernet over fibre.

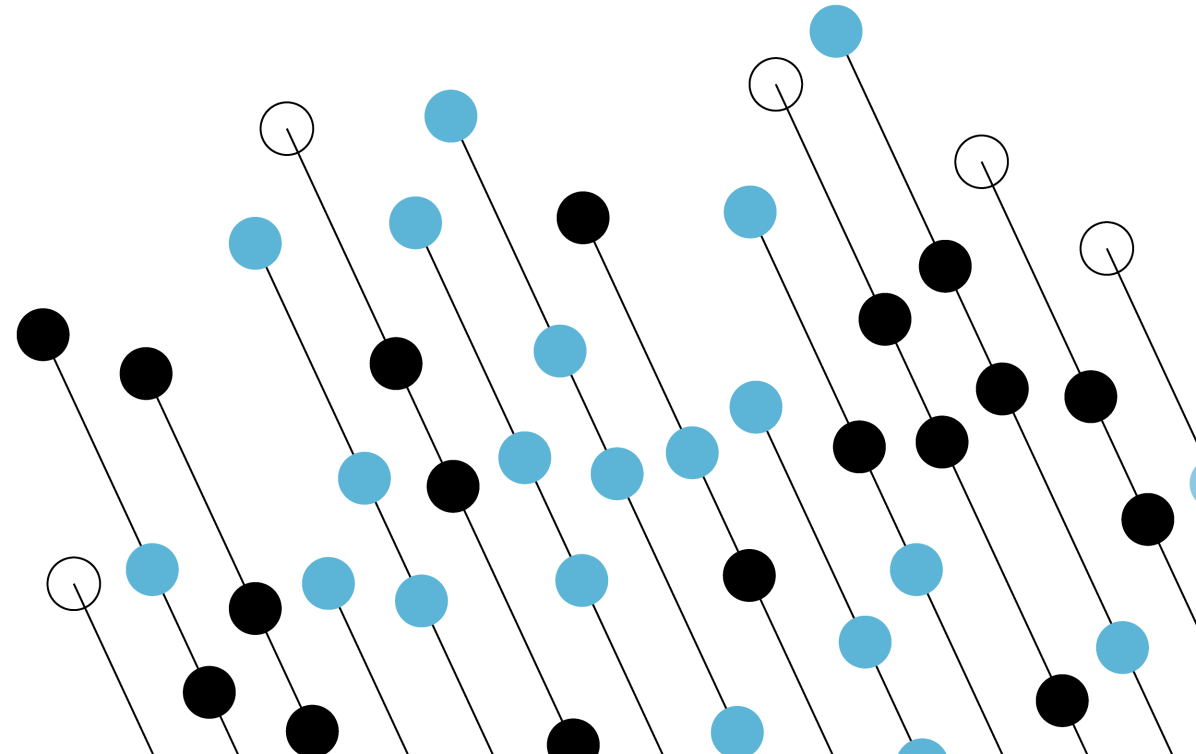
Ethernet media access and contention (CSMA / CD)

Carrier sense multiple access with collision detection.

Used in traditional Ethernet networks to manage how devices access the communication cable:



Cable types and connectors

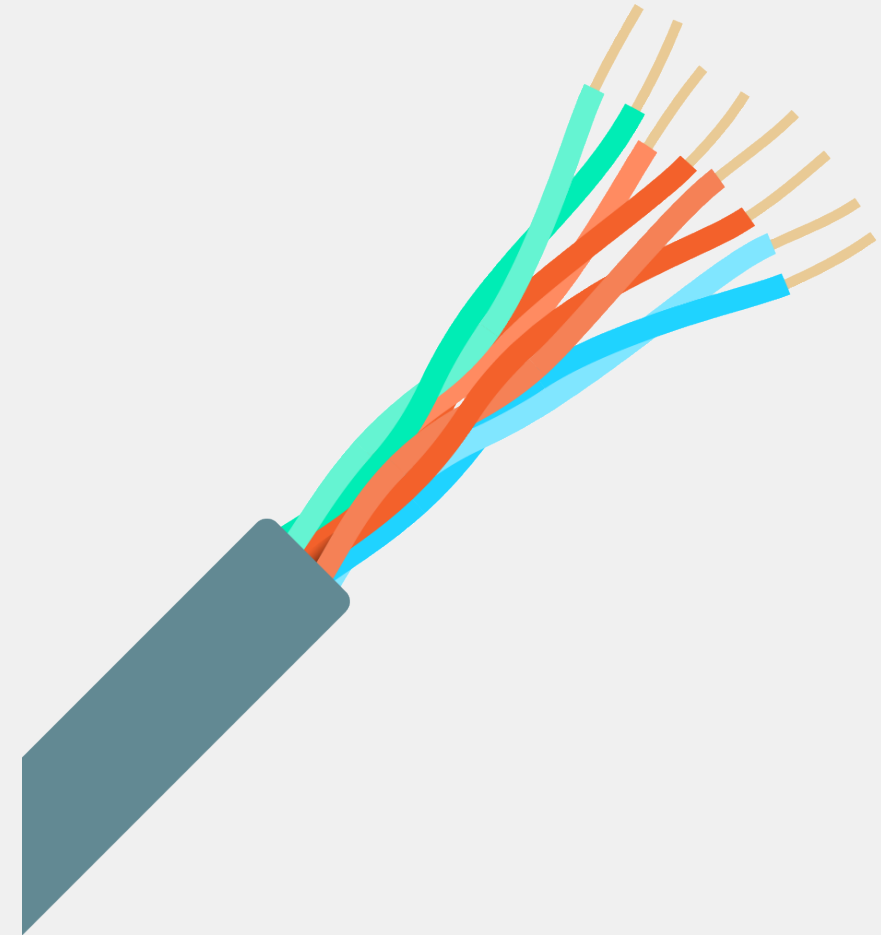


Coaxial cable

- What are the main components of a coaxial cable, and how do they contribute to its performance?
- In what scenarios is coaxial cable commonly used (for example, cable television, internet)?
- What are the benefits and drawbacks of using coaxial cable for network transmission?
- How does coaxial cable handle signal interference and attenuation compared to other media?

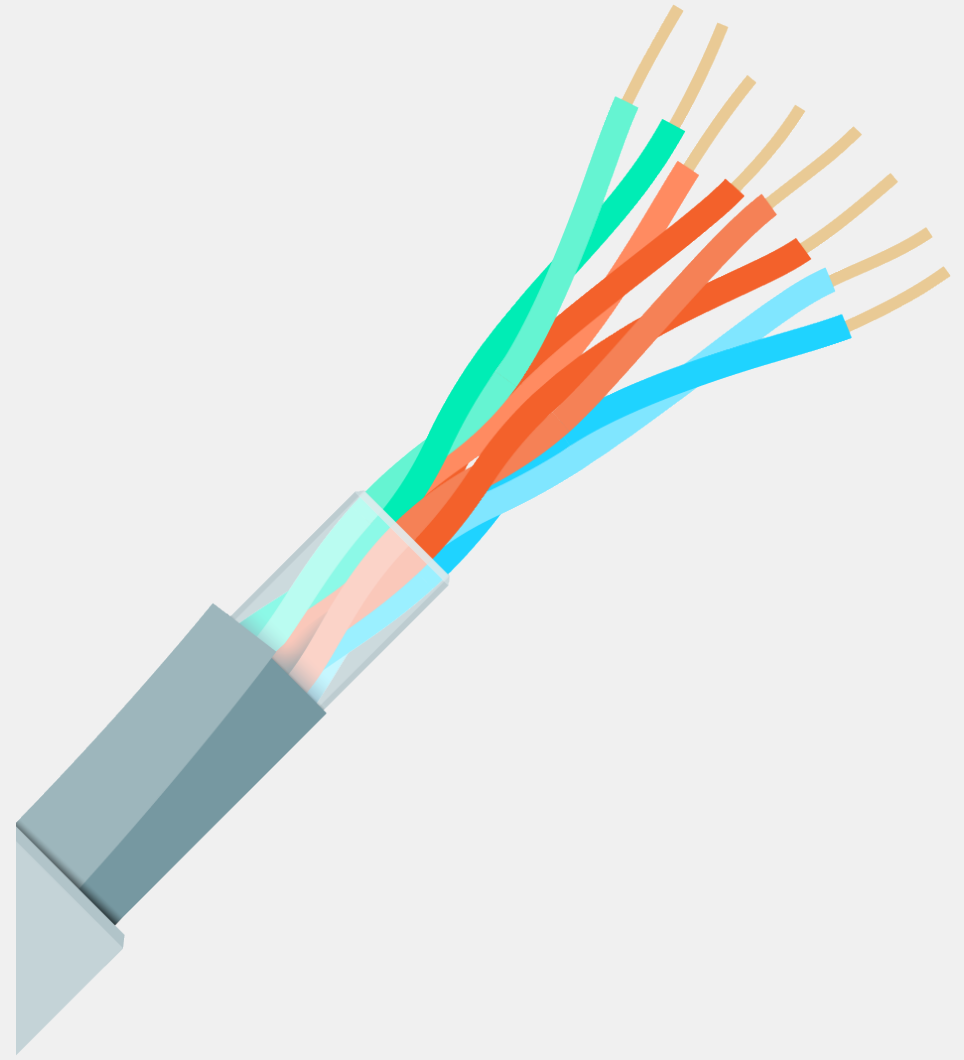
Unshielded twisted pair (UTP) and shielded twisted pair (STP)

- What is UTP cable, and how does it differ from other types of copper cables?
- How does STP cable provide additional protection against electromagnetic interference (EMI)?
- Common uses: Where are UTP and STP cables typically used in networking?
- What are the performance characteristics and limitations of UTP and STP cables?
- What are the best practices for installing and maintaining UTP and STP cables?



STP (shielded twisted pair)

- Includes shielding layer (foil or braid).
- Protection against electromagnetic interference and crosstalk.
- More expensive.
- Requires more care during installation.



Fibre optic

- What are the different types of fibre optic cables and how do they differ in structure and application?
- What are the key advantages of using fibre optic cables over copper cables?
- In what scenarios are fibre optic cables most beneficial (for example, long-distance communication, high-speed data transfer)?
- What are the challenges associated with installing and maintaining fibre optic cables?
- How is the use of fibre optic technology evolving in modern networks?

Single-mode fibre (SMF)

- Has a small core that allows a transmission of a single light mode (one light ray or wavelength at a time).
- Suitable for long-distance communication (up to 80 km or more).
- High data transfer rates.
- Used for long distance and large-scale enterprise networks.
- More expensive.

Multimode fibre (MMF)

- Wider core which allows multiple lights at different angles.
- Best for short to medium distances (up to 500 metres).
- Adequate for most local-area networks design.
- Cheaper than SMF.

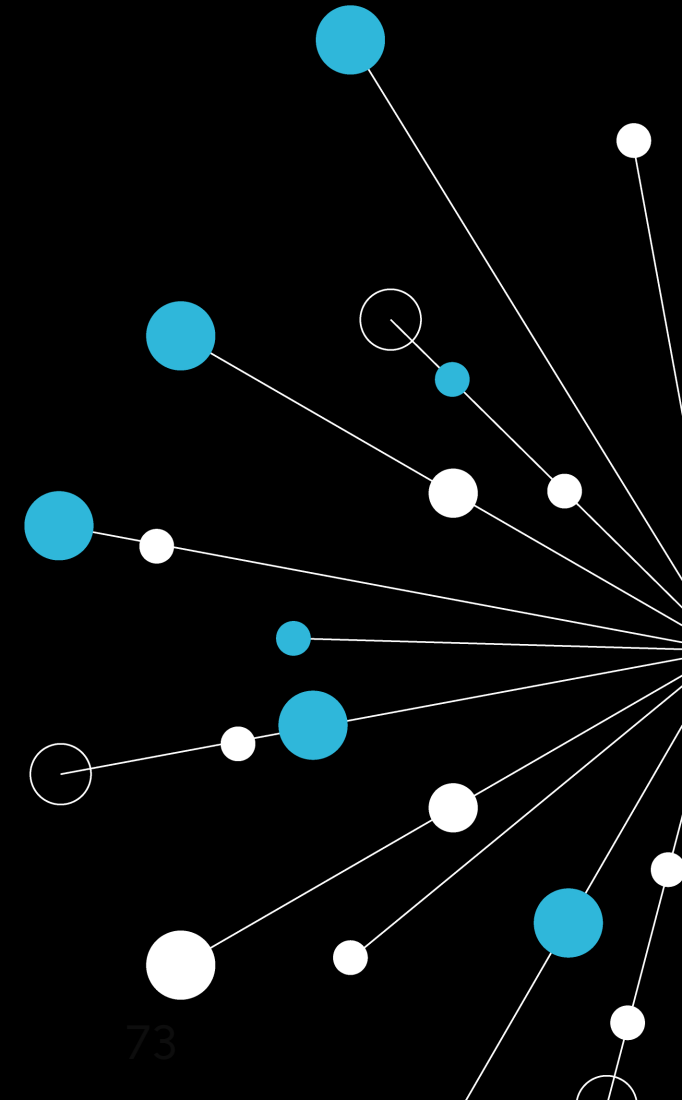
Cable connector and wiring standard

Connector type	Cable type	Use case
RJ45	Twisted pair (UTP / STP)	Standard Ethernet connector for LANs, connecting devices like PCs and switches.
RJ11	Twisted pair (UTP)	Used for telephone lines and DSL internet connections.
SC (subscriber connector)	Fibre optic	Push-pull design for high-speed fibre optic connections, often in data centres.
LC (lucent connector)	Fibre optic	Bayonet-style connector used in multimode fibre networks.
FC (ferrule connector)	Fibre optic	Threaded connector for secure fibre optic connections in industrial settings.

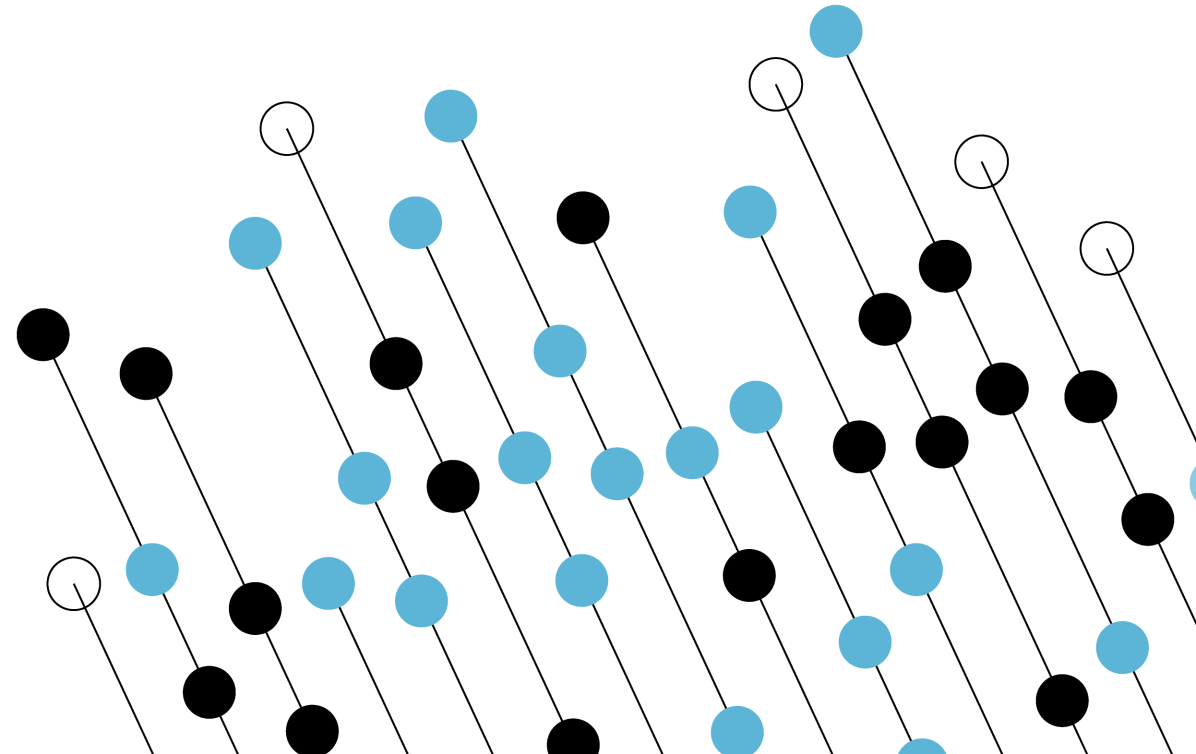
Activity 8: Select appropriate cabling

Find the activity in your **Learner Activity Document**, and read the instructions:

- Examine a scenario and select the appropriate cabling for the network requirements.



Layer 2: Data Link



Data Link layer

- What is the function of the Data Link layer in network communication?
- How does MAC address work, and why is it important?
- What is the difference between collision domains and broadcast domains?
- How does error detection and correction work at this layer?
- How do VLANs improve network segmentation at Layer 2?

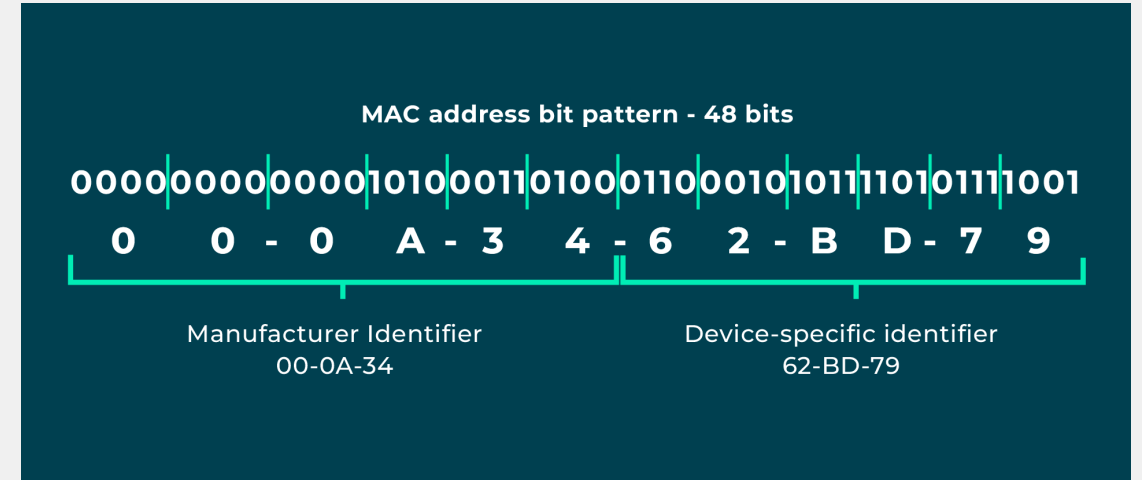
MAC address

Media access control (MAC) layer 2 addressing scheme defined in IEEE 802 standard:

- 48-bit
- MAC address / EUI-48 is 12 hex digits (0-9,A-F)
- 00:60:8c:12:3a:bc

Each MAC address is split into two parts of 24 bits:

- The first 24 bits represents the manufacturer (OUI).
- The last 24 bits is the burned-in or physical address.



Traffic types: Unicast

Unicast

- One-to-one communication method.
- Involves sending data from a network node to a specific receiver.
- Frame is directed to a single end device interface using the MAC address.
- Recipient address is used at this layer to direct frames to the intended device.

Traffic types: Broadcast

Broadcast

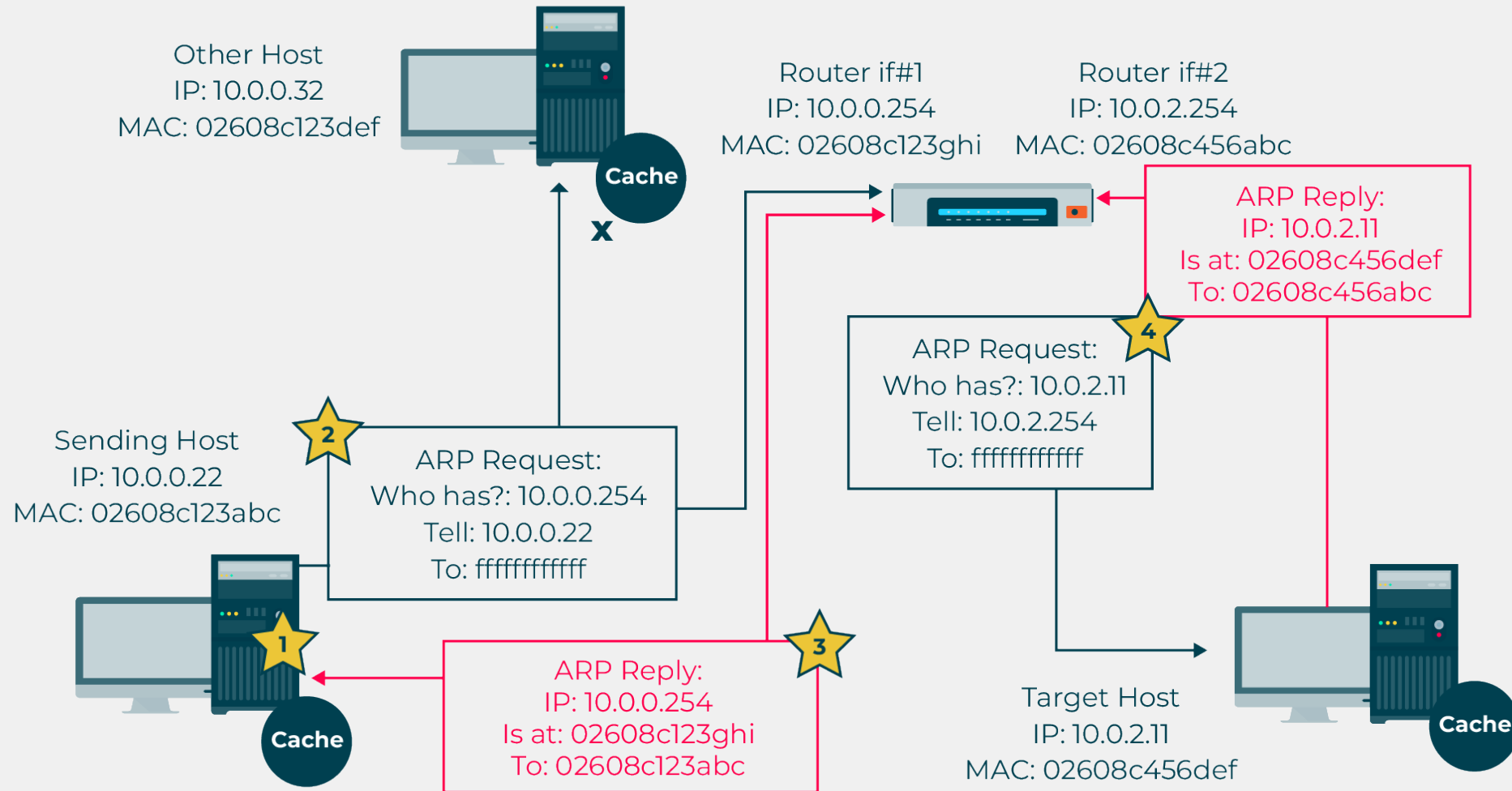
- One-to-all communication.
- Involves sending data from one sender to multiple devices on a network segment.
- Can be used a part of host discovery (ARP).
- Uses the MAC address ff:ff:ff:ff:ff:ff /255.255.255.255.

Address resolution protocol (ARP)

A network protocol used to resolve logical address (IP) to the corresponding physical address (MAC).

APR is sent as a broadcast traffic.

Address resolution protocol (ARP) (continued)



Activity 9: Evaluate and compare network devices

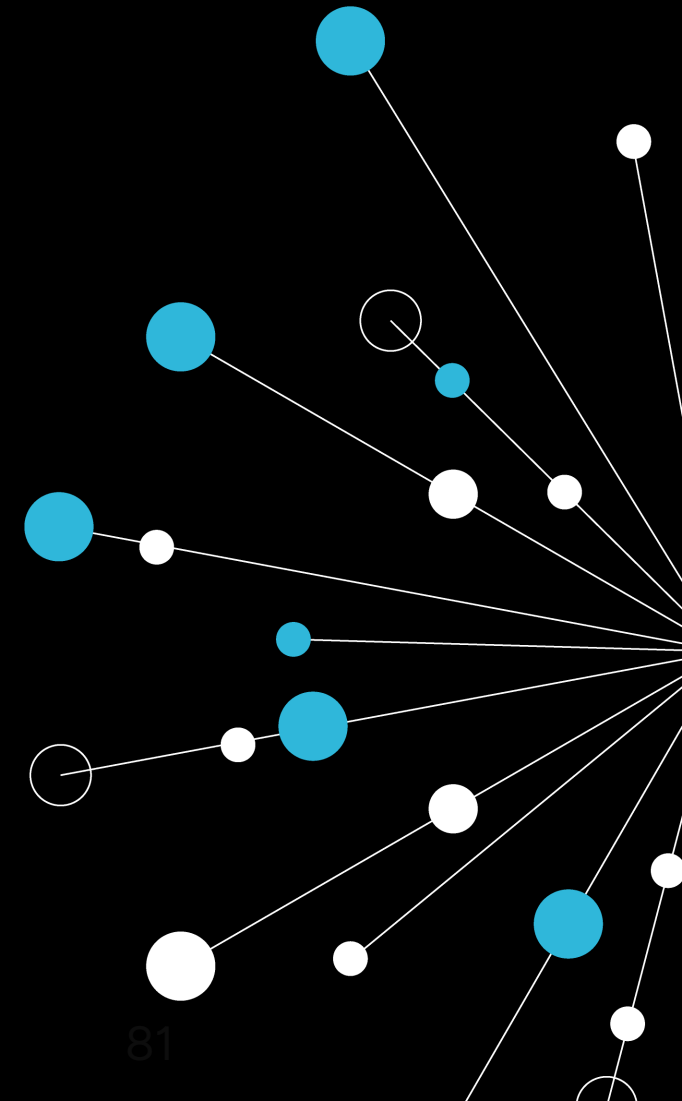
Find the activity in your **Learner Activity Document**, and read the instructions:

- Evaluate different network devices and recommend an appropriate network device for each network segment.

Extension task:

- Create a network design using Packet Tracer and segment each department using your proposed devices.

Be prepared to present and justify your solution to the rest of the group.



End of Day 1 summary

- Network types (LAN, MAN, WAN)
- Client server vs P2P
- Network devices
- Networking models
- OSI model
- Layer 1: Network transmission medium
- Layer 2: MAC addresses and ARP resolution

Network Infrastructure

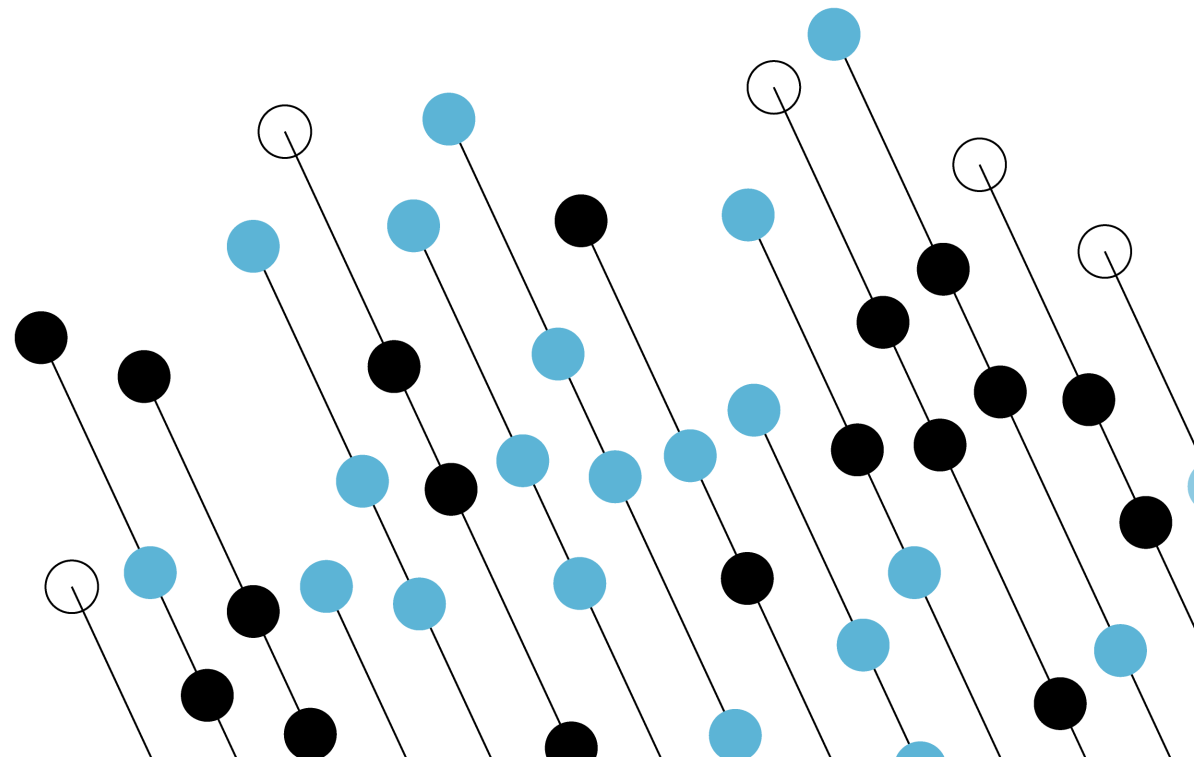
Day 2



Previous lesson recap (Q&A)

- Network types (LAN, MAN, WAN)
- Client server vs P2P
- Network devices
- Networking models
- OSI model
- Layer 1: Network transmission medium
- Layer 2: MAC addresses and ARP resolution

Layers 3-7



Layer 3: Network

- Manages routing between different networks.
- Determines the best path in moving packets from source to destination network.
- Uses network addressing.
- Routing.
- QoS (quality of service) and congestion control are also addressed in this layer.
- Fragmentation / reassembly.

Examples:

- Router
- Layer 3 switch
- Basic firewall

Layer 4: Transport

- Ensure reliable data transfer.
- Connection oriented.
- Provides error recovery and flow control.
- Segments and reassembles data.

Examples:

- TCP
- UDP

Differentiate between TCP and UDP

TCP

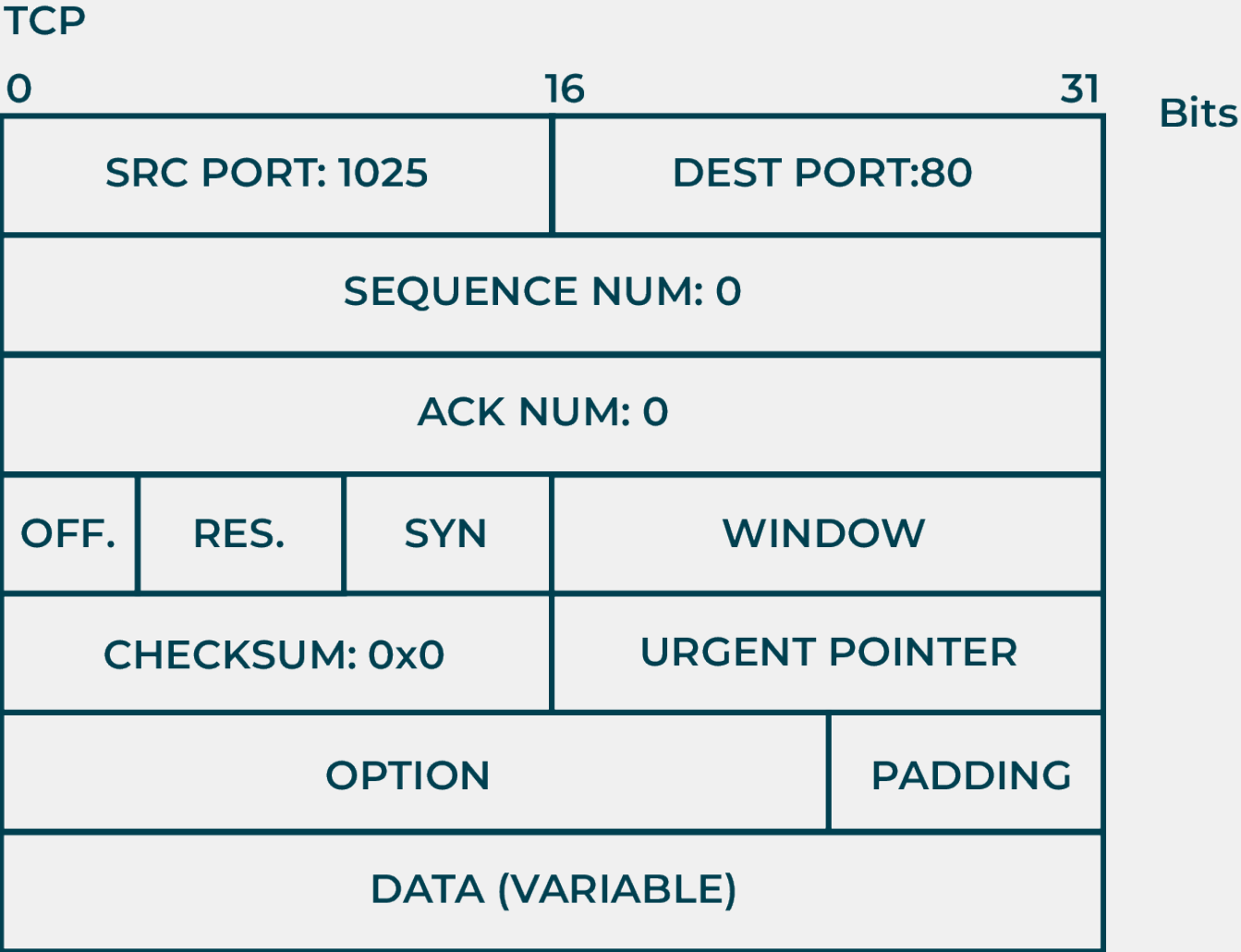
- Connection orientated / reliable.
- Incorporates error correction / retransmission.
- Incorporates flow control / windowing.
- Higher overhead.

Differentiate between TCP and UDP (continued)

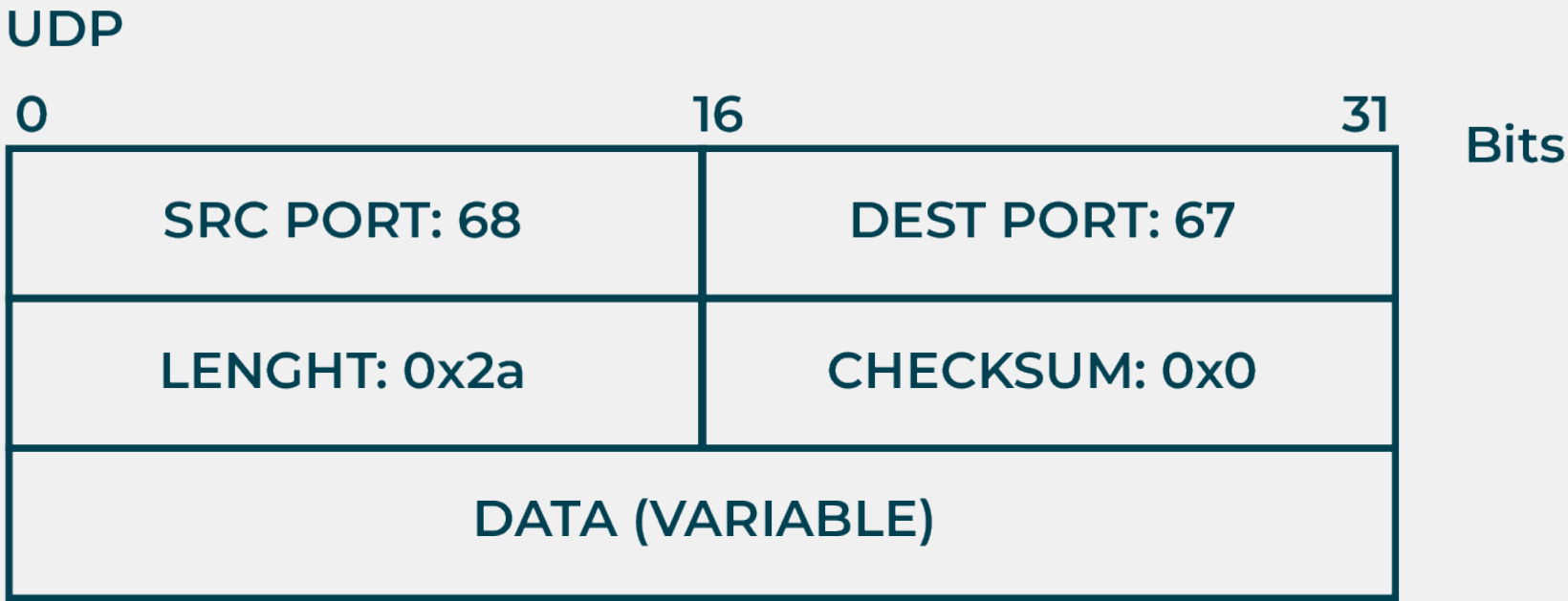
UDP

- Connectionless.
- No error correction / retransmission.
- Checksum checks for integrity but not to correct errors.
- Unreliable.
- Lower overhead than TCP.
- Faster but with no reliability.

TCP headers

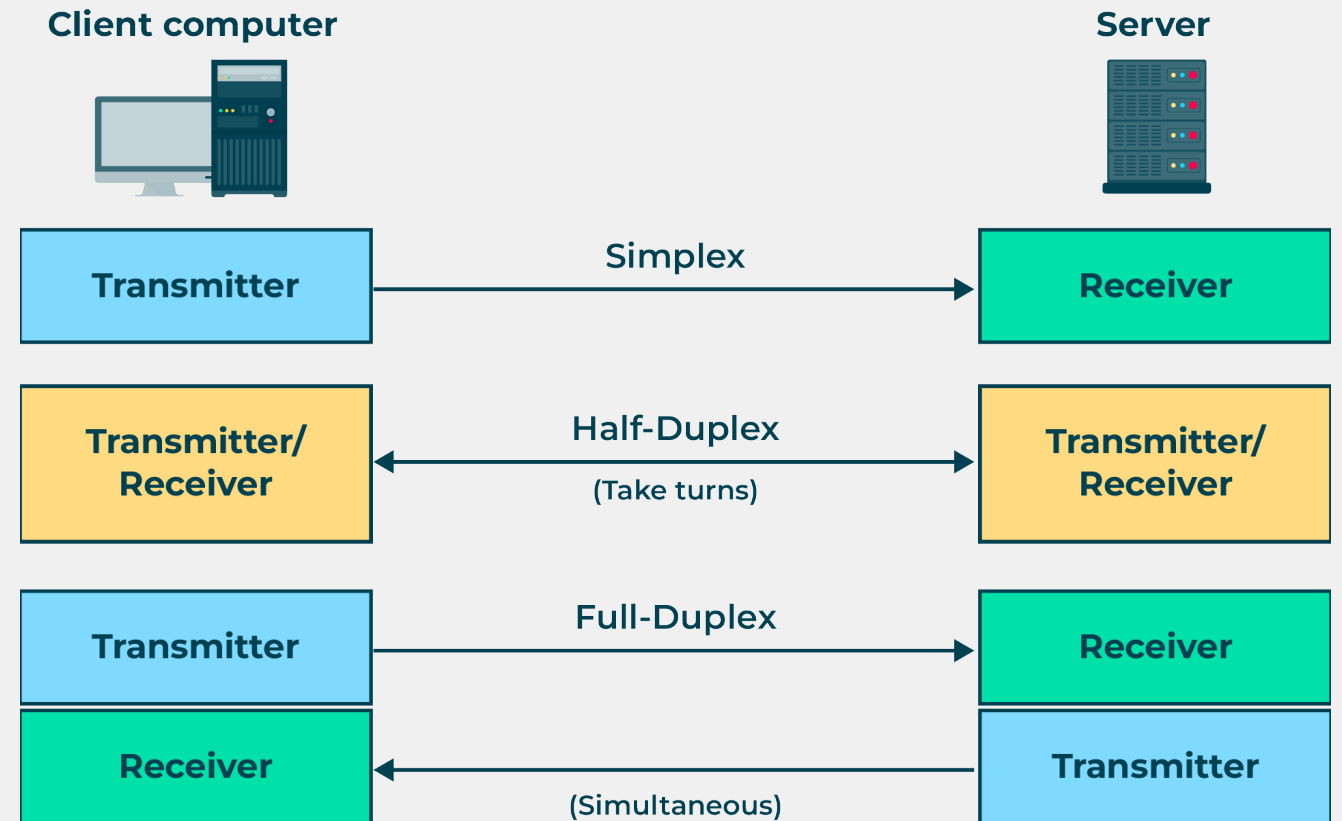


UDP headers



Layer 5: Session

- Manages sessions or connections between applications.
- Establishes, maintains, and terminates communication sessions.
- Identify sequence of messages belonging to the same conversation.
- Dialogue control through simplex, half-duplex and full-duplex.
- Examples:
 - NetBIOS
 - RPC
 - PPTP



Layer 6: Presentation

- Ensures data is in a readable format for the application layer.
- Data formats.
- Data compression.
- Encryption and decryption.
- Data representation.

Examples:

- JPEG
- ASCII

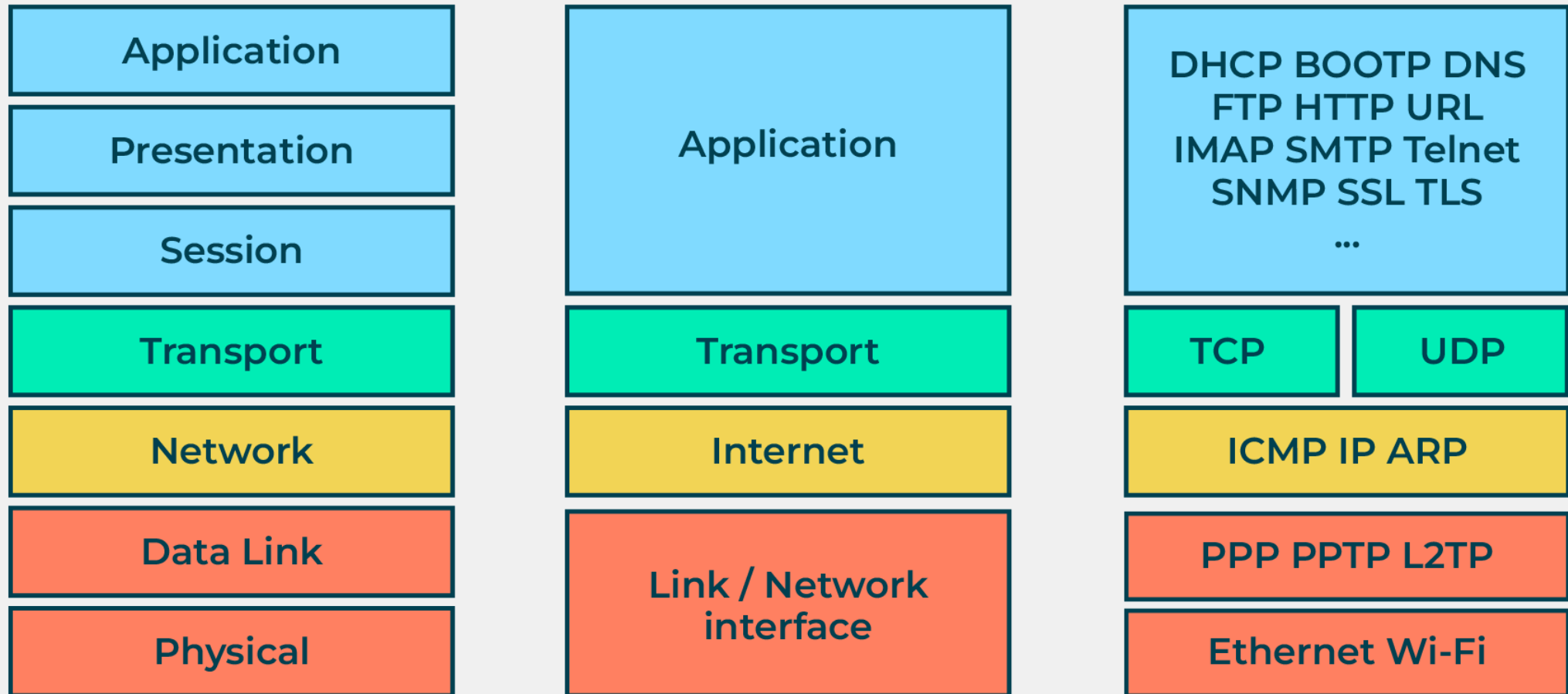
Layer 7: Application

- Interface directly with user application and network services.
- Facilitates communication between applications and the lower layers of the OSI.
- Interprets user input and prepare data for network transfer.

Examples:

- DNS
- DHCP
- HTTP
- SNMP
- SMTP
- HTTPS
- RDP

OSI / TCP comparison

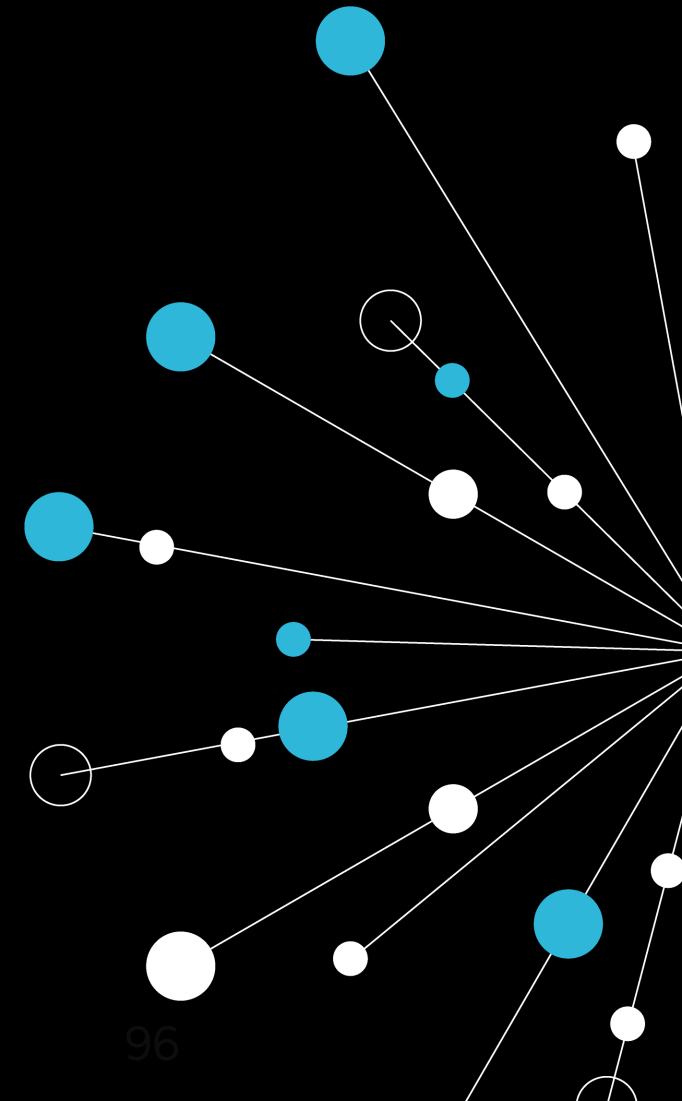


Activity 10: The OSI model

Find the activity in your **Learner Activity Document**, and read the instructions:

Read the NetworkPro Academy scenario and complete the following tasks:

- **Research:** Gather detailed information on each OSI layer, focusing on the functionalities, devices, and protocols.
- **Analysis:** Compare the roles of each layer and how they interact with each other.
- **Identification:** Create a diagram or chart that visually represents the OSI model, including the layers, devices, and protocols.
- **Presentation:** Prepare a presentation or report explaining each layer, the associated devices, and protocols, and how they contribute to network communication.

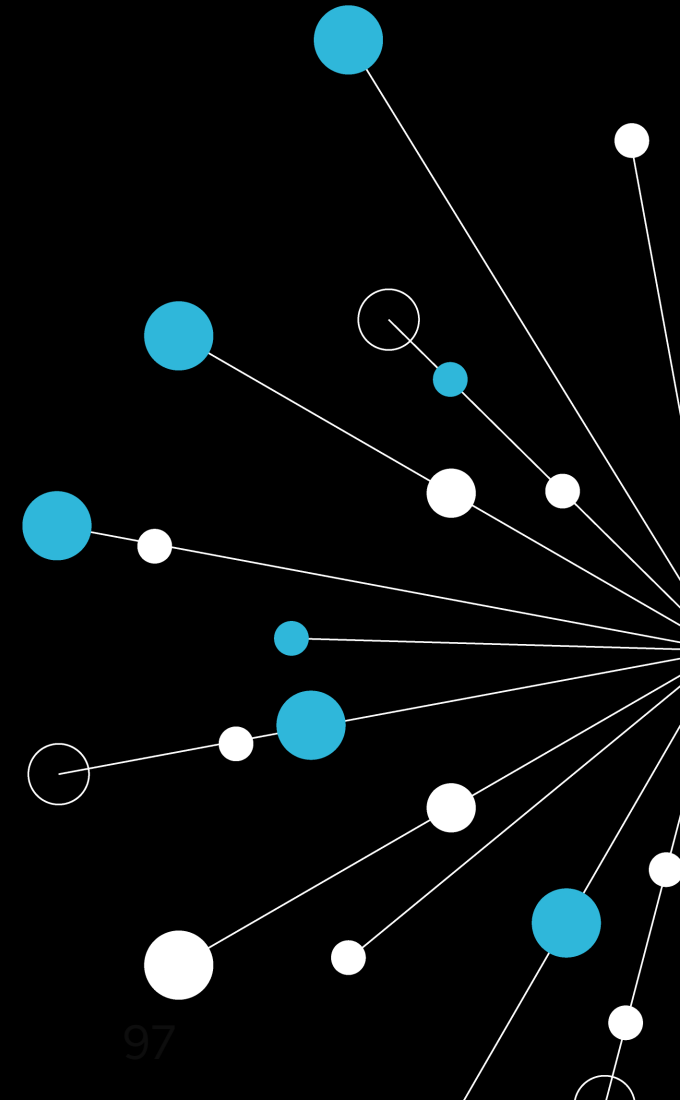


Activity 11: OSI encapsulation

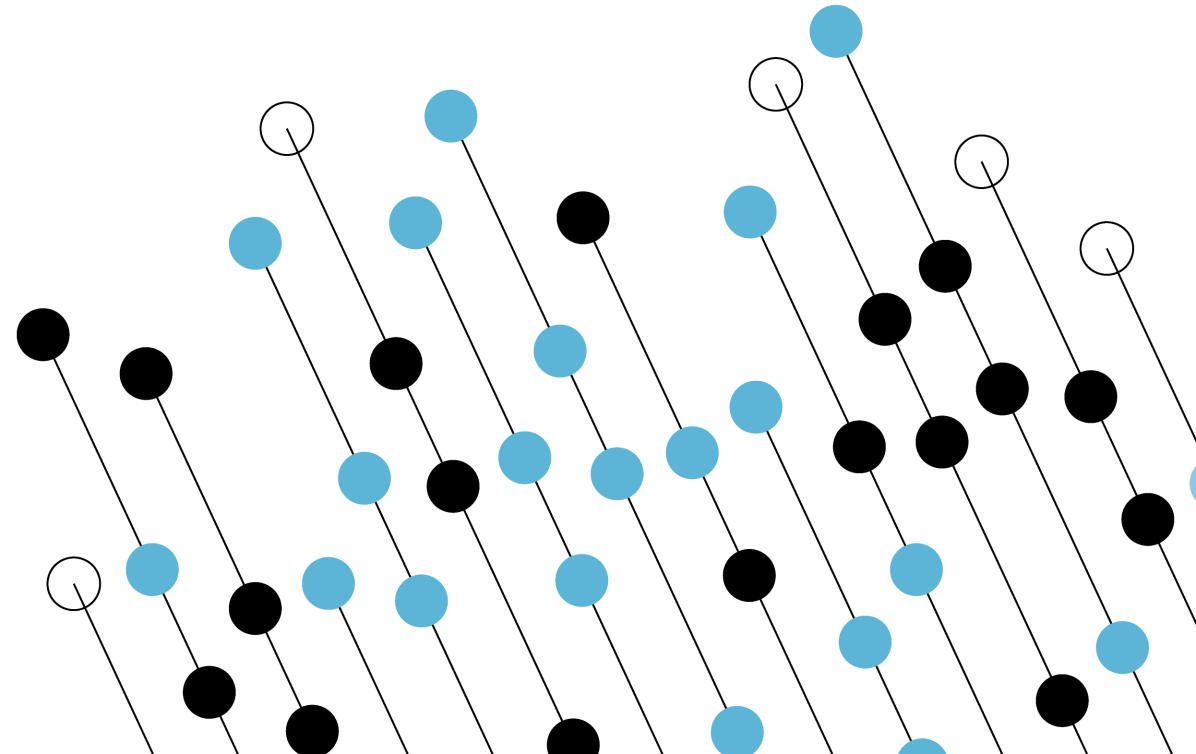
Open the '**Activity11 OSI encapsulation.pkt**' file in Packet Tracer.

Follow your trainer as they demonstrate the flow of data through the layers of the OSI model.

You can also follow the instructions in the **Learner Activity Document**.



Wireless technologies



Wireless technologies

- Enable devices to communicate without physical cable restrictions.
- Use radio frequency (RF) waves.
- Provide flexibility and mobility for users.
- Defined by the IEEE under 802.11 standards.

Wireless bands

The 2.4 GHz band provides the most coverage but transmits data at slower speeds.

The 5 GHz band provides less coverage but transmits data at faster speeds.

Wireless deploy mode

Infrastructure mode:

- Devices communicate through a configured wireless access point (WAP).
- Commonly deployed in businesses, homes and public Wi-Fi networks.

Ad-hoc mode:

- Peer-to-peer-like mode where devices connect directly to each other without a WAP.
- Suitable for a small-scale networks.
- Cannot scale and has range limitations.

Hotspot:

- Often uses cellular data and allows internet access for nearby devices.

Wireless standards

Standard	Frequency band	Maximum speed	Range	Notes
802.11a	5 GHz	54 Mbsp	36m	Less interference, shorter range.
802.11b	2.4 GHz	11 Mbsp	46m	Susceptible to interference.
802.11g	2.4 GHz	54 Mbsp	46m	Backward compatible with 802.11b.
802.11n	2.4 GHz/5 GHz	600 Mbsp	70m	Introduced MIMO (multiple input, multiple output).
802.11ac	5 GHz	1.3 Gbsp	35m	Introduce MU-MIMO (multi-user MIMO).
802.11ax	2.4 GHz/5 GHz	9.6 Gbps	36m	Improved efficiency, lower latency.

Wireless security

Protocol	Description	Level of security
WEP (wired equivalent privacy)	Older protocol, weak encryption, easily broken.	Weak
WPA (Wi-Fi protected access)	Improved security, uses TKIP encryption.	Moderate
WPA2	Stronger encryption using AES (advanced encryption standard).	Strong
WPA3	Latest standard, stronger encryption, protection against brute force attacks.	Very strong

Best practice for wireless security

- Strong encryption.
- Remove default password.
- MAC filtering.
- Disable SSID broadcasting.
- Update firmware.
- Use VPN.
- Additional layer of security (firewall, IDS, IPS).
- Reduce power range.

Network infrastructure

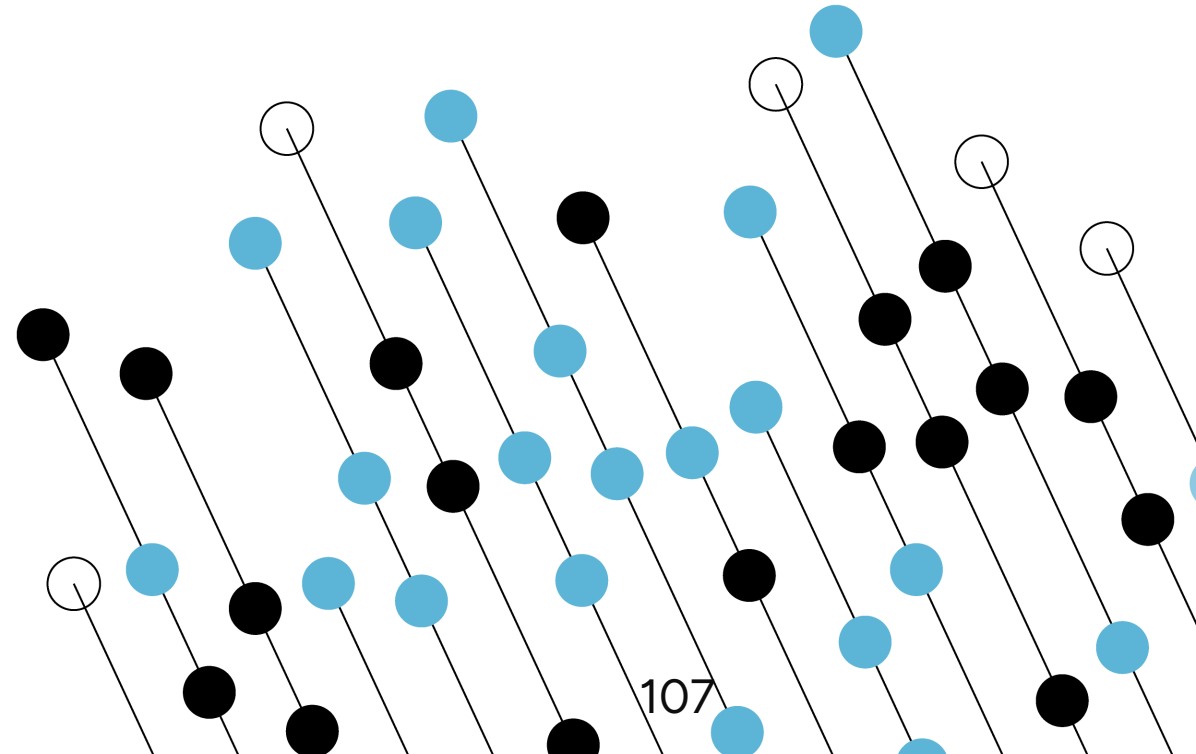
Day 3



Day 3 overview

- Numbering systems
- internet protocol version 4 (IPv4)
- Subnetting
- IPv4 versus IPv6
- Network address translation (NAT)
- Dynamic host configuration protocol (DHCP)
- Domain name system (DNS)

Binary, decimal, and hexadecimal



Numbering systems

A system for representing numbers of a certain type:

- Decimal
- Binary
- Octal
- Hexadecimal

Decimal	Binary	Octal	Hex
00	0000	00	0
01	0001	01	1
02	0010	02	2
03	0011	03	3
04	0100	04	4
05	0101	05	5
06	0110	06	6
07	0111	07	7
08	1000	10	8
09	1001	11	9
10	1010	12	A
11	1011	13	B
12	1100	14	C
13	1101	15	D
14	1110	16	E
15	1111	17	F

Decimal system

The decimal system, also known as the base-10 number system, is the most widely used positional numeral system in the world.

It is based on 10 symbols (digits) and uses positional value to represent numbers.

Key features

Base-10 system:

- The system uses 10 digits: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9.
- It is called base-10 because each digit's position represents a power of 10.

Decimal system (continued)

Positional value:

- Each digit in a number has a value depending on its position (place value).
- Moving left increases the power of 10, while moving right (for decimals) decreases it.

For a number like 5432, each digit is multiplied by 10 raised to the power of its position (starting from 0 on the right):

- $5432 = (5 \times 10^3) + (4 \times 10^2) + (3 \times 10^1) + (2 \times 10^0) = 5000 + 400 + 30 + 2$

Binary system

The binary system, also known as the base-2 number system, is a positional numeral system that uses only two digits: 0 and 1.

It is the foundation of modern digital technology, as computers and electronic systems process data using binary.

Key features

Base-2 system:

- Uses only two digits: 0 and 1.
- Each position represents a power of 2, starting from 2^0 at the rightmost position.

Binary system (continued)

Positional value:

- Similar to the decimal system, the value of a binary number depends on the digit and its position.
- Moving left increases the power of 2. Each binary digit (bit) represents a power of 2.

The binary number 1011 can be converted to decimal as:

- $1011 = (1 \times 2^3) + (0 \times 2^2) + (1 \times 2^1) + (1 \times 2^0) = 8 + 0 + 2 + 1 = 11$

Hexadecimal system

The hexadecimal system (often abbreviated as 'hex') is a base-16 numeral system used widely in computing and digital systems.

It is an extension of the decimal system, using 16 symbols to represent values.

Key features

Base-16:

- The system uses 16 symbols:
 - 0, 1, 2, 3, 4, 5, 6, 7, 8, 9 (for values 0 to 9)
 - A, B, C, D, E, F (for values 10 to 15).

Hexadecimal system (continued)

Positional value:

- Like other positional numeral systems (for example, decimal and binary), each digit's position in a hexadecimal number represents a power of 16.
- Each digit is multiplied by 16^n , where n is its position (starting from 0 on the right).

The hexadecimal number 1A3 can be converted to decimal as:

- $1A3 = (1 \times 16^2) + (10 \times 16^1) + (3 \times 16^0) = 256 + 160 + 3 = 419$

Complete the table for the binary and hexadecimal values

Decimal			Binary							HEX
10s	Units		8s	4s	2s	Units		256s	16s	Units
	0		0	0	0	0				0
	1		0	0	0	1				1
	2									
	3									
	4									
	5									
	6									
	7									
	8									
	9									
1	0									
1	1									
1	2									
1	3									
1	4									
1	5									

Complete binary table

Decimal			Binary							HEX
10s	Units		8s	4s	2s	Units		256s	16s	Units
	0		0	0	0	0				0
	1		0	0	0	1				1
	2		0	0	1	0				
	3		0	0	1	1				
	4		0	1	0	0				
	5		0	1	0	1				
	6		0	1	1	0				
	7		0	1	1	1				
	8		1	0	0	0				
	9		1	0	0	1				
1	0		1	0	1	0				
1	1		1	0	1	1				
1	2		1	1	0	0				
1	3		1	1	0	1				
1	4		1	1	1	0				
1	5		1	1	1	1				

Completed hexadecimal table

Decimal			Binary							HEX
10s	Units		8s	4s	2s	Units		256s	16s	Units
	0		0	0	0	0				0
	1		0	0	0	1				1
	2		0	0	1	0				2
	3		0	0	1	1				3
	4		0	1	0	0				4
	5		0	1	0	1				5
	6		0	1	1	0				6
	7		0	1	1	1				7
	8		1	0	0	0				8
	9		1	0	0	1				9
1	0		1	0	1	0				A
1	1		1	0	1	1				B
1	2		1	1	0	0				C
1	3		1	1	0	1				D
1	4		1	1	1	0				E
1	5		1	1	1	1				F

Decimal to binary: Convert 167 to its binary value

Conversion steps:

- 1. $167 - 128 = 39$
- 2. No 64s
- 3. $39 - 32 = 7$
- 4. No 16s or 8s
- 5. $7 - 4 = 3$
- 6. $3 - 2 = 1$

128	64	32	16	8	4	2	1
128 +	0 +	32 +	0 +	0 +	4 +	2 +	1
1	0	1	0	0	1	1	1

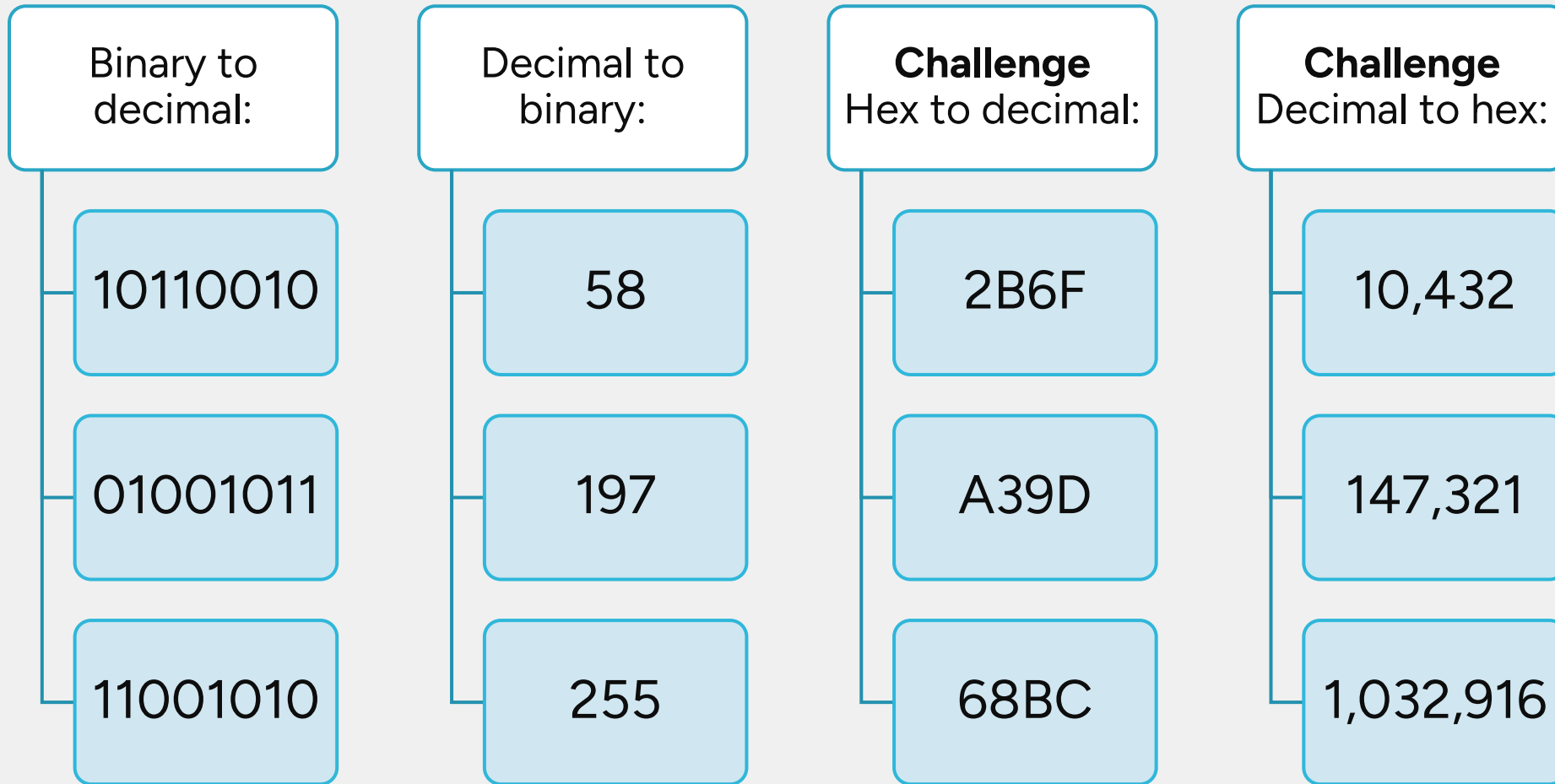
Binary to decimal: Convert 10110101 to its decimal value

128	64	32	16	8	4	2	1
1	0	1	1	0	1	0	1
128 +	0 +	32 +	16 +	0 +	4 +	0 +	1

Converting hexadecimal to binary

4096	256	16	1
1	D	B	9
4096	13 x 256	11 x 16	1 x 9
4096 +	3328 +	176 +	9

Activity: Convert between number systems

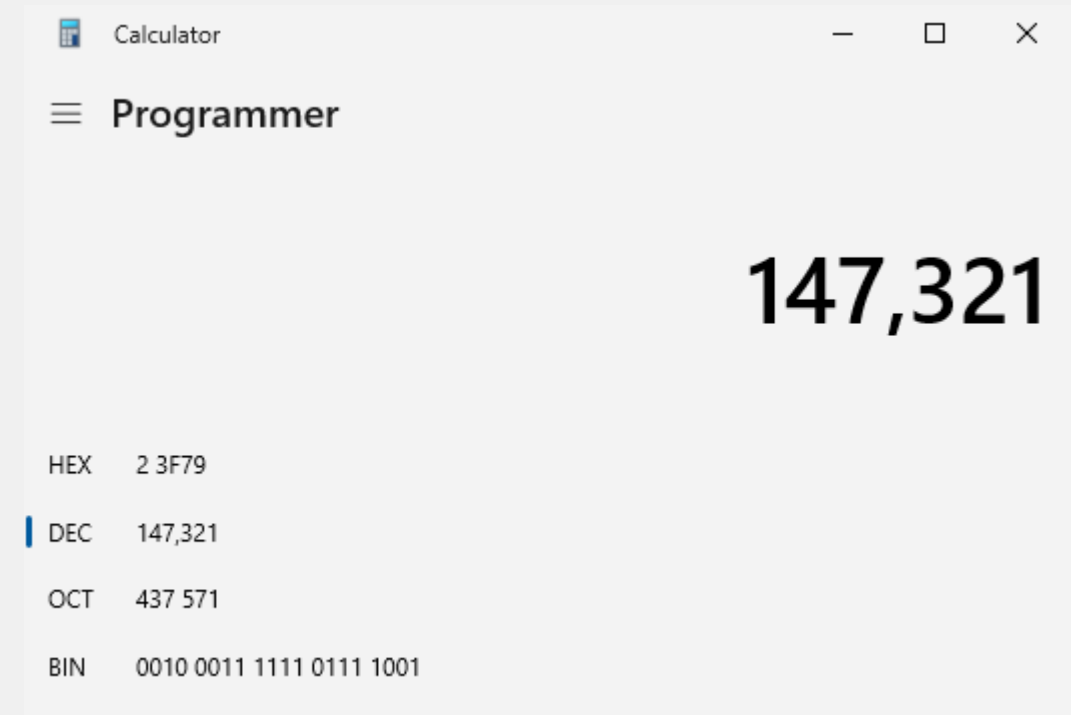


How can we check our answers?

Open the Calculator on your computer.

Open the menu and change to calculator type to **Programmer**.

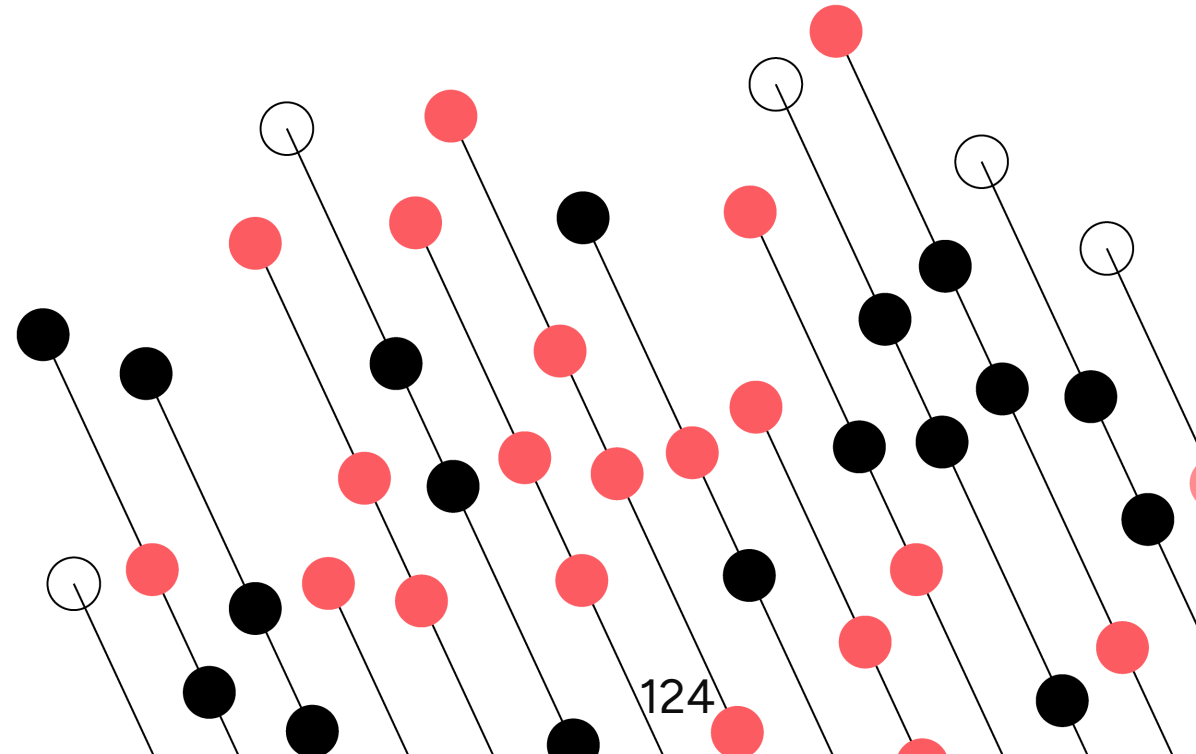
Enter the value and select the different numbering systems to see the equivalent values.



Answers

Decimal	Binary	Hexadecimal
178	10110010	B2h
75	01001011	48h
202	11001010	CAh
58	00111010	3Ah
197	11000101	C5h
255	11111111	FF
11,119	-	2B6F
41,885	-	A39D
26,812	-	68BC
10,432	-	28C0h
147,321	-	23F79h
1,032,916	-	FC2D4h

internet protocol version 4 (IPv4)



IPv4 address structure

How big is the IPv4 address space?

- 32-bits

How many addresses does that provide?

- 0 to 4,294,967,295 ($2^{32} - 1$)

IPv4 address structure (continued)

IPv4 addresses consist of four numbers, each between 0 and 255.

Valid IP addresses:

- 10.80.1.5
- 172.16.5.254
- 192.168.1.4

Why are these invalid IP addresses?

- 10.80.265.5
- 172.16.254
- 0.0.0.0
- 255.255.255.255

IPv4 structure

IP address represents:

- The network number (network ID) — this number is common to all hosts on the same network.
- The host number (host ID) — this unique number identifies a host on a particular network.

IPv4 structure (continued)

32-bit binary value:

- 11000110001010010001000000001001

Expressed in 8-bit octets:

- 11000110 00101001 00010000 00001001

Converted to dotted decimal notation for entry into configuration dialogs:

- 198.41.16.9

IPv4 in binary

IP decimal	192	168	1	5
IP binary	1100 0000	1010 1000	0000 0001	0000 0101
Subnet mask decimal	255	255	255	0
Subnet mask binary	1111 1111	1111 1111	1111 1111	0000 0000
Network ID decimal	?	?	?	?
Host ID decimal	?	?	?	?

IPv4 in binary (continued)

IP decimal	192	168	1	5
IP binary	1100 0000	1010 1000	0000 0001	0000 0101
Subnet mask decimal	255	255	255	0
Subnet mask binary	1111 1111	1111 1111	1111 1111	0000 0000
Network ID decimal	192	168	1	0
Host ID decimal	0	0	0	5

IPv4 class A summary

Range: 1.0.0.0 to 126.255.255.255

First octet: 0–127 (except 127, reserved for loopback).

Default subnet mask: 255.0.0.0 (or /8 CIDR).

Network/host division:

- **1 octet** for the network.
- **3 octets** for hosts.

Total networks: 128 (but only 126 usable).

Total hosts: ~16.7 million per network.

Usage: Large organisations and ISPs.

IPv4 class B summary

Range: 128.0.0.0 to 191.255.255.255

First octet: 128–191.

Default subnet mask: 255.255.0.0 (or /16 CIDR).

Network/host division:

- **2 octets** for the network.
- **2 octets** for hosts.

Total networks: ~16,384.

Total hosts: ~65,536 per network.

Usage: Medium-sized organisations and universities.

IPv4 class C summary

Range: 192.0.0.0 to 223.255.255.255

First octet: 192–223.

Default subnet mask: 255.255.255.0 (or /24 CIDR).

Network/host division:

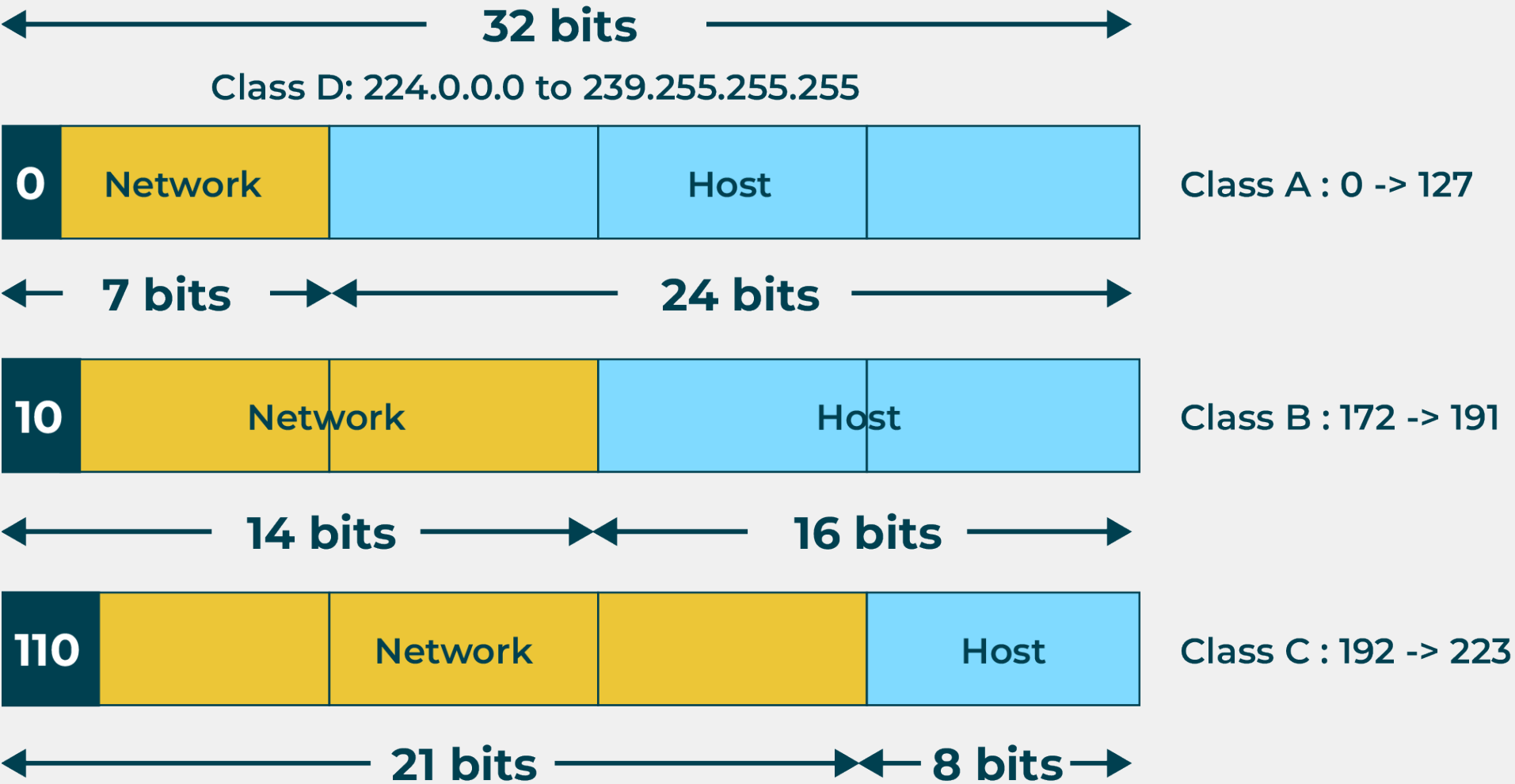
- **3 octets** for the network.
- **1 octet** for hosts.

Total networks: ~2 million.

Total hosts: 254 per network.

Usage: Small networks like private businesses.

IPv4 addresses: Classes A, B, and C



IPv4 classes D and E summary

Class D

Range: 224.0.0.0 to 239.255.255.255

First octet: 224–239.

Purpose: Reserved for multicasting.

Not usable: No subnetting or host addresses.

Usage: Streaming and broadcasting data (for example, video or audio).

Class E

Range: 240.0.0.0 to 255.255.255.255

First octet: 240–255.

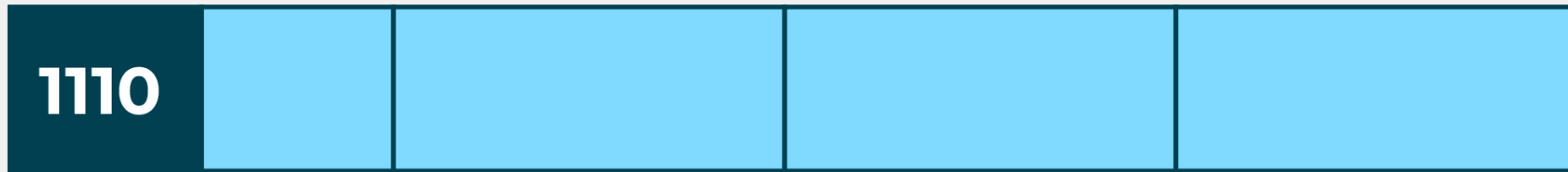
Purpose: Reserved for experimental use.

Not usable: Not assigned to hosts.

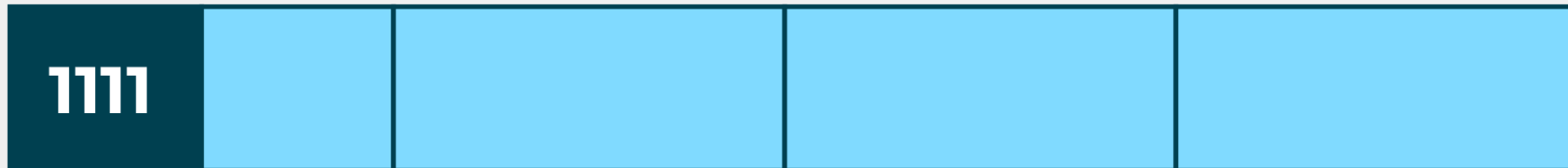
IPv4 addresses: Classes D and E

← 32 bits →

Class D: 224.0.0.0 to 239.255.255.255



Class E: 240.0.0.0 to 255.255.255.255



Subnetting

Subnetting is the act of dividing the range of network IPv4 addresses into smaller blocks.

The numbers will halve or double, due to the nature of binary.

As the number of network IDs goes up, number of host IDs goes down.

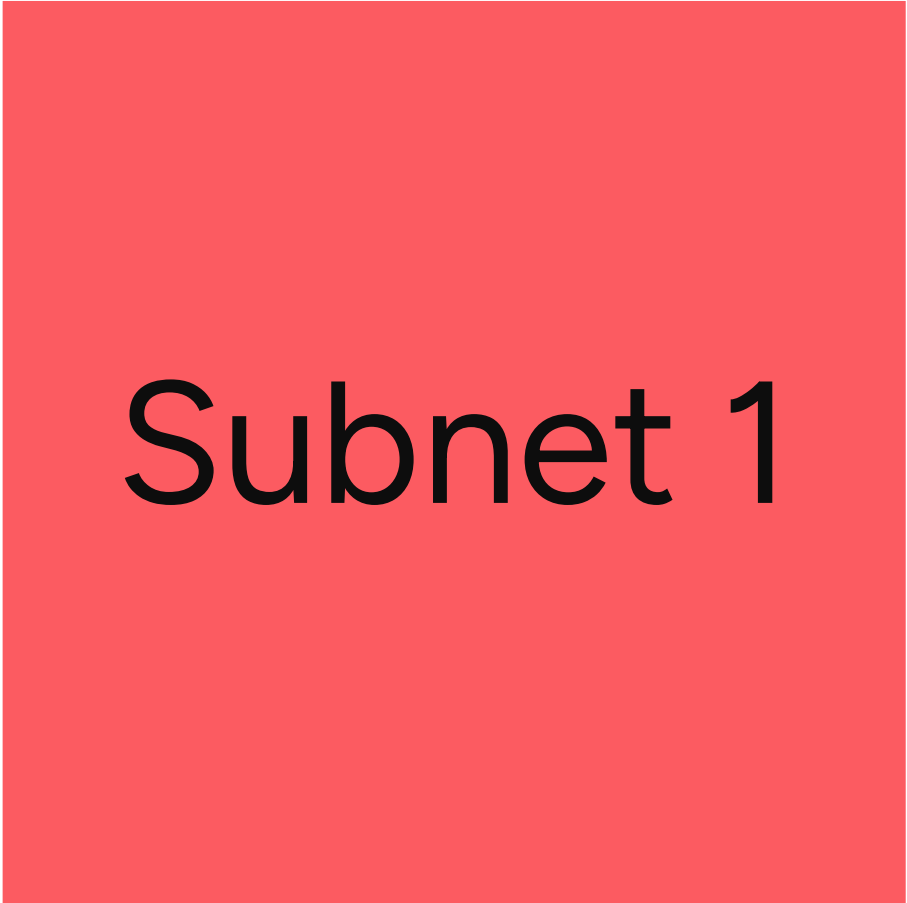
Subnetting (2)

128	64	32	16	8	4	2	1
2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
2x2x2x2x2x2x2	2x2x2x2x2x2	2x2x2x2x2	2x2x2x2	2x2x2	2x2	2	1

Subnetting method

The image represents the class C address space:

- Network ID: 192.168.10.0
- First host address: 192.168.10.1
- Last host address: 192.168.10.254
- Broadcast address: 192.168.10.255



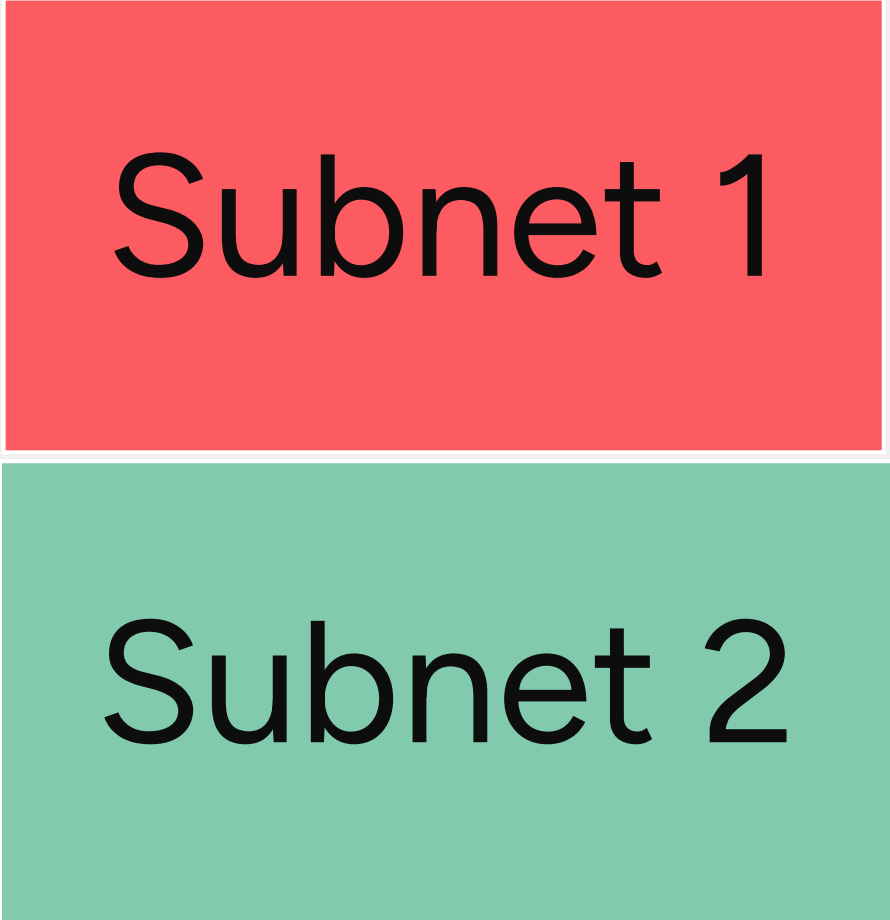
Subnet 1

Subnetting method (2)

By allocating bits from the IP address to represent the Subnet, the address space can be divided into smaller blocks.

By dividing the block in half, you create two subnets.

Can you identify the addresses for each?



Subnet 1

Subnet 2

Subnetting method (3)

Subnet 1

Network ID: 192.168.10.0

First address: 192.168.10.01

Last address: 192.168.10.126

Broadcast address: 192.168.10.157

Subnet 2

Network ID: 192.168.10.128

First address: 192.168.10.129

Last address: 192.168.10.254

Broadcast address: 192.168.10.255

How does subnetting work?

Decimal	Binary	Number of subnets possible	Number of hosts per subnet	Slash notation
0	0000 0000	0	254 (256-2)	/24
128	1000 0000	2	126 (128-2)	/25
192	1100 0000	4	62 (64-2)	/26
224	1110 0000	8	30 (32-2)	/27
240	1111 0000	16	14 (16-2)	/28
248	1111 1000	32	6 (8-2)	/29
252	1111 1100	64	2 (4-2)	/30
254	1111 1110	128 – not used	0	/31
255	1111 1111	0 – not used	0	/32

Create a subnetting 'cheat sheet'

	Subnet mask	Network ID	1st address	Last address	Broadcast address
Class C address	/24	0	1	254	255
2 x IP address range 126 hosts per subnet	/25	0	1	126	127
		128	129	254	255

Subnet mask: 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 0 0 0 0 0 0 0 = 255.255.255.128

Network ID
24 bits

Subnet
ID bit(s)

Host ID
7 bits

Create a subnetting 'cheat sheet' (continued)

	Subnet mask	Network ID	1st address	Last address	Broadcast address
Class C address	/24	0	1	254	255
2 x IP address range 126 hosts per subnet	/25	0	1	126	127
		128	129	254	255
4 x IP address range 62 hosts per subnet	/25	0	1	62	63
		64	65	126	127
		128	129	190	191
		192	193	254	255

/25 subnet mask: 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 0 0 0 0 0 0 0 = 255.255.255.192

Network ID
24 bits

Subnet ID bit(s)
6 bits

Host ID
6 bits

Public and private addresses

- IPv4 addresses are further classified as either public or private.
- Public IP addresses are ones that are exposed to the internet.
- Any other computers on the internet can potentially communicate with them.
- Private IP addresses are hidden from the internet and any other networks.
- They are usually behind an IP proxy or firewall device.

Private IPv4 address ranges

Can you name any of the private IP address ranges?

- 10.0.0.0 to 10.255.255.255
- 172.16.0.0 to 172.31.255.255
- 192.168.0.0 to 192.168.255.255

APIPA:

- 169.254.x.y - FE80

Loopback:

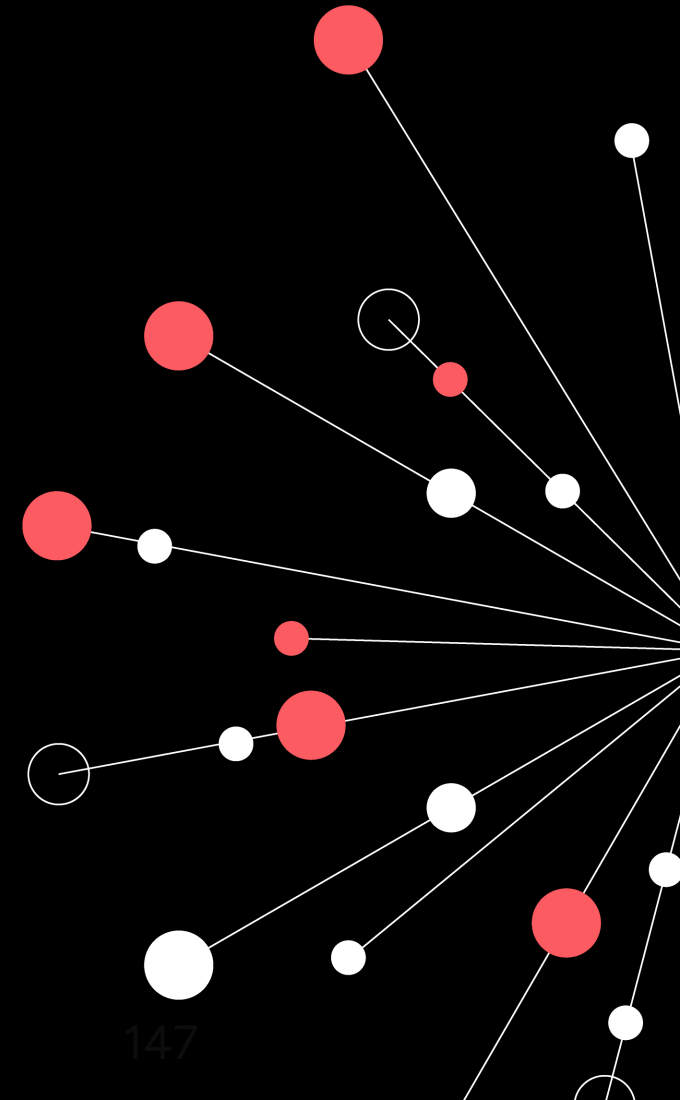
- 127.0.0.1 - ::1

Activity 12: IPv4 addressing

Open the '**Activity 12: IPv4 addressing**' file in Packet Tracer:

- Use the **Check Results** button at the bottom of the Packet Tracer **Activity Instructions** window to check your progress.
- Take a screenshot of the final assessment items to show you have completed the activity successfully.
- The instructions are in the PKA file and repeated in the **Learner Activity Document**.

If you need any assistance, just message the trainer.



Port numbers

A port number in computer networking is a numerical identifier assigned to specific processes or services running on a device. It helps distinguish between different types of network traffic, enabling computers to communicate efficiently.

How many port numbers are there in total?

- 65536

Why is it that number?

- 16-bit address space

Types of port

Well-known ports (0–1023):

- Assigned by IANA (Internet Assigned Numbers Authority).
- Used by standard network services.

Examples:

- 20, 21 – FTP (file transfer protocol)
- 22 – SSH (secure shell)
- 25 – SMTP (email sending)
- 53 – DNS (domain name System)
- 80 – HTTP (web browsing)
- 443 – HTTPS (secure web browsing)

Types of port (2)

Registered ports (1024–49151):

- Assigned for specific applications but not as universal as well-known ports.

Examples:

- 1433 – Microsoft SQL Server
- 3306 – MySQL database
- 3389 – Remote desktop protocol (RDP)

Types of port (3)

Dynamic / ephemeral ports (49152–65535):

- Used temporarily for client connections.
- Assigned dynamically by the operating system.

Examples:

- Used in online gaming, video calls, or temporary sessions.

How port numbers work

When a client initiates communication, it selects a source port from the dynamic range.

It connects to a destination port corresponding to the intended service.

Example:

A web browser accessing <https://example.com>:

- Client: Uses a random ephemeral port (for example, 54321).
- Server: Listens on port 443 (HTTPS).
- Response: Sent back to the client's ephemeral port.

Port number top 20 quiz: Part 1

Port number	Protocol	Service / function
20		FTP (file transfer protocol): Data transfer
21	TCP	
	TCP	SSH (secure shell): Secure remote login
23	TCP	
		SMTP (simple mail transfer protocol): Sending emails
53		
		DHCP (Dynamic host configuration protocol): IP address assignment
80	TCP	HTTP (hypertext transfer protocol): Web browsing
	TCP	POP3 (post office protocol v3): Email retrieval
119	TCP	

Port number top 20 quiz: Part 1 answers

Port number	Protocol	Service / function
20	TCP	FTP (file transfer protocol): Data transfer
21	TCP	FTP: Command control
22	TCP	SSH (secure shell): Secure remote login
23	TCP	Telnet: Remote terminal access (deprecated, insecure)
25	TCP	SMTP (simple mail transfer protocol): Sending emails
53	UDP / TCP	DNS (domain name system): Name resolution
67/68	UDP	DHCP (Dynamic host configuration protocol): IP address assignment
80	TCP	HTTP (hypertext transfer protocol): Web browsing
110	TCP	POP3 (post office protocol v3): Email retrieval
119	TCP	NNTP (network news transfer protocol): Usenet discussions

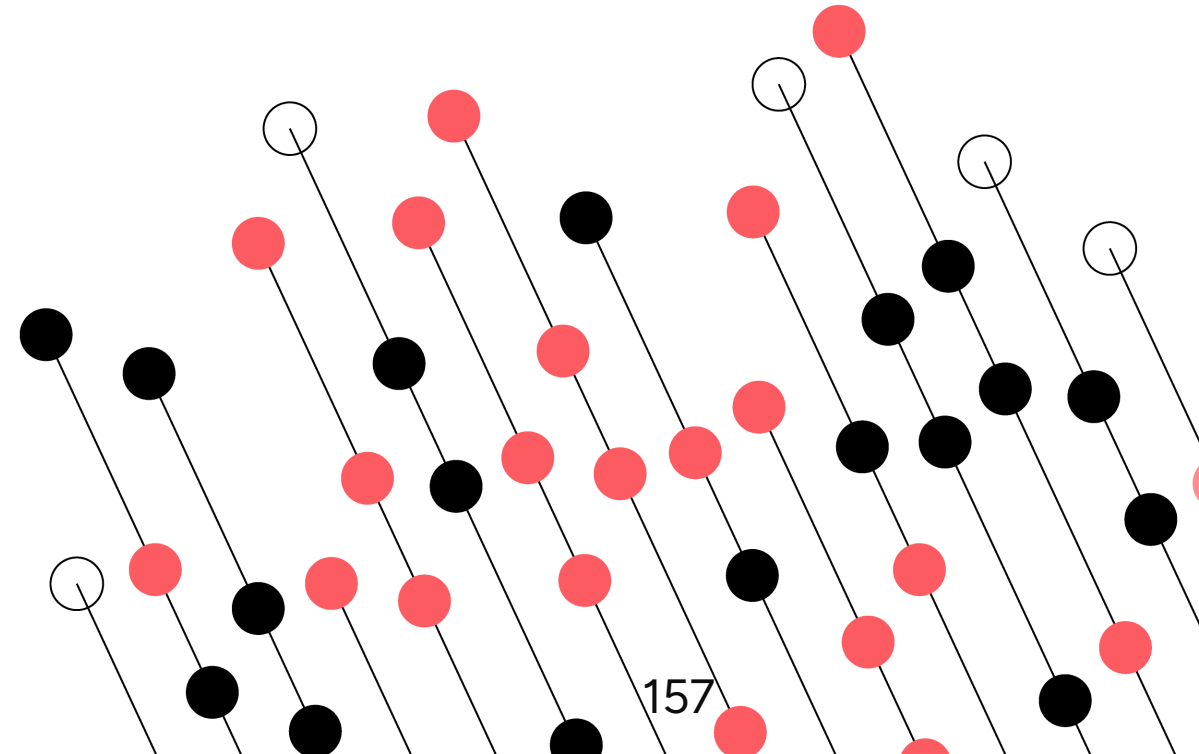
Port number top 20 quiz: Part 2

Port number	Protocol	Service / function
		NTP (network time protocol): Time synchronisation
143		
		SNMP (simple network management protocol): Network monitoring
179	TCP	
		LDAP (lightweight directory access protocol): Directory services
443	TCP	HTTPS (secure web browsing — encrypted HTTP)
465		
		IMAPS (secure IMAP): Secure email retrieval
995	TCP	
3389		

Port number top 20 quiz: Part 2 answers

Port number	Protocol	Service / function
123	UDP	NTP (network time protocol): Time synchronisation
143	TCP	IMAP (internet message access protocol): Email retrieval
161/162	UDP	SNMP (simple network management protocol): Network monitoring
179	TCP	BGP (border gateway protocol): internet routing
389	TCP / UDP	LDAP (lightweight directory access protocol): Directory services
443	TCP	HTTPS (secure web browsing — encrypted HTTP)
465	UDP	SMTPTS (secure SMTP): Secure email sending
993	TCP	IMAPS (secure IMAP): Secure email retrieval
995	TCP	POP3S (secure POP3): Secure email retrieval
3389	TCP	RDP (remote desktop protocol): Remote Windows access

Network address translation (NAT)



Network address translation

Network address translation (NAT) is a method used in networking to modify IP address information in packet headers while they are in transit. It is primarily used to manage and conserve public IP addresses and improve network security.

NAT key concepts

Purpose:

- Enables multiple devices on a private network to share a single public IP address.
- Provides an additional layer of security by masking internal IP addresses.

Function:

- Translates private IP addresses to a public IP address for outgoing traffic.
- Maps public IP addresses back to private IP addresses for incoming traffic.

Location:

- Typically implemented on routers or firewalls at the boundary of private and public networks.

Types of NAT

Static NAT:

- One-to-one mapping of a private IP address to a public IP address.
- Used for servers or devices requiring a consistent public IP.

Dynamic NAT:

- Maps a pool of public IP addresses to private IP addresses.
- Allocation is on a first-come, first-served basis.

Port address translation (PAT) (or NAT overloading):

- Maps multiple private IP addresses to a single public IP using unique port numbers.
- Most common type of NAT.
- Efficient use of public IP addresses.

Advantages of NAT

IP conservation:

- Reduces the need for multiple public IP addresses.

Enhanced security:

- Internal network structure is hidden from external networks.

Flexibility:

- Allows seamless integration of private and public networks.

Disadvantages of NAT

Breaks end-to-end connectivity:

- Can interfere with protocols requiring direct IP-to-IP communication.

Performance overhead:

- Additional processing can introduce latency.

Limited port availability:

- Overloading can exhaust available ports in large networks.

Static NAT (one-to-one mapping)

Description:

- Maps a private IP address to a fixed public IP address.
- Used when an internal device (for example, a web server) needs a permanent public IP.

Use case:

- Hosting internal servers (for example, web, email, or VoIP servers) accessible from the internet.
- Allowing remote access to specific devices.

Static NAT (one-to-one mapping) (continued)

Configuration example (Cisco IOS):

- `ip nat inside source static 192.168.1.100 203.0.113.50`

This maps 192.168.1.100 (internal server) to 203.0.113.50 (public IP).

Dynamic NAT (many-to-many mapping)

Description:

- Uses a pool of public IP addresses to translate internal private addresses dynamically.
- Assigns a public IP from the pool whenever an internal device needs access to the internet.

Use case:

- Suitable when multiple users need internet access but with a limited pool of public IPs.
- Enterprise environments that require outbound internet connectivity without exposing internal addresses permanently.

Dynamic NAT (many-to-many mapping) (continued)

Configuration example (Cisco IOS):

- `ip nat pool NAT_POOL 203.0.113.100 203.0.113.200 netmask 255.255.255.0`
- `ip nat inside source list 1 pool NAT_POOL`
- `access-list 1 permit 192.168.1.0 0.0.0.255`

This allows dynamic allocation of public IPs from 203.0.113.100 to 203.0.113.200 for internal users in the 192.168.1.0/24 subnet.

Port address translation (PAT) or NAT overload (many-to-one)

Description:

- Maps multiple private IP addresses to a single public IP by using unique port numbers.
- Most commonly used NAT method due to its efficiency in conserving public IPs.

Use case:

- Used in home networks, corporate offices, and ISPs where many users share a single public IP.
- Ideal when an enterprise has limited public IPs.

Port address translation (PAT) or NAT overload (many-to-one) (continued)

Configuration example (Cisco IOS):

- `ip nat inside source list 1 interface GigabitEthernet0/1 overload`
- `access-list 1 permit 192.168.1.0 0.0.0.255`

Here, all private IPs in 192.168.1.0/24 will share the GigabitEthernet0/1 interface's public IP.

Policy-based NAT

Description:

- Translates IP addresses based on specific conditions like destination, protocol, or application.
- More flexible than static or dynamic NAT.

Use case:

- Used when certain traffic requires different NAT rules.
- For enterprises that need to route traffic differently based on internal policies.

Policy-based NAT (continued)

Configuration example (Cisco IOS):

- route-map NAT_POLICY permit 10
- match ip address 2
- set ip next-hop 203.0.113.254
- ip nat inside source route-map NAT_POLICY interface GigabitEthernet0/1 overload
- access-list 2 permit 192.168.1.100 0.0.0.0

This applies NAT only for specific traffic (for example, traffic from 192.168.1.100).

NAT64 (IPv6 to IPv4 translation)

Description:

- Used in networks transitioning from IPv4 to IPv6.
- Allows IPv6-only clients to communicate with IPv4 services.

Use case:

- Large enterprises migrating to IPv6 while maintaining access to IPv4-only systems.
- ISPs that provide IPv6 connectivity while ensuring compatibility with IPv4 services.

NAT64 (IPv6 to IPv4 translation) (continued)

Configuration example (Cisco IOS):

- interface GigabitEthernet0/1
- nat64 enable
- nat64 prefix 64:ff9b::/96

This enables NAT64, using 64:ff9b::/96 as the prefix to translate IPv6 addresses to IPv4.

Dual NAT (inside and outside NAT)

Description:

- Applies NAT in both incoming and outgoing directions.
- Allows enterprises to handle both private-to-public and public-to-private translations.

Use case:

- Used by service providers or enterprises with multiple routing domains.
- Networks that need both inbound and outbound NAT.

Dual NAT (inside and outside NAT) (continued)

Configuration example (Cisco IOS):

- `ip nat inside source static 192.168.1.10 203.0.113.50`
- `ip nat outside source static 203.0.113.50 192.168.1.10`

This ensures bi-directional NAT for internal and external users.

Choosing the right NAT method

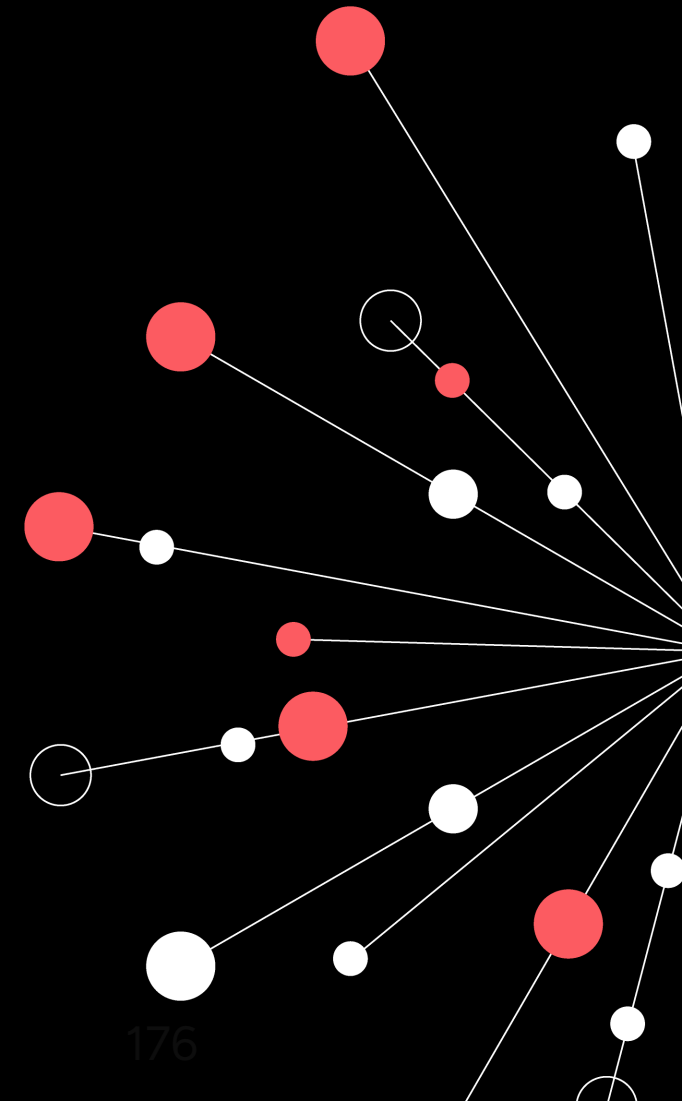
NAT type	Best for	Pros	Cons
Static NAT	Servers needing fixed public IP	Permanent mapping	Uses many public IPs
Dynamic NAT	Limited pool of public IPs	Flexible allocation	Runs out of IPs if demand is high
PAT (NAT overload)	Large networks with limited public IPs	Saves public IPs, efficient	Port exhaustion in high-traffic networks
Policy-based NAT	Conditional NAT rules	Granular control	Complex setup
NAT64	IPv6-only clients communicating with IPv4	Bridges IPv6 and IPv4 networks	Doesn't support full bidirectional communication
Dual NAT	Complex networks with inside and outside NAT needs	Full control of NAT rules	Can be difficult to manage

Activity 13: NAT and PAT lab

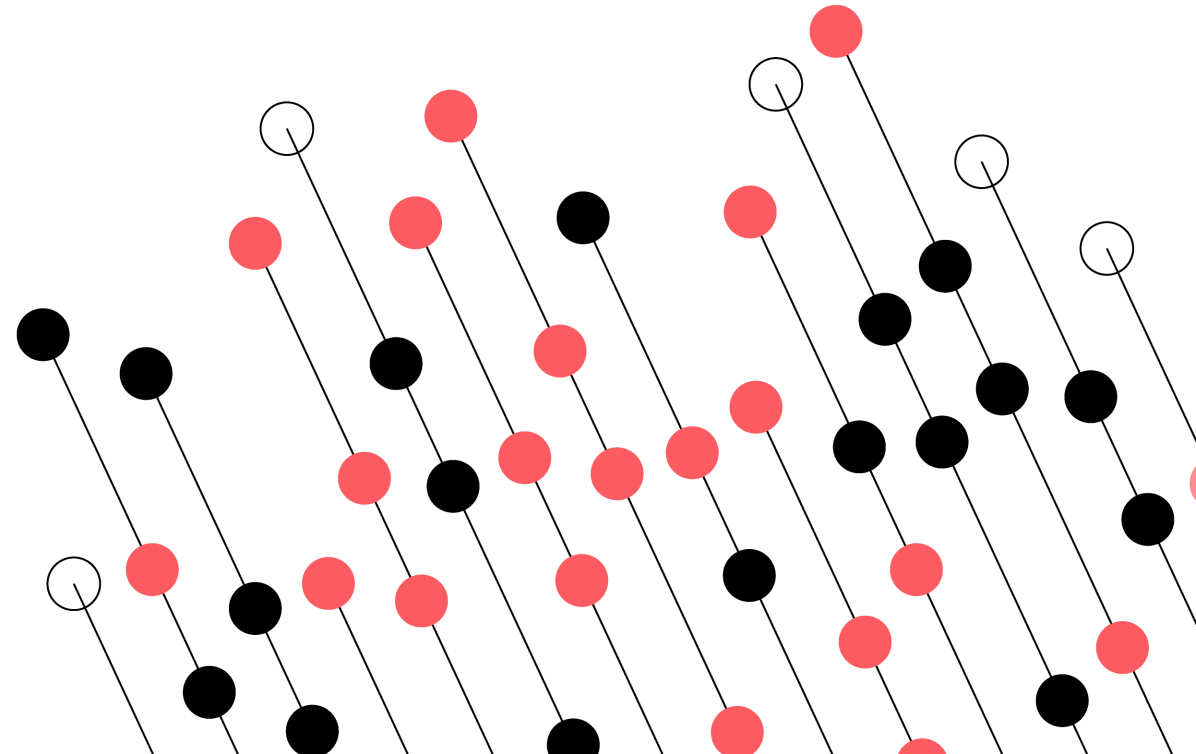
Open the '**Activity 13 NAT and PAT lab.pka**' file in Packet Tracer:

- Use the **Check Results** button at the bottom of the Packet Tracer **Activity Instructions** window to check your progress.
- Take a screenshot of the final assessment items to show you have completed the activity successfully.
- The instructions are in the PKA file and repeated in the **Learner Activity Document**.

If you need any assistance just message the trainer.



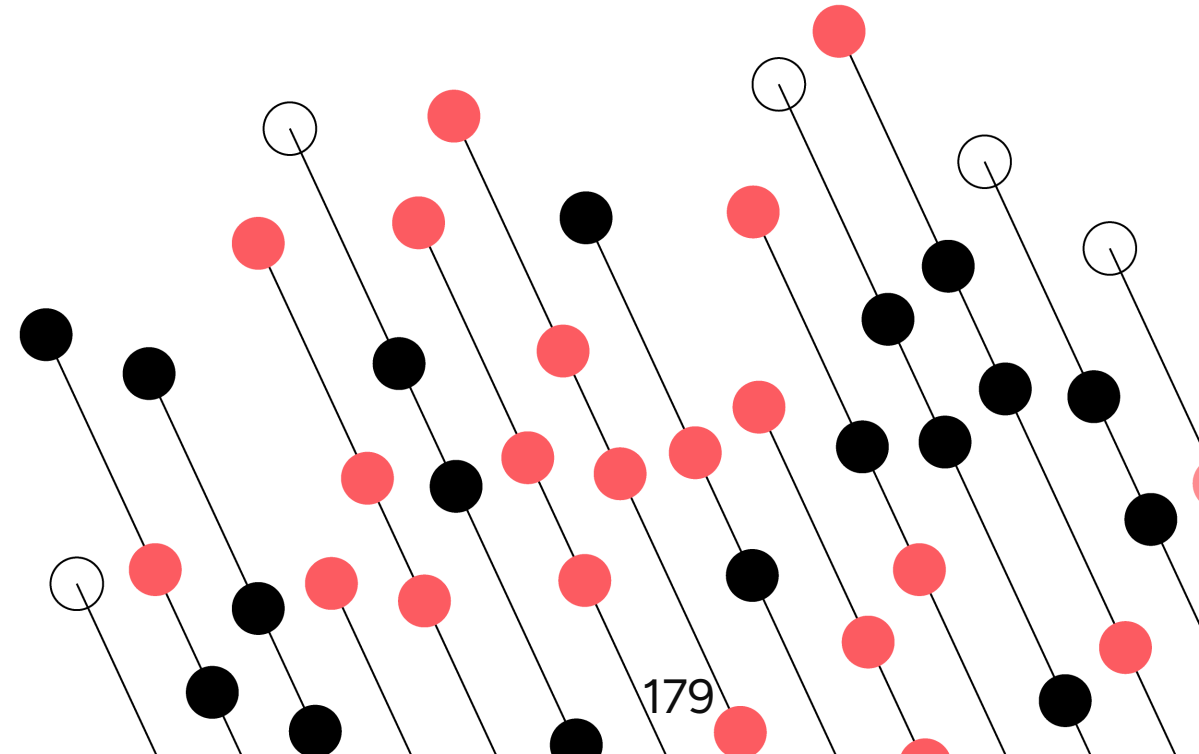
Troubleshooting IPv4



Troubleshoot IPv4

- What are the main questions to ask when presented with a networking issue?
- What are the main points to look for in the initial diagnosis?
- What tools can you use to gather the correct information?
- What deeper diagnosis steps could you take if the problem is not a simple one?

IPv4 vs IPv6



IPv6 recap

IPv4 and IPv6 are the two main versions of the internet protocol (IP) used for identifying and addressing devices on networks.

IPv4 is the older and widely used version, while IPv6 is the newer protocol designed to address the limitations of IPv4.

IPv6 recap (continued)

What address space have they defined for IPv6?

- 128-bit

How many addresses has that made available?

- $2^{128} = 340$ undecillion. 3.4×10^{38} An undecillion = 10^{36}
- 340 trillion trillion trillion or 340 billion billion billion billion
- 340,282,366,920,938,463,463,374,607,431,768,211,456

Comparing IPv4 to IPv6: Address structure and format

Feature	IPv4	IPv6
Address length	32-bit	128-bit
Address format	Dotted decimal (for example, 192.168.1.1)	Hexadecimal (for example, 2001:db8::1)
Address space	~4.3 billion	340 undecillion
Subnetting	Uses subnet masks (for example, 255.255.255.0)	Uses CIDR notation (/64, /128)

Comparing IPv4 to IPv6: Performance and efficiency

Feature	IPv4	IPv6
Routing efficiency	More complex, requires NAT and subnetting	Simplified due to hierarchical addressing
Header size	Variable, complex	Fixed, streamlined
Packet processing	Requires checksums, more overhead	No checksums, optimised routing
Quality of service (QoS)	No built-in QoS support	Built-in 'flow label' for prioritisation

Comparing IPv4 to IPv6: Security features

Feature	IPv4	IPv6
Security built-in	No, must use add-ons (for example, IPSec)	Yes, supports IPSec by default
End-to-end encryption	Not required	Mandatory for secure communication
Address spoofing prevention	Possible	Harder due to secure neighbour discovery (SEND)

Comparing IPv4 to IPv6: Address configuration

Feature	IPv4	IPv6
Address assignment	DHCP (dynamic host configuration protocol)	SLAAC (stateless address autoconfiguration) or DHCPv6
Network address translation (NAT)	Required due to address shortage	Not needed (every device has a unique address)
Private addressing	Uses private IP ranges (for example, 192.168.0.0/16)	Unique local addresses (fc00::/7)

Comparing IPv4 to IPv6: Compatibility and deployment

Feature	IPv4	IPv6
Adoption rate	Still dominant	Growing but slower adoption
Backward compatibility	Fully supported	Not directly backward-compatible with IPv4
Transition mechanisms	N/A	Tunnelling (for example, 6to4, Teredo), dual stack, translation

Comparing IPv4 to IPv6: Use cases and future outlook

Feature	IPv4	IPv6
Best for	Legacy systems, small networks	Future-proofing, IoT, large-scale networks
Internet of Things (IoT)	Limited due to address exhaustion	Ideal for IoT with a virtually unlimited address pool
Cloud and data centres	Supported but complex	More efficient and scalable

Match the IPv6 address to its type

Address types:

- Loopback address
- Global unicast address
- Link-local address
- Unique-local address
- Multicast address

IPv6 address	Address type
2001:0F32:1:1357::FD63:ABCD:C012	
FF03::5	
FF00::	
::1	
FF00::E37:4D49:B286:27B	
FE80::6891:BCA1:A439	
FC00:89:F:6::DE4:A7D3:789A	
2042:DC8:A:D4:28:EF3D:12E2:AC4A	

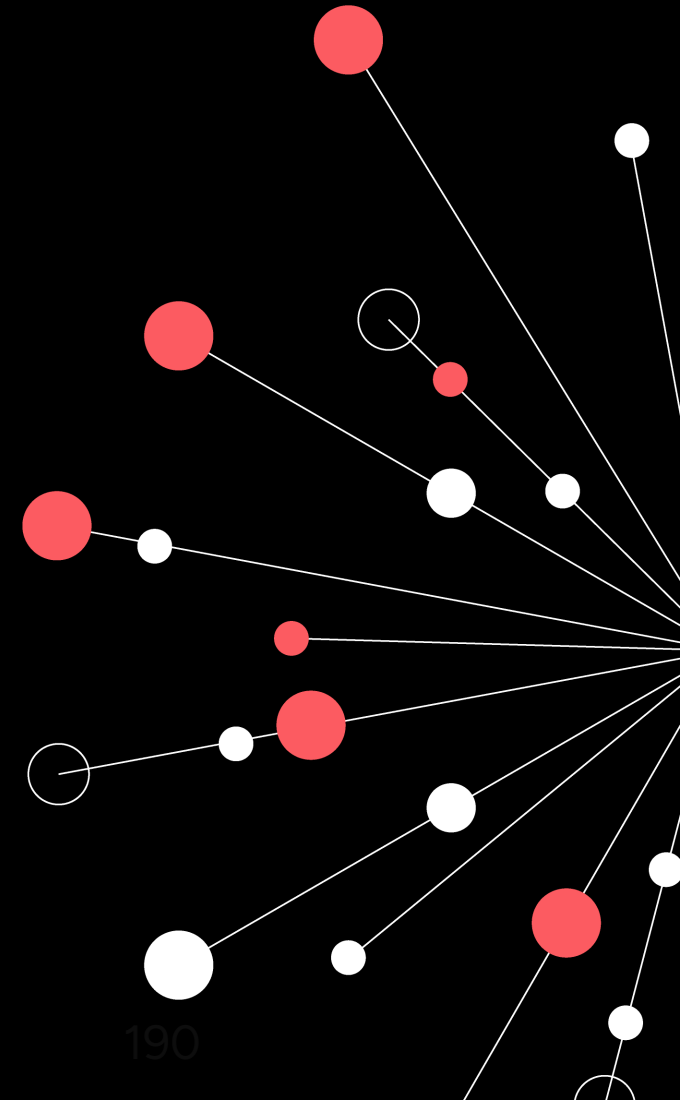
Match the IPv6 address to its type (answers)

IPv6 address	Address type
2001:0F32:1:1357::FD63:ABCD:C012	Global unicast address
FF03::5	Multicast address
FF00::	Multicast address
::1	Loopback address
FF00::E37:4D49:B286:27B	Multicast address
FE80::6891:BCA1:A439	Link-local address
FC00:89:F:6::DE4:A7D3:789A	Unique-local address
2042:DC8:A:D4:28:EF3D:12E2:AC4A	Global unicast address

Activity 14: Subnetting practice

Find the activity in your **Learner Activity Document**, and read the instructions:

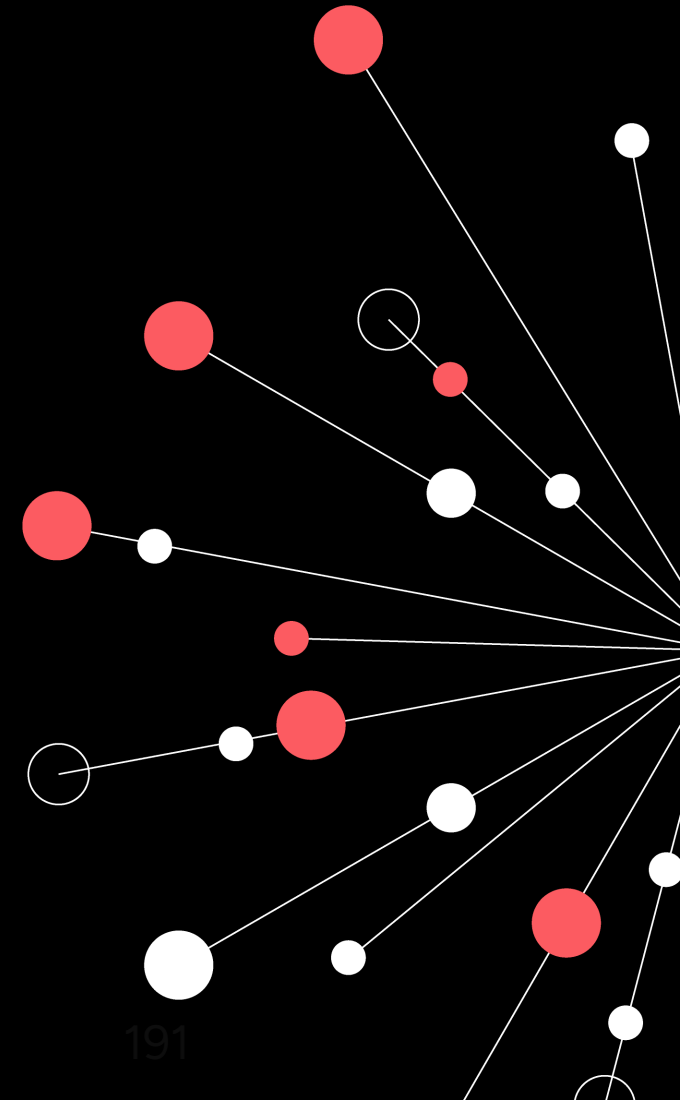
- Identify subnetting requirements for the network topologies.



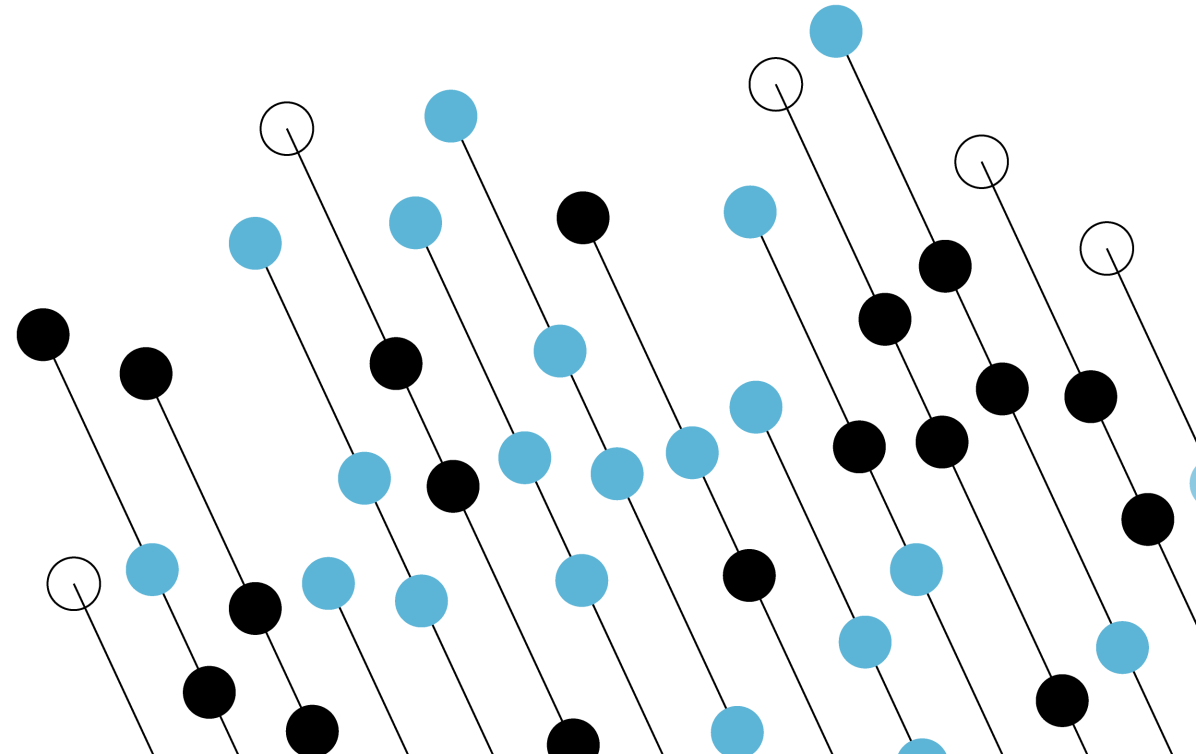
Activity 15: Variable length subnet masks practice

Find the activity in your **Learner Activity Document**, and read the instructions:

- Analyse the scenario and design a VLSM scheme to meet the given requirements.



DHCP



What is DHCP?

What does it stand for?

- Dynamic host configuration protocol.

What does it do?

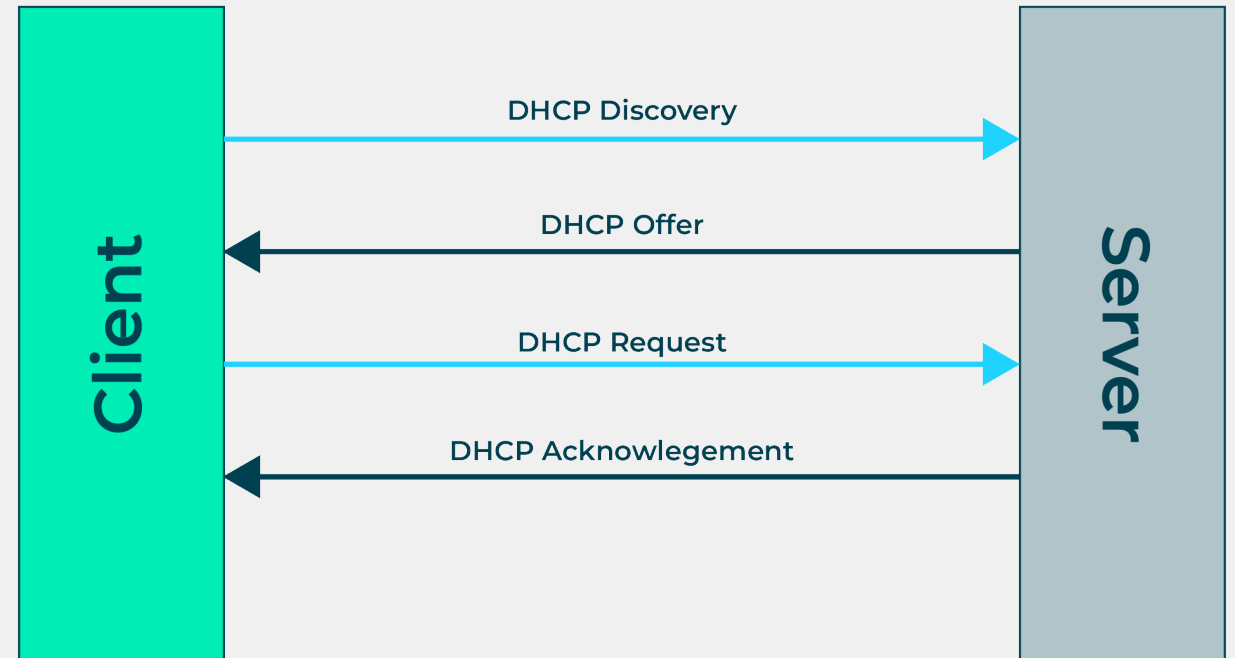
- Issues an IP address to clients that request one.

How many steps in the process?

- Four

What are the names of the steps?

- Discover, offer, request, acknowledge

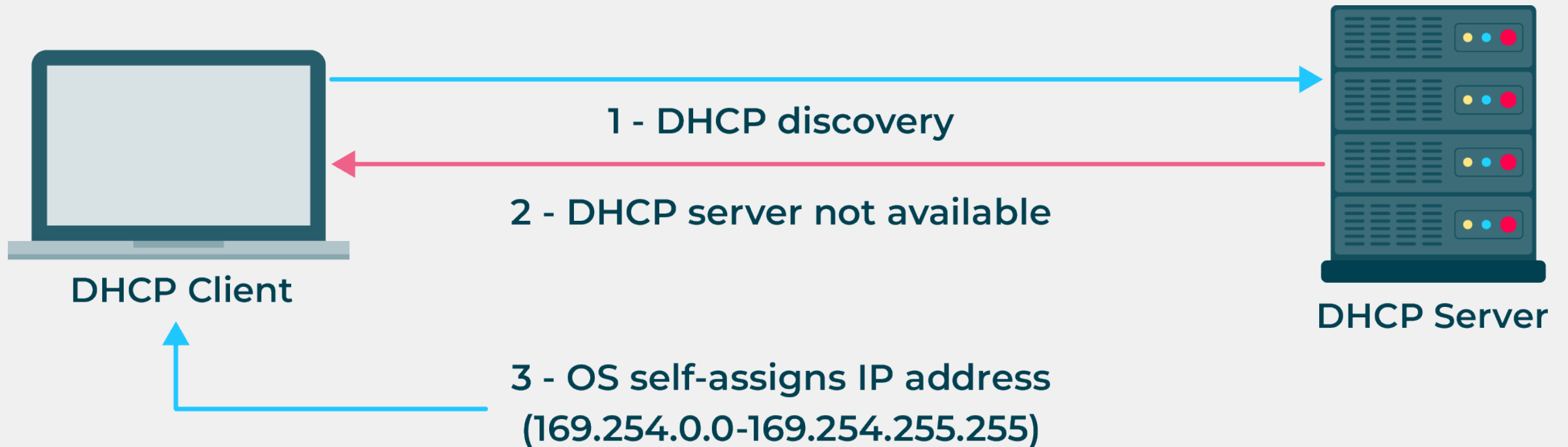


Automatic private IP addressing (APIPA)

If a DHCP server isn't available, DHCP clients can automatically self-configure an IP address and subnet mask using APIPA.

A client uses the self-configured IP address until a DHCP server becomes available.

APIPA process

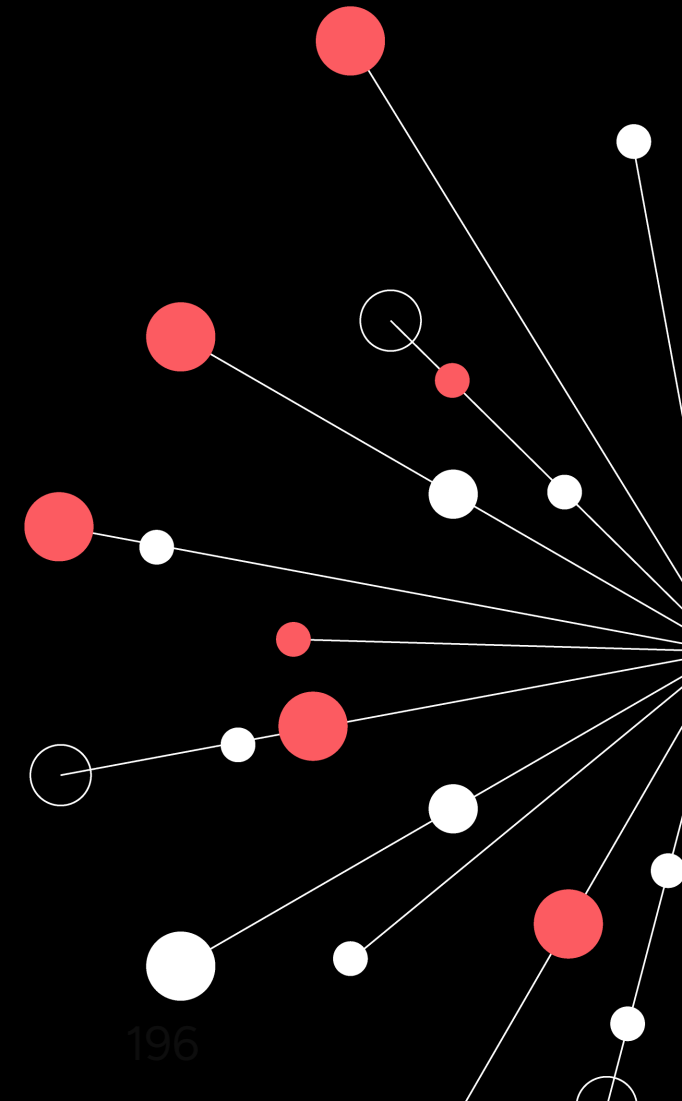


Activity 16: DHCP and APIPA lab

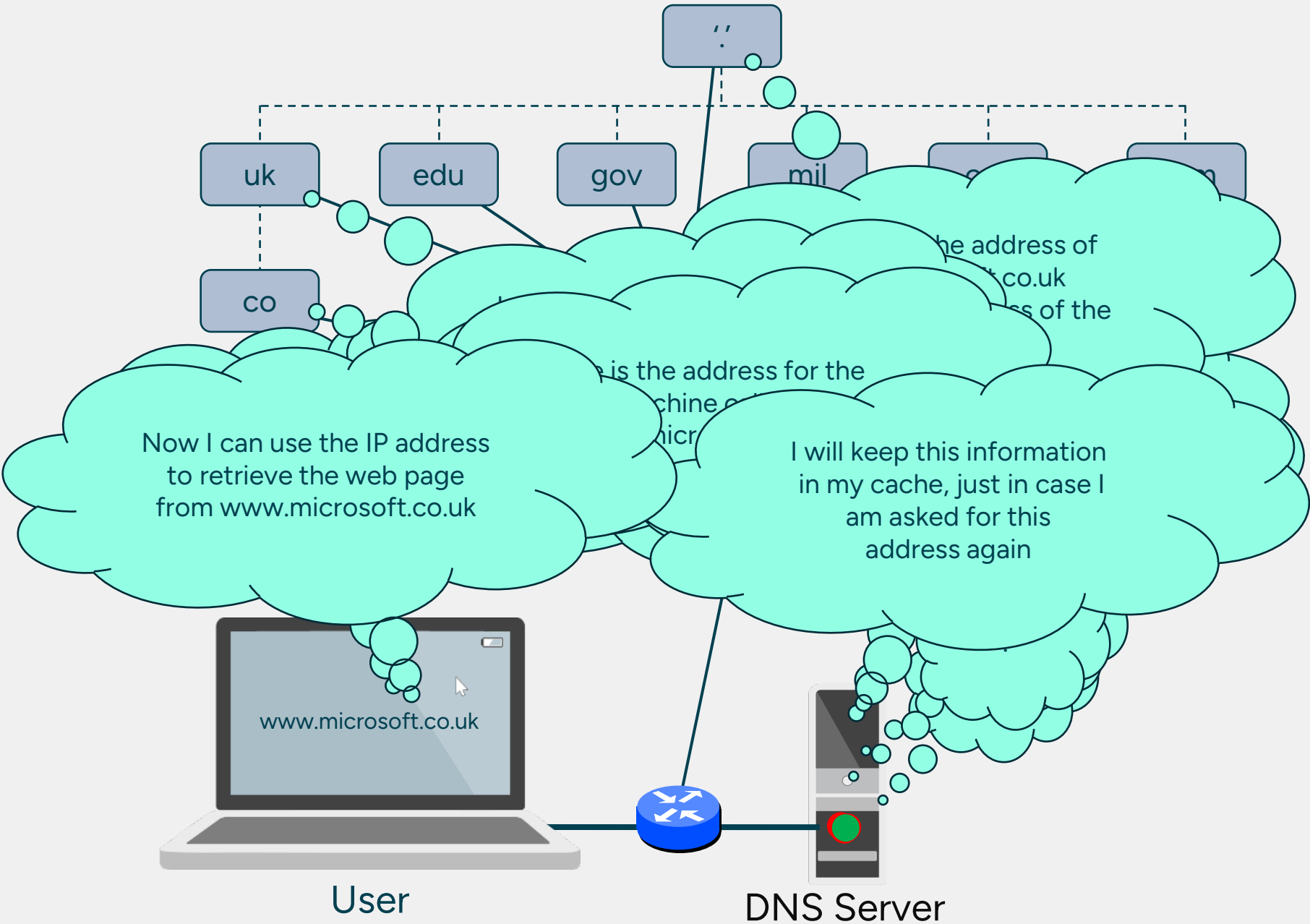
Open the '**DHCP and APIPA.pkt**' file in Packet Tracer.

Find the activity instructions in the **Learner Activity Document**.

Follow the instructions to complete the lab.



DNS



DNS records

What the purpose of each type of DNS record:

- NS
- A
- AAAA
- PTR
- CNAME
- MX

DNS records (continued)

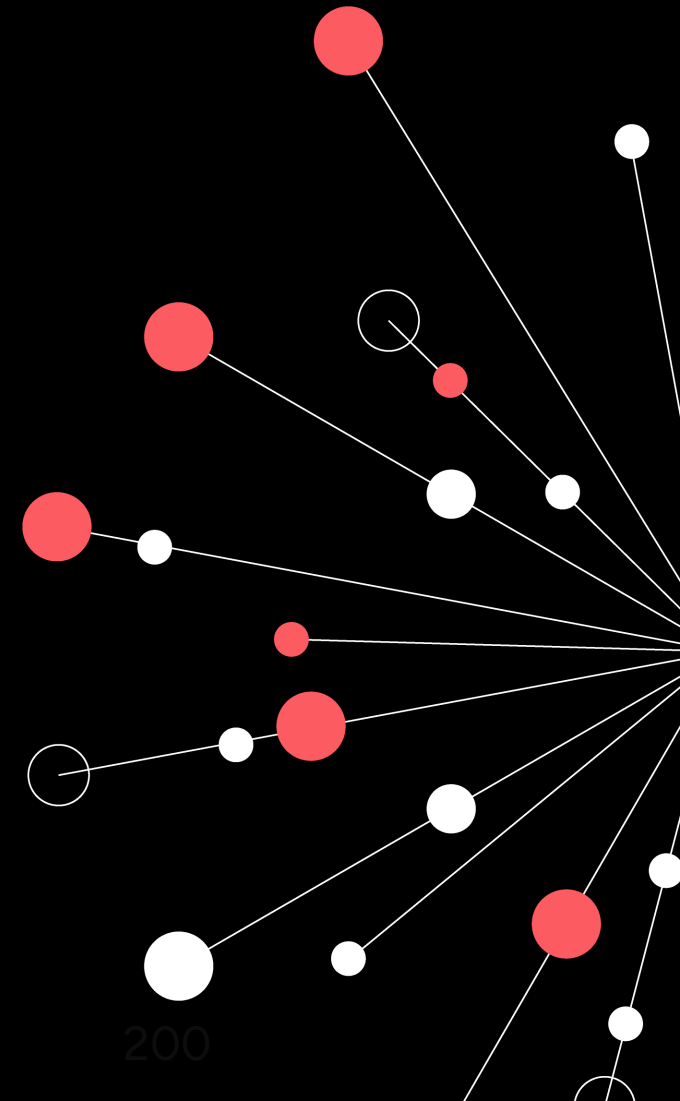
Name of DNS record	Description
SOA (start of authority)	Contains administrative information about the zone, including the primary name server, the email of the domain administrator, the domain's serial number, and timers relating to refreshing the zone.
NS (name server)	Specifies the authoritative name servers for the domain.
A (address)	Maps a domain name to an IPv4 address.
AAAA (IPv6 address)	Maps a domain name to an IPv6 address.
PTR (pointer)	Maps an IP address to a domain name, typically used for reverse DNS lookups.
CNAME (canonical name)	Alias for one name to another: the DNS lookup will continue by retrying the lookup with the new name.
MX (mail exchange)	Specifies the mail servers responsible for receiving email on behalf of the domain.

Activity 17: DNS lab

Open the '**DNS lab.pkt**' file in Packet Tracer.

Find the activity instructions in the **Learner Activity Document**.

Follow the instructions to complete the lab.



End of Day 3 summary

- Numbering systems
- Internet protocol version 4 (IPv4)
- Subnetting
- IPv4 versus IPv6
- Network address translation (NAT)
- Dynamic host configuration protocol (DHCP)
- Domain name system (DNS)

Network infrastructure

Day 4

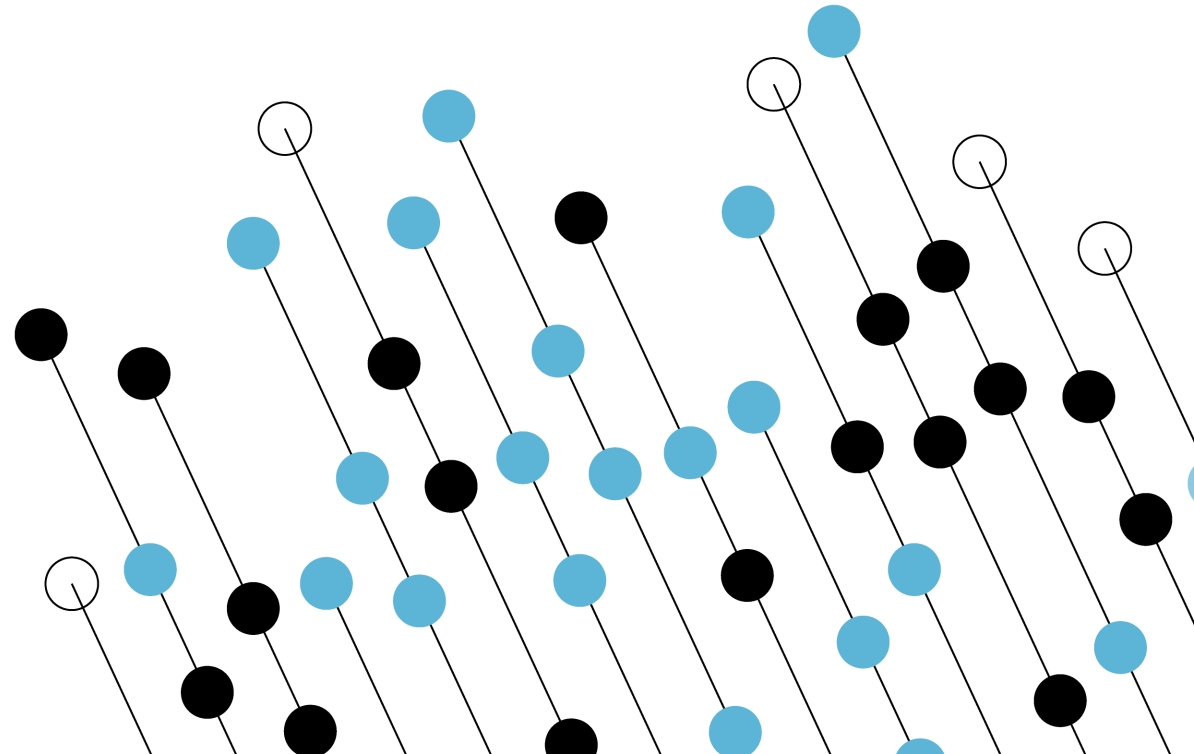


Overview

Day 4

- Routing principles
- Static routing
- The default route
- Dynamic routing protocols:
 - RIP
 - EIGRP
 - OSPF
 - BGP

Routing



Routing

- How do advanced routing protocols like OSPF, BGP, and EIGRP differ in their design and use cases?
- What factors determine the choice of a specific routing protocol in an enterprise network?
- How do hybrid routing protocols (for example, EIGRP) combine the best features of distance-vector and link-state protocols?
- What is route convergence, and why is it critical in large networks?

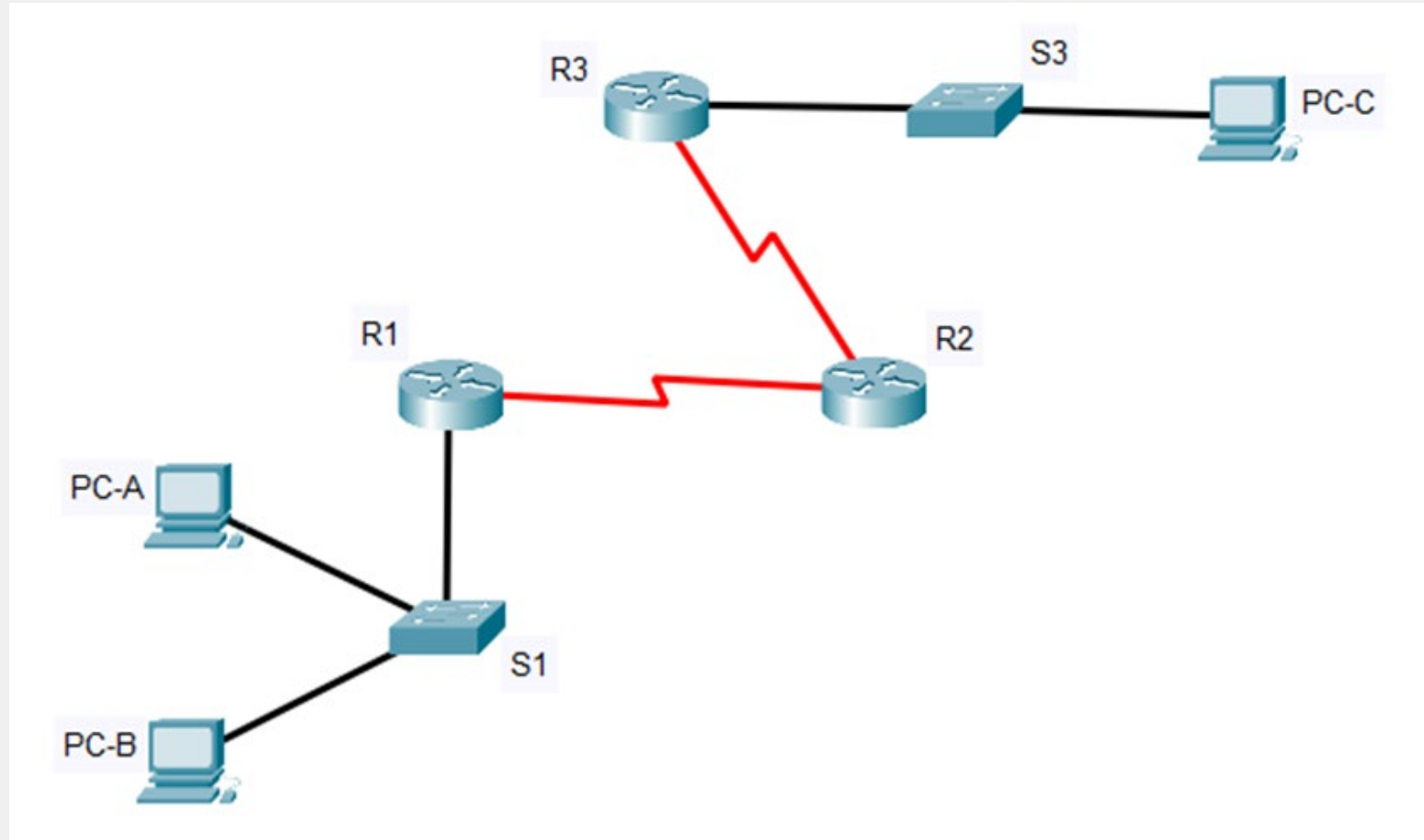
Routing protocols

- How do different routing protocols achieve convergence, and which of the protocol has the fastest convergence time?
- What challenges arise in scaling routing protocols in global networks like the internet?
- What is route redistribution, and why is it necessary in multi-protocol environments?
- What challenges arise when redistributing routes between different protocols?

Static routing

- What are the primary advantages of static routing in small or stable networks?
- What challenges arise when using static routing in large or dynamic networks?
- In what scenarios is static routing preferred over dynamic routing?
- How does static routing contribute to network security?
- What are the key considerations when configuring static routes?
- How does manual configuration of static routes impact network management and troubleshooting?

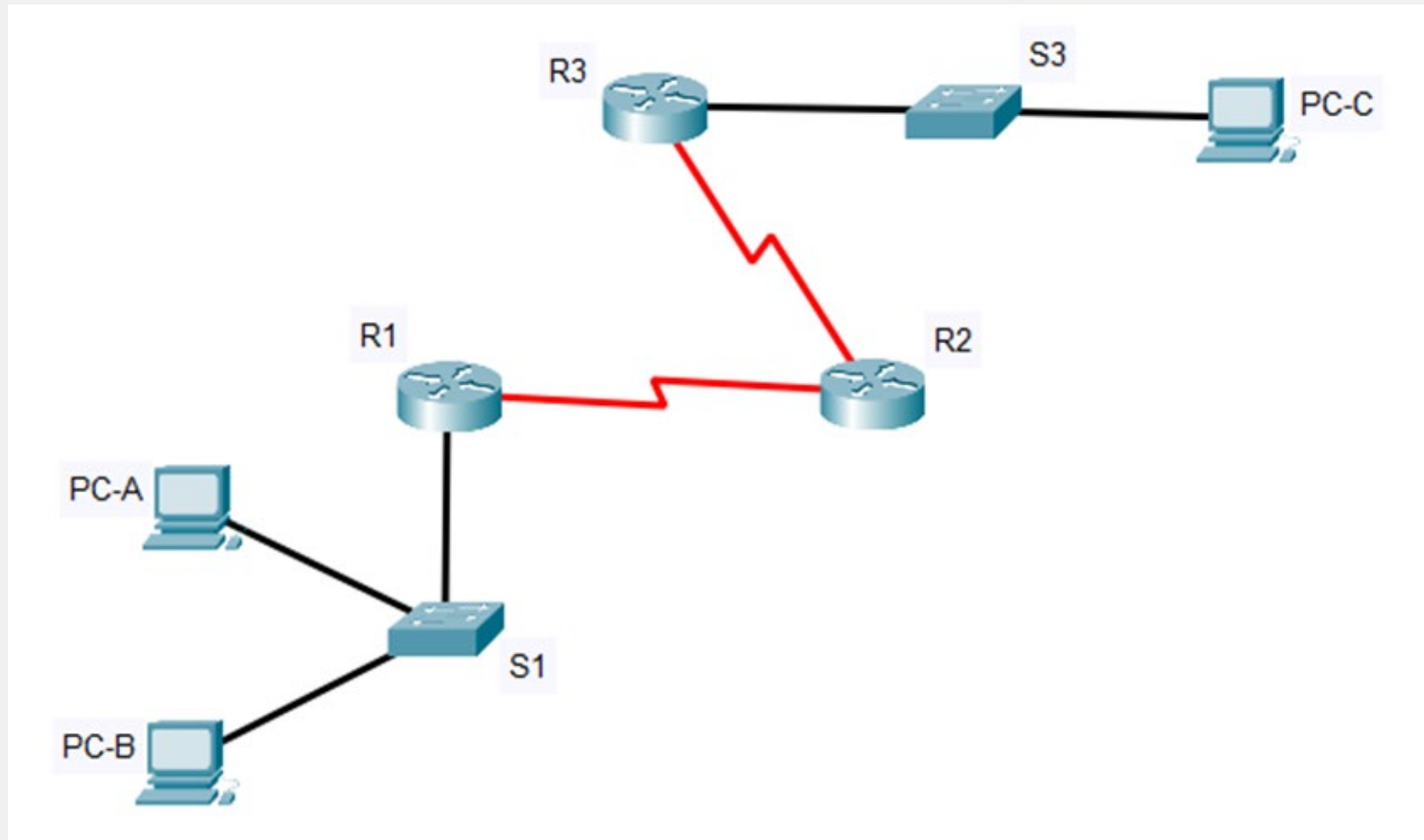
Static route configuration demo



The default route

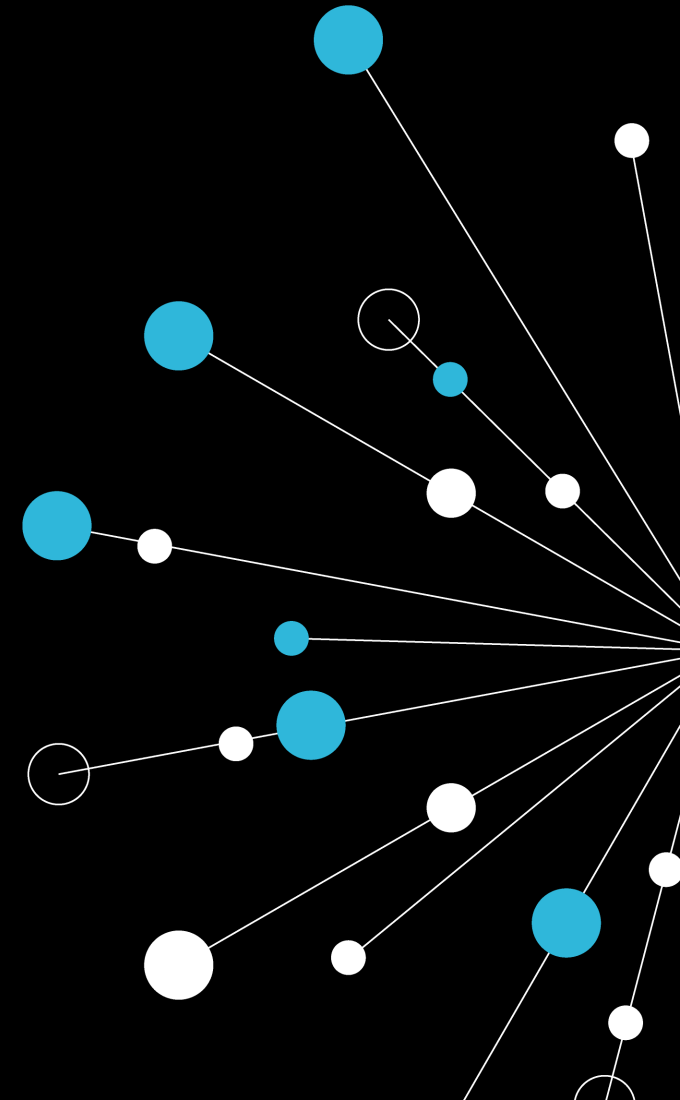
- What is the purpose of a default route in a network?
- How does a default route simplify routing in smaller or branch office networks?
- How does a router decide to use a default route when multiple routes are available?
- What are the risks of relying solely on default routes in a network?
- How are default routes used in connecting a LAN to the internet?
- What are the differences between default routes in IPv4 and IPv6?

Default route configuration demo



Activity 18: Static routing

Open the '**Activity 18 Static routing.pka**' file in Packet Tracer.
Find the activity instructions in the **Learner Activity Document**.
Follow the instructions to complete the lab.



Dynamic routing

- How does dynamic routing address the limitations of static routing?
- What are the trade-offs between the control of static routing and the adaptability of dynamic routing?
- How do network size and complexity influence the choice of a dynamic routing protocol?
- Why are some dynamic routing protocols more suitable for enterprise networks than others?
- How does dynamic routing handle changes in large-scale networks?
- What are the challenges of maintaining scalability with dynamic routing protocols?

Distance vector routing

- How does the distance-vector algorithm calculate the best path to a destination?
- Why is the 'count to infinity' problem a significant issue in distance-vector routing?
- How do protocols like RIP and EIGRP implement distance-vector routing?
- Why is RIP rarely used in modern networks compared to other protocols?
- What are the scalability and convergence limitations of distance-vector protocols?
- How do split horizon and poison reverse mitigate routing loops in distance-vector protocols?

Link-state routing

- How does link-state routing overcome the limitations of distance-vector protocols?
- Why is link-state routing more suitable for large and complex networks?
- How do routers using link-state protocols maintain and share information about the network topology?
- What is the role of Dijkstra's algorithm in link-state routing?
- What are the differences between OSPF and IS-IS in terms of design and deployment?

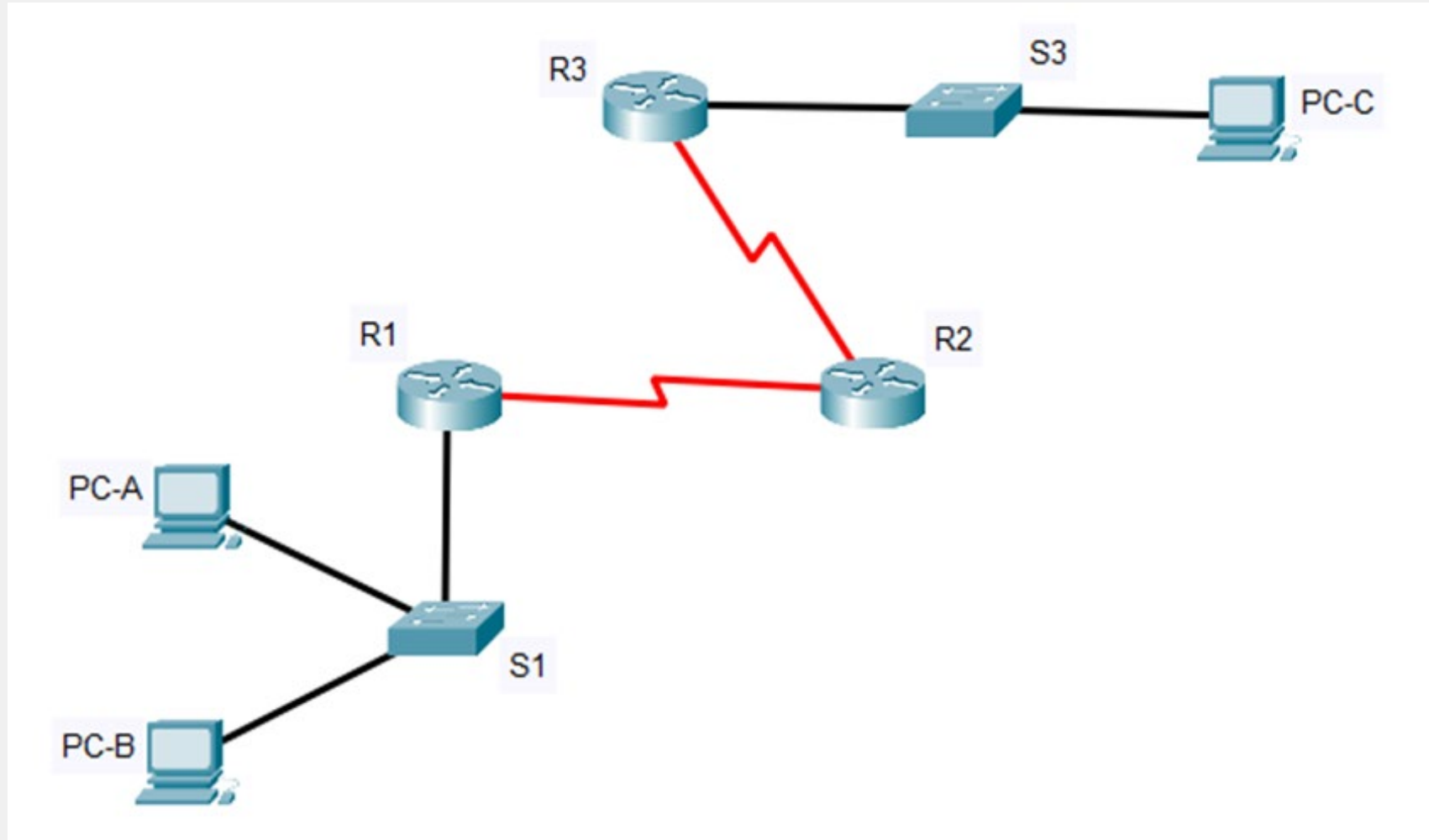
Summary and quiz

- Advanced routing
- Types of routing
- Static
- Default
- Dynamic
 - RIP
 - EIGRP
 - OSPF
 - BGP

Routing information protocol (RIP)

- What are the key differences between RIPv1 and RIPv2 in terms of functionality and features?
- Why is RIPv2 considered an improvement over RIPv1?
- What are the scalability limitations of RIP protocols, and how do they impact large networks?
- How does RIP's maximum hop count of 15 affect its usability in enterprise environments?
- In what scenarios is RIP still relevant, despite the availability of more advanced protocols?
- How does the addition of authentication in RIPv2 improve network security?

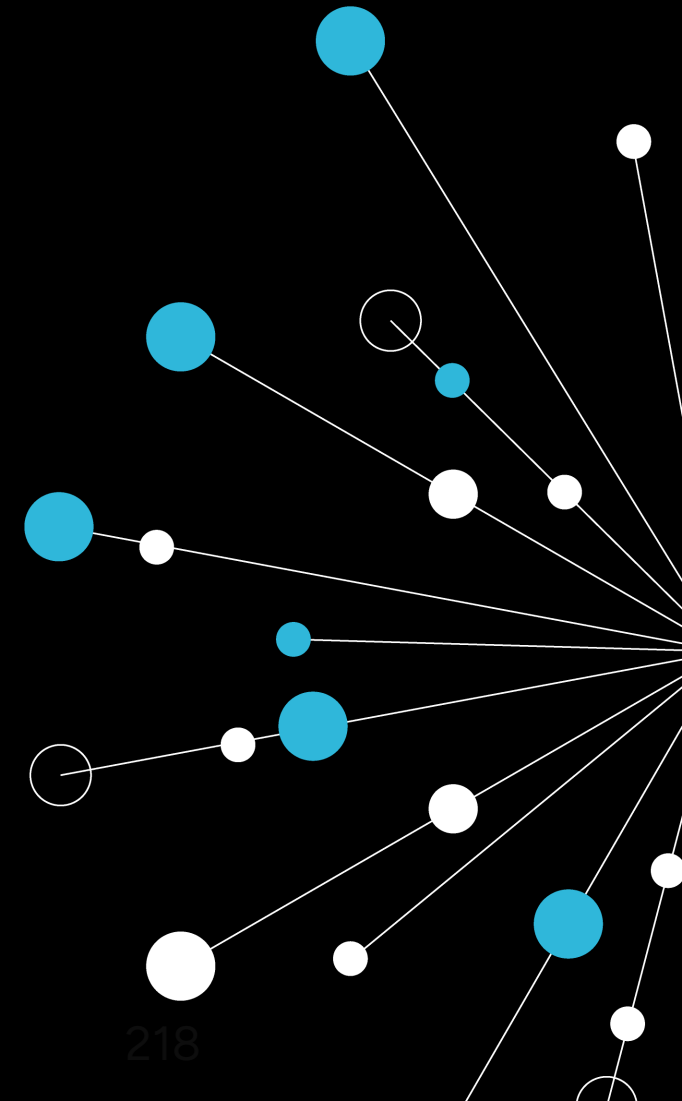
RIP configuration demo



Activity 19: Configure RIPv2 lab

Find the activity instructions in the **Learner Activity Document**.

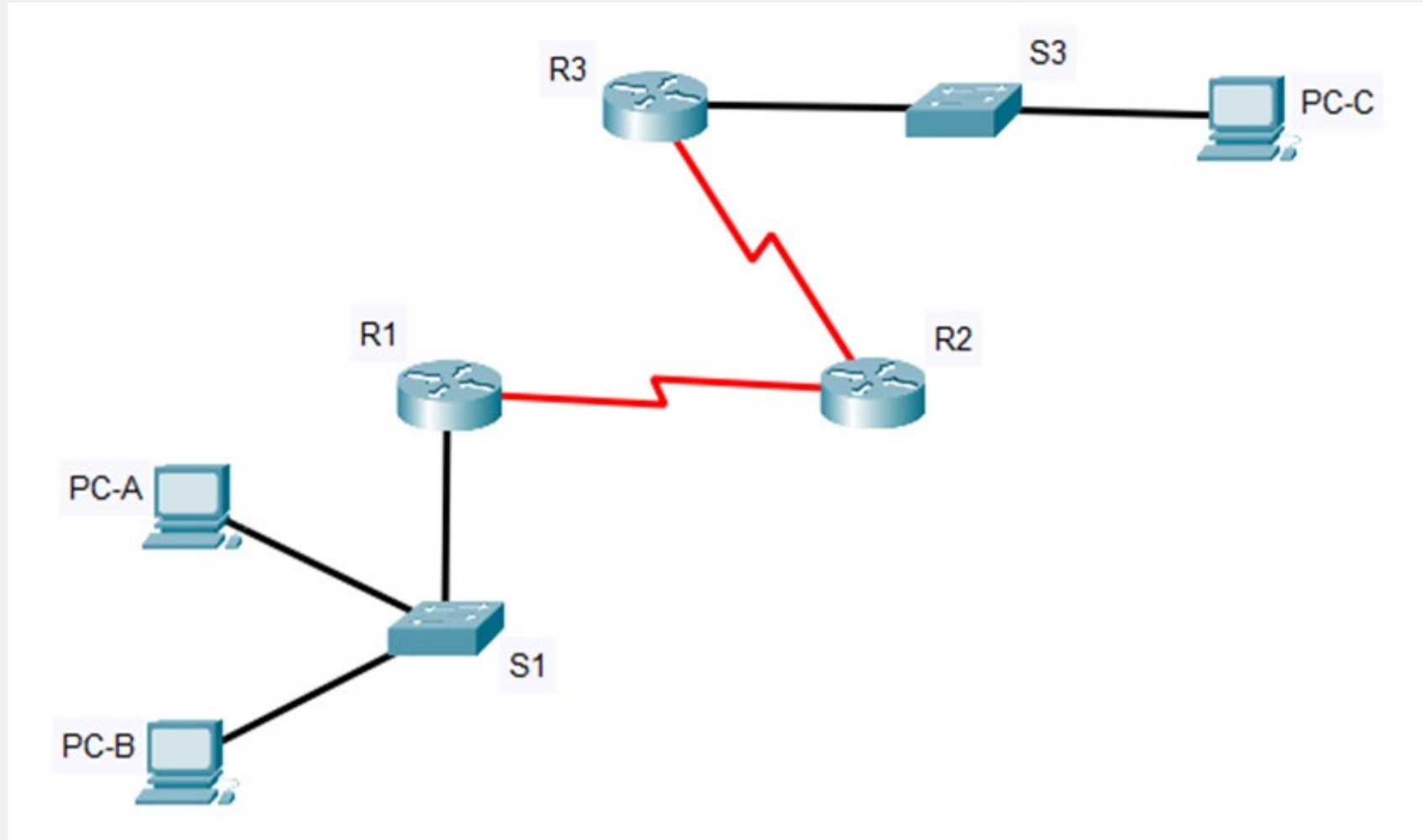
Follow the instructions to complete the lab.



Enhanced interior gateway routing protocol (EIGRP)

- How does EIGRP differ from traditional distance-vector protocols like RIP?
- What is the role of the diffusing update algorithm (DUAL) in EIGRP?
- What makes EIGRP a popular choice for enterprise networks?
- How does EIGRP support both IPv4 and IPv6 networks?
- What are the limitations of EIGRP in multi-vendor environments?
- How does EIGRP's reliance on Cisco proprietary technology affect its adoption?

EIGRP configuration demo

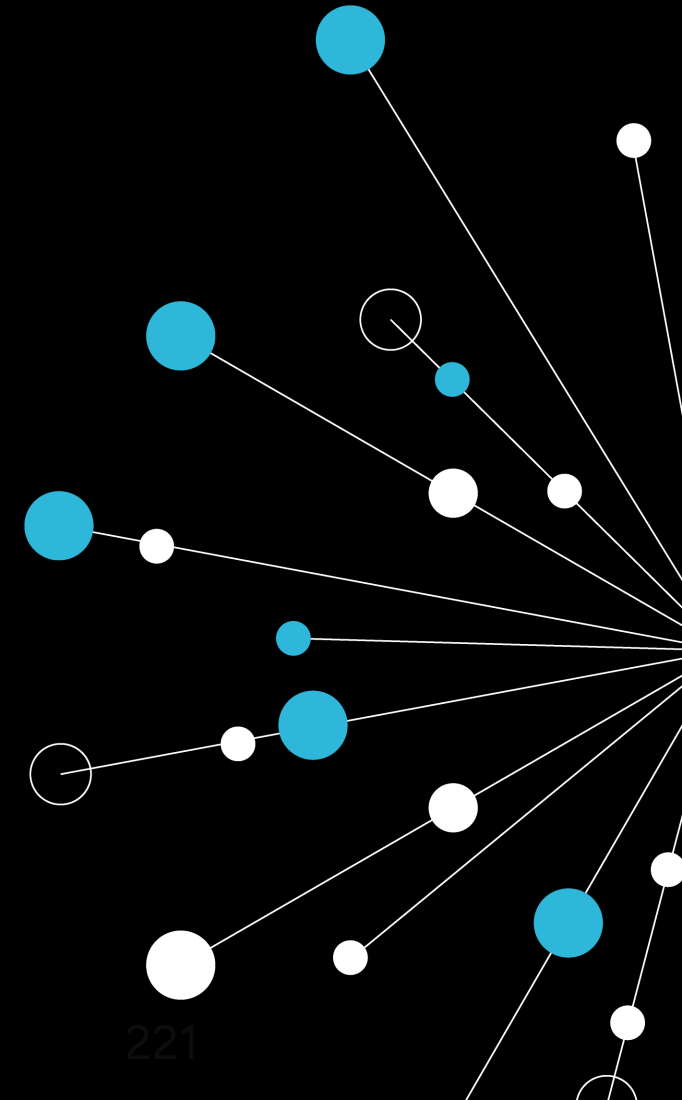


Activity 20: EIGRP lab

Open the '**Activity 20 EIGRP lab.pka**' file in Packet Tracer.

Find the activity instructions in the **Learner Activity Document**.

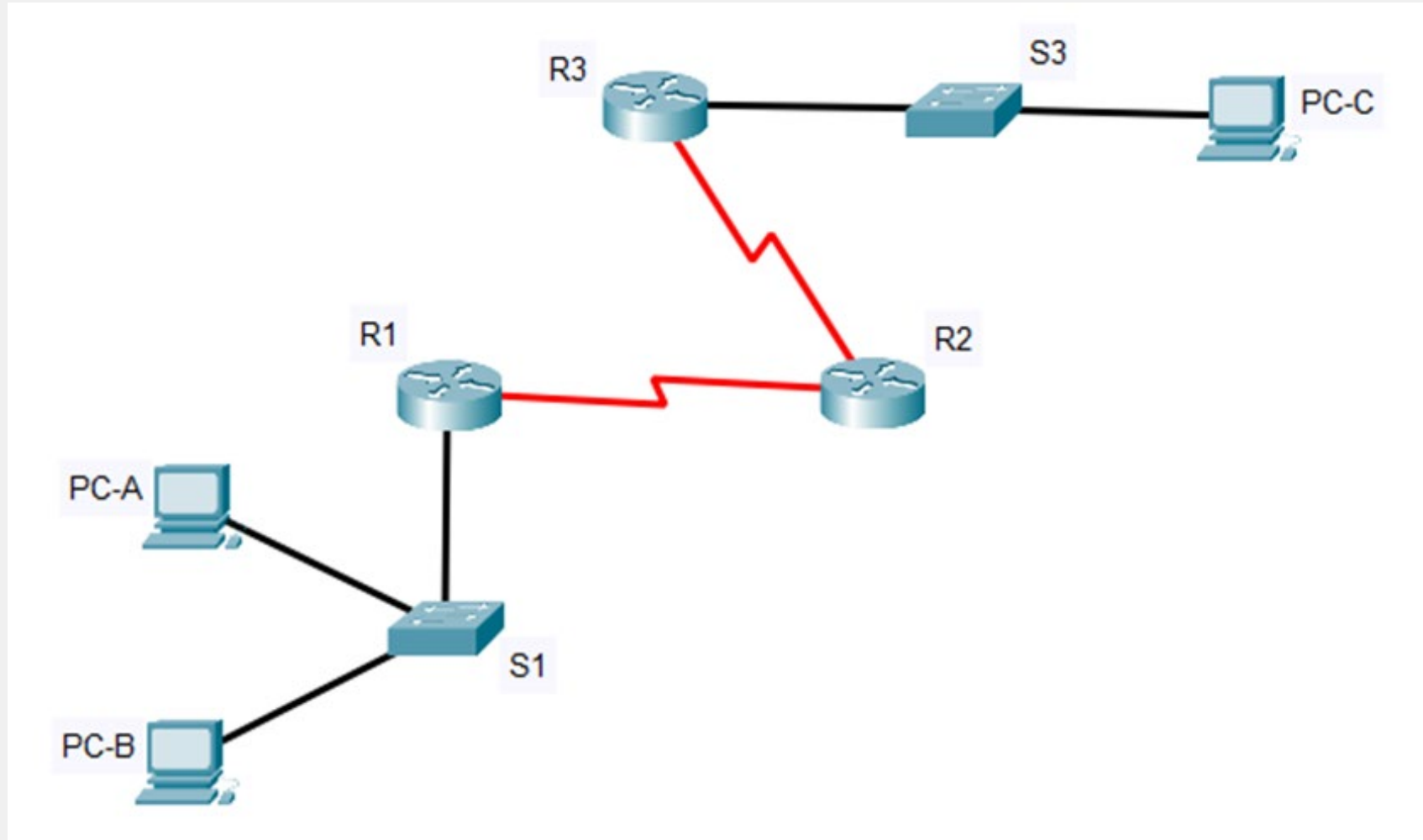
Follow the instructions to complete the lab.



Open shortest path first (OSPF)

- How does OSPF achieve faster convergence compared to distance-vector protocols?
- What is the significance of OSPF's hierarchical design with areas?
- How does OSPF handle large-scale networks using areas and LSAs (link-state advertisements)?
- What is the role of OSPF's cost metric in path selection?
- How does OSPF compare to EIGRP in terms of scalability, performance, and complexity?
- Why is OSPF preferred over RIP in dynamic and large networks?

OSPF configuration demo



Intermediate system to intermediate system (IS-IS)

- How does IS-IS differ from OSPF in terms of design and implementation?
- What makes IS-IS a popular choice for service provider networks?
- Why is IS-IS considered simpler and more flexible in large-scale deployments?
- How do OSPF and IS-IS differ in their use of areas and metrics?
- Why is IS-IS widely used in carrier networks and not as common in enterprise networks?
- How does IS-IS support both IPv4 and IPv6 routing?

Border gateway protocol (BGP)

- Why is BGP referred to as the 'backbone of the internet'?
- How does BGP handle routing between autonomous systems (AS)?
- How does BGP differ from IGP (interior gateway protocols) in terms of route advertisement?
- What are the security risks associated with BGP, such as route hijacking and prefix leaks?
- How can technologies like RPKI and BGP monitoring improve BGP security?

Day 4 summary

Day 4

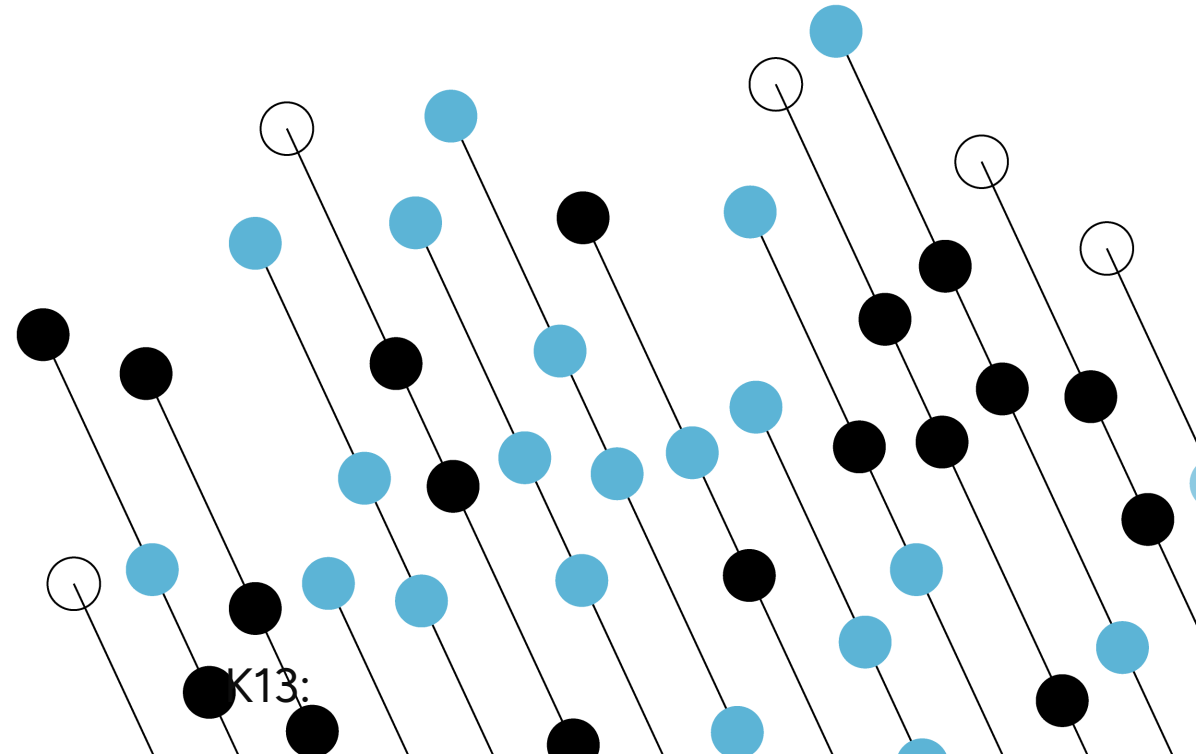
- Routing principles
- Static routing
- The default route
- Dynamic routing protocols:
 - RIP
 - EIGRP
 - OSPF
 - BGP

Network Fundamentals

Day 5



Concepts of cloud and cloud services



Cloud computing

- What is cloud computing and how does it differ from traditional computing?
- How does cloud computing differ from traditional on-premises IT infrastructure?
- What are the essential characteristics of cloud computing, such as on-demand self-service, resource pooling, and scalability?
- How does the concept of elasticity in cloud computing support businesses?
- What are some common examples of cloud computing services we use in daily life?

Benefits of cloud computing

- How does cloud computing help businesses scale operations without significant upfront costs?
- In what ways does the pay-as-you-go pricing model benefit organisations?
- How does cloud computing enhance disaster recovery and business continuity?
- What role does cloud computing play in remote work and collaboration?

Benefits of cloud computing (continued)

- How does cloud computing enable faster development and deployment of applications?
- How can businesses leverage the cloud for data-driven decision-making?
- What are the primary benefits of using cloud computing for businesses?
- How does cloud computing enhance scalability and flexibility?
- In what ways does cloud computing contribute to cost savings?

Drawbacks of cloud computing

- What are the potential security risks associated with cloud computing?
- How can data privacy be a concern in cloud environments?
- What are the challenges related to cloud service outages and downtime?
- How does vendor lock-in impact businesses that rely heavily on a single cloud provider?
- How can the costs of cloud services spiral out of control without proper management?
- What challenges arise in estimating the total cost of ownership (TCO) for cloud solutions?

Cloud deployment models

- What are the different cloud deployment models (public, private, hybrid, and community)?
- When would an organisation choose a hybrid cloud over other deployment models?
- What is a community cloud, and in what scenarios is it beneficial?
- How do shared responsibilities work in a community cloud environment?
- What factors should a business consider when selecting a cloud deployment model?
- How do regulatory and compliance requirements influence the choice of deployment model?
- What are the advantages and disadvantages of each deployment model?

Cloud computing (2)

- How do shared responsibilities work in a community cloud environment?
- What factors should a business consider when selecting a cloud deployment model?
- How do regulatory and compliance requirements influence the choice of deployment model?
- What are the advantages and disadvantages of each deployment model?

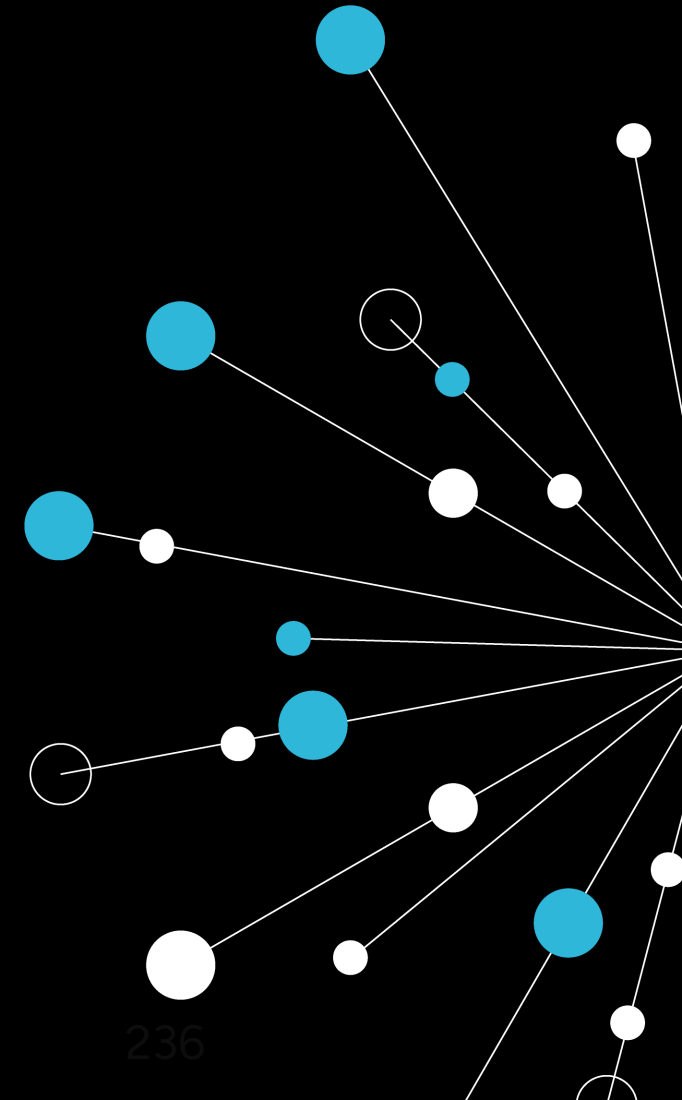
Cloud service models

- What are the key differences between infrastructure as a service (IaaS), platform as a service (PaaS), and software as a service (SaaS)?
- How do these service models differ in terms of user control and responsibility?
- Can you provide examples of popular services for each model?
- How do businesses use these models to address specific needs?
- How can an organisation combine IaaS, PaaS, and SaaS to build a complete solution?

Activity 21: Recommend a cloud deployment model and service type

Find the activity in your **Learner Activity Document**, and read the instructions:

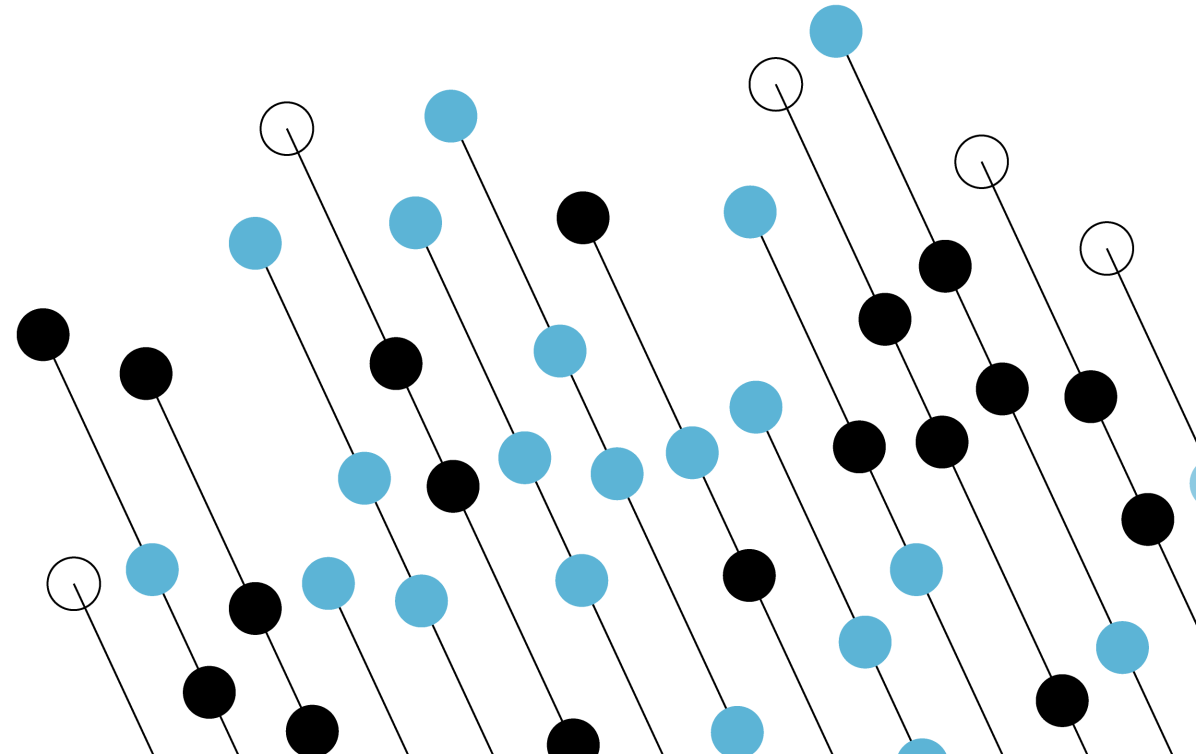
- Review the case study scenario.
- Select an appropriate cloud deployment model and service type.
- Prepare a justification for your choices to share with the group.



Summary

- Cloud computing
- Cloud deployment models:
 - Public
 - Private
 - Hybrid
- Cloud service models:
 - IaaS
 - PaaS
 - SaaS

Virtualisation



Virtualisation

- What is virtualisation and how does it relate to cloud computing?
- How does virtualisation improve hardware utilisation and efficiency?
- What are the different types of virtualisation (server, storage, network)?
- What are the benefits of using virtualisation in data centres?
- How does virtualisation simplify IT management and deployment?
- What are some common challenges faced when implementing virtualisation?
- How does virtualisation impact system performance in resource-intensive environments?

Hypervisors

- What is a hypervisor, and why is it a critical component of virtualisation?
- How does a hypervisor manage virtual machines on physical hardware?
- What are the two main types of hypervisors (Type 1 and Type 2)?
- How do Type 1 and Type 2 hypervisors differ in terms of architecture and performance?
- What are the advantages and disadvantages of using Type 1 vs Type 2 hypervisors?
- What are some examples of popular hypervisors for each type?
- How do different hypervisors compare in terms of performance, features, and compatibility?

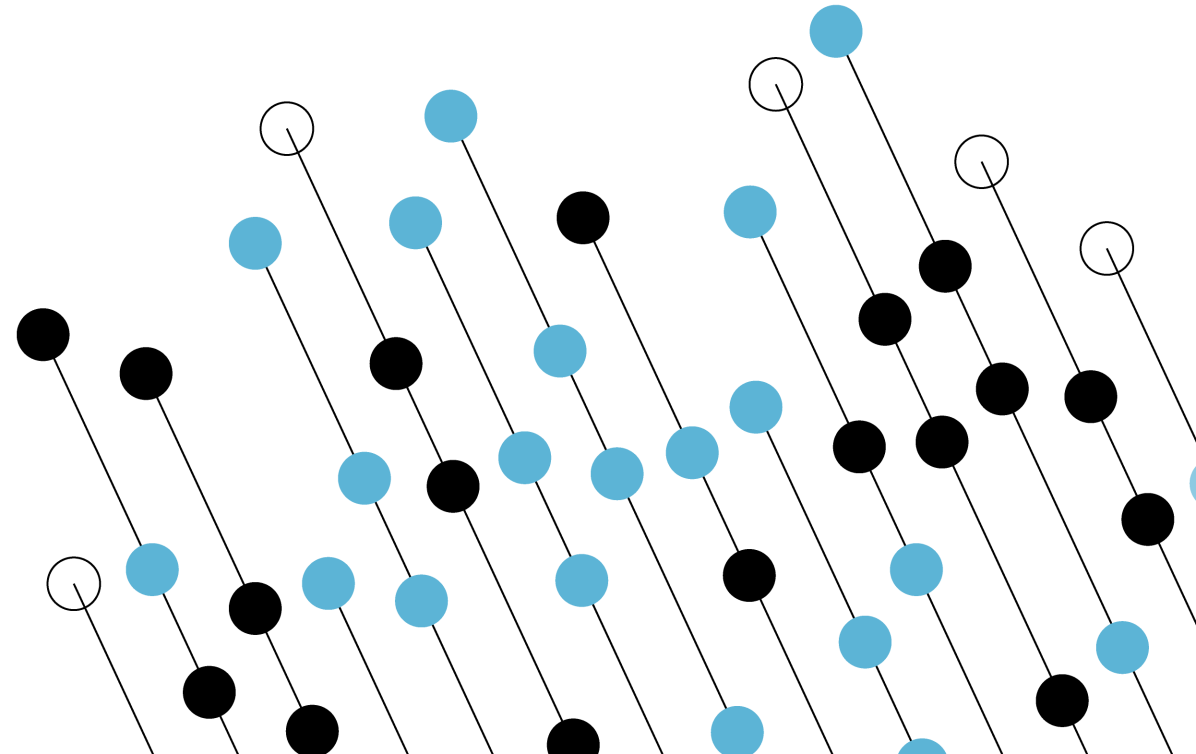
Virtual machines (VMs)

- How do virtual machines differ from physical machines?
- How are virtual machines (VMs) used in cloud computing?
- In what scenarios are virtual machines preferred over physical hardware?
- How are virtual machines used in application development, testing, and deployment?
- How do VMs contribute to disaster recovery and business continuity?
- What are the performance and resource limitations of virtual machines?
- How do VMs compare to container-based virtualisation (for example, Docker)?

Virtualisation summary

- Virtualisation
- Hypervisor
- Types of hypervisor:
 - Type 1
 - Type 2
- Virtual machines

Basic network troubleshooting



The importance of troubleshooting

- Why is network troubleshooting a critical skill for network engineers?
- What are the potential consequences of unresolved network issues?

Common tools for network troubleshooting

- What are the most commonly used tools for network troubleshooting?
- How can network monitoring tools (for example, SNMP, SolarWinds) assist in identifying problems?

Troubleshooting methodologies

- How does the OSI model help in identifying and isolating network issues?
- What are the steps in a structured troubleshooting approach?

Switching troubleshooting

- What are some common issues with network switches, such as port failures or misconfigurations?
- How do you identify and troubleshoot problems caused by faulty switch hardware?

Switch configuration and performance

- How can incorrect duplex settings or speed mismatches affect network performance?
- What are the best practices for verifying and troubleshooting switch port configurations?

Spanning tree protocol (STP)

- How does STP help prevent switching loops, and what are the signs of STP-related issues?
- What steps can you take to troubleshoot a network with frequent topology changes?

VLAN troubleshooting

- How can VLAN mismatches cause connectivity issues between switches?
- What tools or commands can you use to verify VLAN assignments on switches?

Inter-VLAN communication issues

- What are common problems that arise in inter-VLAN routing configurations?
- How do you troubleshoot devices that fail to communicate across VLANs?

Trunking problems

- How do native VLAN mismatches on trunk links cause network issues?
- What are the key steps to troubleshooting 802.1Q trunking between switches?

IPv4 troubleshooting

- What are the symptoms of an IP addressing conflict in a network?
- How do incorrect subnet masks or gateway configurations affect device communication?

Automatic IP troubleshooting

- What are common problems associated with DHCP servers and clients?
- How can you diagnose and resolve issues with DHCP lease allocation?

Ping and connectivity tests

- How does the ping command help identify connectivity issues in an IPv4 network?
- What does it mean if a device responds to a ping but still cannot communicate with others?

Routing: Static route issues

- How do misconfigured static routes lead to connectivity problems?
- What tools can you use to verify and troubleshoot static routing issues?

Dynamic routing protocols

- What are the common causes of routing table inconsistencies in dynamic routing protocols?
- How do you troubleshoot issues in protocols like RIP, OSPF, or EIGRP?

Default gateway and route

- What happens if a default gateway is missing or misconfigured?
- How can you diagnose problems caused by a missing or incorrect default route?

Activity: Build a network based on a case-study scenario

In this activity, you will consolidate and demonstrate what you have learnt throughout the week, by designing a network and including the following:

- Reference to OSI model in relation to:
 - Network type
 - Devices used
 - Cables used
 - Physical / logical topology
- IPv4 address scheme (subnetting / VLSM)
- Routing configured
- Services deployed
- Troubleshooting approach

End of session summary

- Compare and contrast various devices, their features, and their appropriate placement on the network.
- Explain the characteristics of network topologies and network types.
- Summarise the types of cables and connectors and explain which is the appropriate type for a solution.
- Compare and contrast the Open Systems Interconnection (OSI) model layers and encapsulation concepts.

End of session summary and quiz

- Configure a subnet and use appropriate IP addressing schemes.
- Explain common ports and protocols, their application, and encrypted alternatives.
- Explain the use and purpose of network services.
- Explain basic corporate and datacentre network architecture.
- Summarise cloud concepts and connectivity options.
- Install and configure the appropriate wireless standards and technologies.